JOPE DIME / PRIVATE MAINTAINER EDITION

Algorithm Synthesis Model

Technology, operations and the evidence behind the claims

Learns to generate and optimize executable algorithms from data.

User & maintainer manual
23 September 2026

Implementation: reviewed PR83 / main a75e543d2166
30 frozen datasets · 522 feature records · full gap checklist
Historical rankings and current measurements are identified separately.

How to use this book

Start with the status chapter to distinguish what has shipped, what has been demonstrated, and what remains pending. Operators can then use installation and training procedures; maintainers can follow the architecture, language matrix, benchmark contract, detailed roadmap and evidence index.

The appendix contains development statistics for all thirty frozen datasets. The adjacent data directory supplies complete machine-readable statistics, historical ranking inputs, metadata inventories and source pins. The property list supplies the actionable checklist.

Snapshot: implementation reviewed at 68f56404c3e199e06413bfd645d3fe0803d7b96f; merged main a75e543d2166017cb0139e0a251d51d9a0c98e08. Both have tree 176b75bc36d76c7ce4eaeb7234677ec013468889. Individual observations retain their own timestamps.

This private manual explains internal methods at the user's request. Public inference bundle allowlists remain separate. No sealed label bodies, model weights or credentials are embedded in this book.

Evidence vocabulary

Implemented means source exists. **Qualified** means a named execution was measured. **Pending** means acceptance remains. **Unknown** is not zero. Historical results do not establish current capability.

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1 / Reading the system and its current status

Purpose and audience

An **Algorithm Synthesis Model (ASM) learns to generate and optimize executable algorithms from data.** JopeDime turns that objective into a bounded, inspectable workflow: validate original data and row assignments, encode a fitting-only context, propose executable recipes, fit them, measure them under a fixed protocol, evolve measured recipes, retain fitted artifacts, and eventually admit qualified examples into another round of model training.

This manual serves operators, scientists, implementation maintainers and reviewers. Operators need installation, preparation, execution, reporting and recovery procedures. Scientists need dataset identities, split boundaries, metrics, model lineage, comparison budgets and limits of the evidence. Maintainers need the workspace map, command contracts, fitted-state formats, feature boundaries and tests. Reviewers need to distinguish code that exists, behavior that has been demonstrated, changes that have merged, and claims still requiring evidence.

The implementation snapshot is **68f56404c3e199e06413bfd645d3fe0803d7b96f**, dated 23 September 2026. The lifecycle appendix records the separately refreshed reviewed-main state. A rebase merge can produce a different commit with the same reviewed tree. Older plans and historical PASS entries do not establish current capability.

This is a maintainer document for the private product repository. Its explicitly requested GP examples and internal operational paths explain the implementation; they do not expand the allowlist for public inference bundles.

How to read a completion claim

Evidence level	Establishes	Does not establish
Source implemented	The named source and command path exist at that revision	Successful execution on this host
Static or fixture verification	Parsing, linting, schema checks or named synthetic tests passed	Real corpus behavior, GPU execution or competitive performance
Bounded local qualification	Named real processes, artifacts, limits and receipts were observed	Other datasets, backends, checkpoints or protocols
Exact-head CI	Required current-attempt jobs and aggregate passed	Skipped tests, all optional features, GPU qualification or promotion
Reviewed merge	Eligible independent native review and guarded merge were verified	Installation qualification or serving promotion
Measured campaign	Admitted and unadmitted outcomes are accounted for under one window	Promised win rate or unbiased sealed-test performance
Release qualification	Exact installation, artifact and backend passed acceptance	Other runtimes, artifacts or numerical policies

A checkpoint file, a nonempty journal, a successful exit and a genuine optimizer update are different observations. A skipped test is not a pass. A missing measurement is unknown, not zero. Cancelled, failed and unavailable fits remain outcomes rather than scores.

Working implementation

The frozen benchmark foundation consumes actual row assignments, original labels, declared categorical columns, missing values, groups and class order. It uses regression RMSE, binary AUROC and multiclass log loss from a complete probability matrix. The method chapter inventories executable native recipes and the four external adapter families.

The shared execution layer has twenty bounded fit slots, resource accounting, independent deadline enforcement, process-tree cleanup, durable journals and original-window resume. Its ten real Linux process/cgroup tests use synthetic workloads to exercise completion, stalled controllers, cancellation, forced process death, memory failure, concurrency and setup failure.

The regression neural path has a qualified native context bridge and a measured proposal-to-fit-to-evolution proof. Policy-origin metadata is checked against pinned producer output, completed inner-fold measurements and measured ancestry. A candidate labelled neural without those links is insufficient. Canonical duplicates retain proposal slots and origins but do not justify fitting the same executable twice.

Observed generations survive cancellation at the arm cutoff and interruption before the final receipt. Resume reauthenticates completed folds, including completed work inside a partially failed candidate. Only jobs positively proven never admitted can acquire a different expiry status. That exception cannot create a new fit, replace fitted state or extend the clock.

Legacy fit, emit and pipeline surfaces also exist. Their recipe emitters can train a recipe again. That is distinct from the pending portable fitted-state export work; consult the language matrix before assuming an emitted artifact needs no refitting.

Latest measured operational proof

This is **one preselected regression-dataset operational qualification**, not the unchanged thirty-dataset demonstration. Its scores are development scores.

Item	Observation
Dataset	`230_machine_cpu`
Initial neural arm	64 neural slots plus 16 exploration slots
Initial generic arm	80 generic slots
Arm / fit allowance	120 seconds per arm; 30 seconds per worker fit
Global proof window	600 seconds, including 30 seconds for reporting
Host limits	20 CPUs/workers, 64 GiB, zero swap, no GPU tasks
Started and finished worker commands	964
Completed / failed commands	834 / 130
Unique fitted predictors / reusable recipes	832 / 164
Final refits / predictions per refit	2 / 34

Item	Observation
Run-command wall time	239.143121970 seconds
Independent report replay	18.718997325 seconds
Neural development RMSE	166.3181717454848
Generic development RMSE	155.25909998168862
Score ordering	Generic has lower RMSE in this comparison
Genuine neural initial fits and fitted descendants	Verified
Teachers admitted from this proof	None
Full frozen-panel conclusion	Not established

The recovered report SHA256 is `6e0c20720256b0735a6969c45032f44082c92abacadf2c4eb21ef5c390f249dd`. The experiment identity is `646095badab6706168c7e869118d62fddd399faa2a9d6bc7f484c95a021a3944`; its original start/deadline are Unix milliseconds `1790180210613` and `1790180810613`. Plan identity: `a7befb386344bd0cbffa866ad8793077dead185f64e2ae0fb4035a4ed3dcfcc2`.

This proves execution and evidence continuity. It does not prove ASM superiority, ten wins per category, learned improvement after training, an under-sixty-second warm demo, installation speed or portable inference for every winner.

Cost: interpret the units first

Exact supervisor CPU was **270.889403 CPU-seconds**; summed worker CPU was **181.220773 CPU-seconds**. The supervisor aggregate contains worker use: **do not add them**. Summed worker wall durations were **202.266164543 seconds**; concurrent durations are not elapsed campaign time.

Amortized over 832 unique predictors, these observations yield approximately 0.217813 worker CPU-seconds, 0.325588 aggregate CPU-seconds and 0.287432 run wall-seconds per predictor. Over 164 recipes, they yield approximately 1.105005, 1.651765 and 1.458190 seconds. These are campaign accounting ratios, not isolated latency predictions for arbitrary models.

GPU work, tokens, money, isolated controller CPU, downloads and installation were not measured by this proof. The supplied one-hour/ten-million-token scenario provides a 3,600-second contextual reference: this proof's run wall time is approximately 6.64% of that duration. This is not a matched-workload speedup. No scenario hardware allocation, predictor count, token trace or cost ledger exists here, so ten million tokens cannot be converted into a measured productivity or currency comparison.

Pending acceptance

Workstream	Current boundary	Remaining acceptance
Regression neural consumption	Qualified contexts, proposals, actual fits and descendants	Full panel and newly trained checkpoint consumption
Classification neural consumption	Experimental context/invariance support; follow-on producer prepared	Task-correct live consumption and clearly labelled fixture scope
HWM and teaching eligibility	Follow-on source prepared	Full identity, finite OOF improvement, four-of-five-fold gate and no-signal refusal

Workstream	Current boundary	Remaining acceptance
Training	Guarded source and test plans prepared	Positive updates, changed parameters, reload, continuation, resume compatibility and bounded GPU proof
Trained regression export use	Explicit experimental consumer prepared	Compile and execute exported-checkpoint-driven fits
Portable fitted inference	Native fitted artifacts; export packet prepared	Full state parity and actual selected-winner backend qualification
Installation and warm demo	Source packets prepared	Checksum install, CLI-only build, preparation, fitted artifact/predictions and separate cold/warm timing
Thirty-dataset comparison	Frozen panel and protocol	Unchanged two-hour campaign, every loss/refusal/failure and gap
Motifs	Separate typed library branch	Eligible-state reader, mine command, umbrella wiring and validated reuse
Broad full-main tests	Historical failures inventoried	Reproduce and repair; fast CI is not full qualification
Serving promotion	Not established	Exact release/backend/model acceptance and reviewed promotion

The detailed checklist and property-list expand these rows into actionable items. A prepared patch or planned command is not an installed or executed capability.

Failure records are retained

Two failed cancellation/crash reproductions preceded the resume correction. A partial-fold fixture initially failed on a noncanonical path, then passed after fixing the fixture without weakening verification. A live run at `d0f5748b7db1ac419110289f912c7eface0022bc` completed its worker inventory but failed report verification because live and recovered scoring populated different fold-evidence fields. Its journal remains unchanged. The shared scorer correction and successful new-head proof are separate evidence.

Final local suites at the snapshot passed **53 benchmark and 106 pipeline tests**. One privileged test was ignored in that invocation and qualified separately through **ten actual host cases**. The numerical encoder proof passed **eight fixed cases** under unchanged tolerances and thirty-second per-job limits. The lifecycle appendix records CI and merge state independently.

2 / ASM architecture, encodings and neural models

Scope and reading rules

Algorithm Synthesis Model (ASM) learns to generate and optimize executable algorithms from data. The implementation combines a learned proposal policy with actual fitted evaluation and genetic variation. A generated program is a candidate recipe; a fitted predictor is that recipe plus the state learned from a specific training partition. A checkpoint is the proposal model. Those three artifacts have different identities and different evidence requirements.

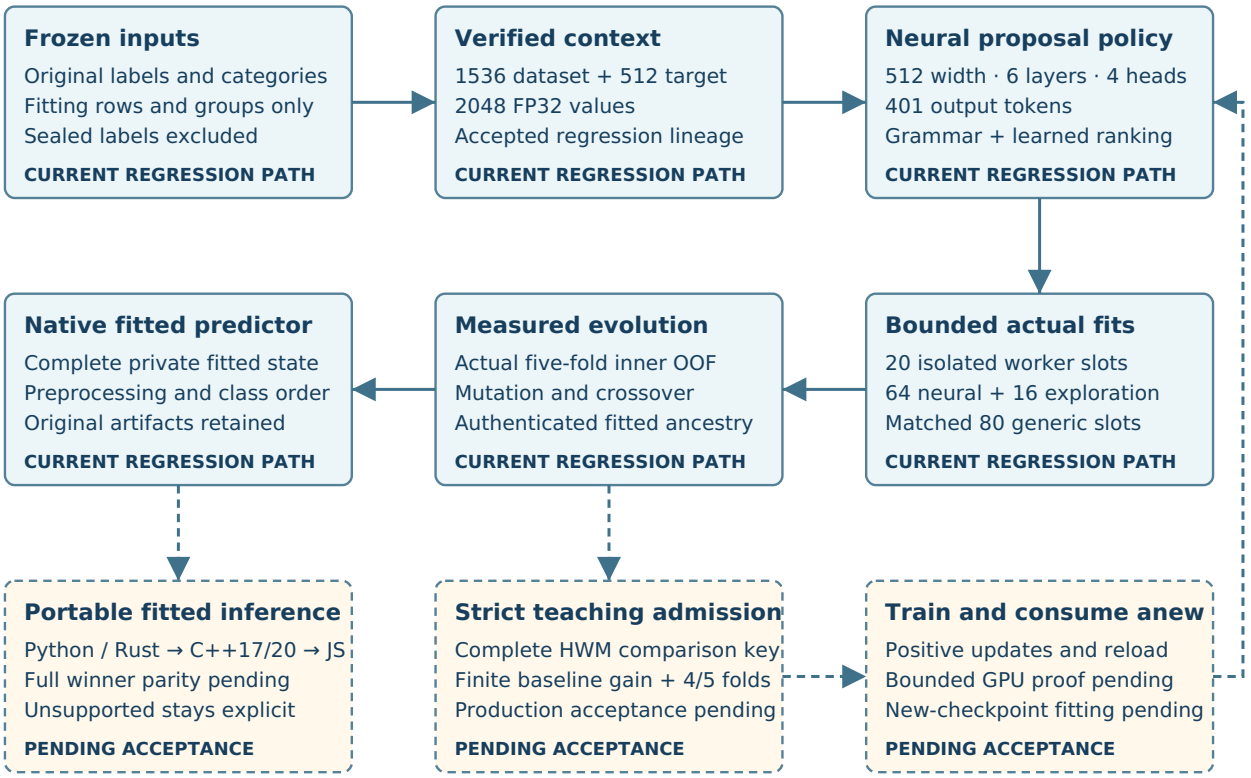
This chapter describes source commit `68f56404c3e199e06413bfd645d3fe0803d7b96f`, the frozen PR83 candidate on 2026-09-23. Changes described as **pending training overlay** refer to the stopped 64-file custody patch `595824ef6eb8e15872be058668503cedad1dd9a5ff98ac4700c6f713251d384c`. That patch includes experimental regression consumption; it is not in this source commit. Current-source references below are repository paths with source line numbers. Overlay references are explicitly marked and use the stopped overlay's line numbers.

This is a maintainer document containing internal model and discovery design. The user's request authorizes this explanation and the companion private GP examples. It does not enlarge the allowlist for distributed inference artifacts or authorize publishing private weights, discovery histories, vocabulary assets or training corpora. Hashes identify dependencies; they are not the dependencies themselves.

The most recent bounded origin proof at this source completed its run command in 239.143 seconds; independent report replay took 18.719 seconds. It demonstrated neural-origin fitting and evolution on one regression dataset. Its neural result lost to the matched generic result. It did not execute the complete thirty-dataset demonstration, create current eligible teachers, train a new policy, or qualify a newly trained policy for serving. The performance chapter gives its exact receipts and denominators.

System at a glance

ASM: the implemented path and the pending learning loop



Current path: one regression operational proof, not a full-panel win. Dashed boxes need runtime acceptance.

Source: reviewed PR83 / main a75e543d; architecture, training and language chapters carry exact identities.

The solid path has bounded regression execution evidence. Dashed stages require the named runtime acceptance; this diagram does not claim an already qualified self-improving training loop.

Component map

Component	Main responsibility	Durable boundary
`jope-datasets`	Frozen dataset metadata, source and split identity	Content-addressed manifests, source hashes and exact row assignments
`jope-bench::panel`	Development-data preparation, methods, metrics, screen/freeze/ confirm protocol, ASM orchestration	Campaign plan, prepared data, fits, frozen selections and reports
`jope-pipeline`	Search identity, bounded execution, original deadlines, process and output authentication	Started/finished records, arm windows, observations and receipts
`jope-policy`	Accepted regression checkpoint binding, CPU context producer, grammar-constrained proposals and learned ranking	Checkpoint receipt, vocabulary, qualified context and proposal evidence
`feat-math`	Executable recipe representation, variation and fitting	Canonical recipe identity, fitted state and prediction output

Component	Main responsibility	Durable boundary
`jope-learning`	HWM and score ledgers; pending stricter teaching admission	Comparison identity, immutable blobs, hash-chained events and frozen teaching manifests
Python policy package	Existing trainable policy model; pending bounded training campaign and experimental proposal consumer	Model/optimizer checkpoints, architecture, task/vocabulary binding and training proof
Native CLI	Current planning/execution front doors; pending `train` orchestration	Pinned requests, source/runtime inventories and supervised execution

The ordinary CLI and `asm` feature graphs in the stopped follow-on were inspected as manifest text: they have no edge to the engine, Torch, CUDA bindings or a local dependency cycle. That is a source graph check, not a compilation result. Python training runs as a supervised external process; native orchestration does not require linking Torch into the CLI.

The complete proposed learning cycle

1. Preserve the frozen data schema, original target strings, category strings, missingness and row/group assignments.
2. Derive the proposal context using only the allowed fitting rows. Record preprocessing and encoder identities, then verify actual context production and the numerical qualification required by that context lineage.
3. Use the chosen checkpoint to generate valid executable recipes, and rank the neural pool using learned reward predictions. This stage has no access to held-out scoring labels or fitted GP scores.
4. Place 64 neural and 16 exploration candidates into the neural arm's generation zero. Compare with an 80-candidate generic arm under the same declared budgets and executable grammar policy.
5. Actually fit candidates on inner training folds and score their out-of-fold predictions. The measured outcomes, not model-predicted quality, determine the search's parents and winner.
6. Generate mutation and crossover descendants with explicit parent identities. Preserve duplicate/cache handling, rejected variations, unavailable methods and every fit failure.
7. Freeze a recipe and evaluate the unchanged development protocol. Its fitted fold artifacts are separate from its reusable recipe.
8. In the pending teaching path, admit a recipe only after its independently preregistered train-side OOF evidence beats a trivial baseline under the frozen numerical rule.
9. Freeze the eligible corpus, train an explicitly experimental policy, prove real updates and continuation, and export a pinned checkpoint.
10. Consume that exact new checkpoint in a subsequent bounded experiment. A successful export alone is insufficient evidence for this final step.

Steps 1–7 have current bounded regression evidence at the scope described above. The stricter teaching/training/experimental consumption integration is stopped source work awaiting the next reviewed lifecycle and actual runtime acceptance.

Accepted regression model

The accepted lineage is the recovered lineage-B step-60000 checkpoint. The binding verifies the state, original model, exported tensors, configuration, parameter digest and vocabulary. The rejected step-69000 identity is explicitly refused. A self-consistent receipt around unrelated weights cannot impersonate the accepted model. The model loader rejects missing, extra, incorrectly shaped or non-FP32 tensors.

Accepted-model property	Value or behavior
Dataset context	1536 FP32 values
Target context	512 FP32 values; all 512 retained
Combined context	2048 FP32 values; 8192 bytes in little-endian vector form
Decoder width	512
Decoder depth	6 layers
Attention heads	4
Feed-forward width	2048
Token vocabulary	401 output tokens
Embedding rows	657, including the configuration's 256 extra rows
Maximum configured sequence	128 tokens; positional table includes the context position
Training dropout field	0.1; disabled for inference
Reward ensemble	4 heads
Tensor inventory	136 tensors for this configuration, including unused inference heads and scalar loss weights
Current Rust arithmetic	Candle CPU FP32
Native CUDA BF16 policy parity	Not established by the current CPU inference implementation

Sources: `crates/jope-policy/src/config.rs:21`, `config.rs:39`, `crates/jope-policy/src/model.rs:1`, `model.rs:41`, `crates/jope-policy/src/lib.rs:133`.

The decoder and reward paths each encode context with a two-layer MLP: 2048→512, GELU, LayerNorm, 512→512, GELU, LayerNorm. The decoder prepends a projected context token to the token embeddings, adds positional embeddings, applies causal self-attention with padding-key masking through pre-normalized layers, normalizes the result and projects through the tied token embedding table narrowed to the output vocabulary.

The reward path pools program-token embeddings, combines them with a separately encoded context, and evaluates four reward MLPs. The combined representation contains the program embedding, its elementwise interaction with a context gate, and an absolute-difference block. A separate runtime head emits a positive runtime estimate through softplus. Learned reward is a proposal-ranking signal; it is not a measured dataset loss.

The source records these accepted identities:

Identity	SHA256
State	`8ac80c269cc47c253b6f53fa3bc1eb75bb2fcda0bdace3ff8554be0175c400d6`
Original model	`053df4f53b84d2ebe7c14fbf907a149760af3d11de4a57b5ff89be8c2d3e8129`
Canonical parameter values	`9c7057e7f6a9e348f030da73f03522977b8c08915d3a221d7d5ab2e4213fcd92`
Configuration	`2508fc8b8268ea000141c9b119ba3da4e777c506c366168912a88d03482939bf`
Original checkpoint vocabulary	`8d55fee4366acd2c175cf55bfd769cebc904770f5227a6fd50a613723dc85787`
Executable product restriction	`570b2e5e8d859a73b888f5d0bee27f59da0e2376536babce24c241604f818e81`

The product restriction preserves the 401-token table while limiting executable heads. A token-compatible restriction is different from inventing a new vocabulary or silently snapping parameters to nearby bins.

Regression context production

Original data and preprocessing

The context input binds prepared-data hash, exact training-content hash, ordered fitting-row hash, cut identity and preprocessing hash. It requires at least eight training rows and a nonempty declared feature schema. Numeric versus categorical kinds come from the schema rather than guesses based on strings. Numeric nulls are imputed using training-only medians, with a zero fallback for an entirely missing column; the recorded training transform contains its fitted location/scale. For categories, every present string is a value, including empty text or strings spelling `NA` or `null`. Only an absent value is missing. Original target strings remain separately authenticated before numerical regression parsing.

This context preprocessing is for the proposal encoder. It does not authorize discarding the original schema or replacing a fitted predictor's own training-only preprocessing. Source: `crates/jope-policy/src/context_producer/input.rs:84`.

Component dimensions and assembly

The current CPU producer uses three components with fixed pretrained assets and fixed transforms:

Component	Production	Raw output	Contribution to policy context
Hyperion	Training features quantized from fitted quantile bounds, three embedding sweeps, scales 4/7/11/15, batch 1024, one target chunk per sweep	128 values per row	Mean, population standard deviation, 10th and 90th percentile per coordinate: 512
Foundation	Feature/target table prepared with the native deterministic recipe; target normalization computed in f64 after the native f32 cast	2048	Frozen mean-centering, learned 2048→1024 projection, then frozen normalization: 1024

Component	Production	Raw output	Contribution to policy context
Target	Training target values scaled into the training range, two native target windows with seed 17	512	Frozen normalization: 512

Assembly concatenates normalized Hyperion512 + projected Foundation1024 + Target512 and clips the full 2048-vector to [-32,32]. It checks all component dimensions and finite values. The source explicitly checks that the Hyperion row width is 128; confusing row width with pooled512 changes the model input meaning. Sources: `crates/jope-policy/src/context_producer.rs:188`, `:278`, `:356`, `:399`, `:422`.

The optimizations already in current source preserve the numerical contract: embeddings-only Hyperion execution avoids unrelated work, and the foundation path uses the ordinary qualified evaluator. Cached decoder operation is separately tested. These are execution changes, not permission to alter transforms or claim native GPU policy parity.

Qualification and admission

The CPU regression encoder has a stronger boundary than “the vector has 2048 values.” Eight fixed native-reference cases span two families and cuts 0–3. Qualification checks finite FP32 component and final-context values with maximum absolute difference at most 0.0003, exact asset/source/program identities, and actual bounded executions. Consumption binds those eight jobs plus the actual dataset context job: nine independently authenticated process proofs. The reference registry/index and transform hashes are pinned in source.

`VerifiedCpuContext` has private fields and can only be constructed by the verification path outside explicit test support. A syntactically valid JSON vector cannot create admission authority. Sources: `crates/jope-policy/src/context_producer/admission.rs:4`, `:115`, `:234`, `:297`; `crates/jope-pipeline/src/cpu_context_bridge.rs`.

Stored model assets and the active encoder distinction

The metadata-only S3 audit on 2026-09-23 fetched the three bundle manifests below and pinned their returned bytes, ETags and lengths. No model bodies were downloaded for this chapter. These are declared object sizes from the retrieved manifests; they do not measure resident memory, compute cost or a new numerical qualification.

Bundle asset	Bytes	Role in this chapter
`policy-step-60000/policy.safetensors`	111,199,088	Accepted policy export identity, consumed only with its checkpoint binding
`policy-step-60000/context_transforms.safetensors`	8,413,808	Accepted fixed context assembly transforms
`jope-prime-gen31-v2/model-bundle/hyperion_v5_arm_b.safetensors`	6,650,692	Exact Hyperion weights pinned by the current CPU context producer
`jope-prime-gen31-v2/starforge/starlight_core_regressor.safetensors`	233,183,532	Foundation component; manifest describes512 hidden features pooled into2048

Bundle asset	Bytes	Role in this chapter
`jope-prime-gen31-v2/model-bundle/target_encoder_fused_v1…safetensors`	78,615,176	Target component; manifest describes normalized64+64+384 fused representation
`hyperion-v7/model.safetensors`	246,456,988	Separate stored bundle, not the accepted context producer's pinned weights
`jope-prime-gen31-v2/model-bundle/jope_policy.safetensors`	108,846,812	Separate gen31 policy artifact; not step-60000

The storage name `hyperion-v7` must not be confused with the CPU implementation's `V7Model` Rust type. The actual current producer pins weight hash `88b9e834…`, which matches the v5 arm-B asset, with row width128. The separate stored v7 manifest describes dimension512, eight heads,12 layers and six bands. Replacing the pinned asset based on its newer-looking name changes the encoder identity and invalidates compatibility.

The retrieved manifests also contain internal/redistribution/provenance declarations that vary by artifact. They are retained evidence about the bundle metadata, not proof that every object is approved for every distribution. The active private run's exact release/input inventory remains the authority for which files it used; S3 presence alone does not establish serving eligibility. Sources: `crates/jope-policy/src/context_producer.rs:41`; custody `metadata-receipts/veox-jopedime-models-use2--bundles--policy-step-60000--manifest.json`, `…--hyperion-v7--manifest.json`, and `…--jope-prime-gen31-v2--manifest.json`.

Experimental classification context

Current source contains the pure class-partition context representation. The pending training overlay adds the bounded feature producer, process admission and operational proposal wiring. Classification does not parse class names as numbers or reuse the regression target encoder by ordinal coding.

The feature blocks remain Hyperion512 and projected Foundation1024. The pending producer explicitly conditions the foundation path on neutral zero targets and labels this an experimental feature-only probe. Each class gets a centroid over the 128-dimensional training row encodings. The target block contains, per coordinate, the mean, standard deviation, minimum and maximum across class centroids, totaling512. Classes are visited in first-row order, so a bijective renaming preserves even floating-point summation order. Original labels still appear in authenticated input identity, even though their spelling cannot influence numerical pooling.

The resulting lineage is `asm.experimental-class-partition-pool/v1`; `policy_compatible` is false. The false marker means it cannot enter the accepted regression policy loader. A declared binary input must have exactly two classes; multiclass requires at least three, and the fitting data must contain the declared classes. Missing classes are an explicit refusal. Sources: `crates/jope-pipeline/src/search_context.rs:278`; pending `crates/jope-policy/src/context_producer/classification.rs:1`.

Proposal generation, ranking and evolution

Accepted proposals are generated under grammar constraints. Ranking orders the neural pool by the mean of four reward heads, descending; exact ties use program SHA256 then text. Runtime estimates are reported without subtracting an undocumented runtime penalty. Ranking binds the checkpoint, context and vocabulary and records that scoring labels and fitted GP scores were not read. Source: `crates/jope-policy/src/ranking.rs:22`.

The search starts with80 candidates per arm, of which64 are neural in the neural arm. The remaining16 are exploration candidates, distinct from a failed neural proposal silently being counted as a neural success. Receipts retain initial-slot origin and descendant ancestry. Actual completed inner OOF losses select the best four parents. New generations alternate mutation and crossover attempts, canonicalize recipes, reject duplicate executable identities, and retain variation attempts in the journal. Source: `crates/jope-pipeline/src/neural_search.rs:27, :1380`.

A recipe may be evaluated on multiple folds or re-used in later fits. Therefore proposal count, recipe count, unique fitted predictors, launched commands and physical fit starts are different denominators. A cache hit does not become another new predictor; a partial/unknown attempt does not become a zero-cost fit. The performance chapter explains the report accounting.

Experimental trained models and compatibility

The pending training campaign uses the existing Python `JopePolicyModelV3` with task-routed reward/stability behavior. Real admitted training uses width256, two layers, four heads and feed-forward1024; synthetic qualification fixtures use width8, one layer, two heads and feed-forward16. Both consume the explicit context schema. Training uses one-epoch passes, learning rate5e-4, batch256 for admitted rows or2 for fixtures, AMP disabled and automatic batch-size tuning disabled. These are the bounded campaign's settings, not the accepted step-60000 model's architecture.

The export records the complete architecture, vocabulary, task, source, input identities, model state and training proof. The pending experimental consumer requires all architecture fields explicitly; the general-purpose Python loader's backward-compatible defaults cannot reinterpret a saved experimental checkpoint. It compares the reconstructed model's architecture and full state to the export. Sources: pending `python/src/jope/policy/training_campaign.py:780`, `python/src/jope/policy/experimental_proposal.py`, `crates/jope-pipeline/src/experimental_producer.rs`.

The final regression follow-on closes a specific gap: `Configuration.experimental.regression` can explicitly select a newly trained regression export. Omission keeps the accepted baseline. This operational regression route requires **real admitted** training, the same source commit and the exact native encoder identity used by training. It first admits the qualified native2048 regression context, then wraps those exact values in the separate `asm.experimental-regression-context/v1` lineage with zero classes and `policy_compatible:false`. It does not promote the new checkpoint to accepted serving or weaken the accepted hard pins.

Binary and multiclass use their separate experimental context path; a synthetic classifier fixture can qualify mechanical training/consumption behavior only when explicitly reported as synthetic. Synthetic regression training is permitted as a CPU test but cannot be substituted into the operational real-regression consumer. There is no silent accepted-policy fallback after an explicitly chosen experimental checkpoint fails compatibility.

Development exposure and limitations

The frozen protocol screens methods on development fold0, freezes the top five recipes, and confirms those recipes on the same five development folds; ASM is additionally confirmed if it missed the top five. The synthesized recipe was learned from the outer fold0 fitting set, containing development folds1–4. Thus those later confirmation folds are exposed to recipe learning. Every frozen recipe's fold0 labels influenced screen selection. The report derives row intersections and records them per recipe/fold; it labels the selected five-fold summary adaptive development and leaves sealed evaluation unopened. This is the chosen protocol, not an unbiased outer generalization estimate. Source: `crates/jope-bench/src/panel/selection_exposure.rs:1`.

Remaining architectural acceptance includes real eligible-teacher collection, actual new-parameter training/reload/continuation, subsequent fitted use of that newly trained checkpoint, classification operational qualification, complete fitted-state parity across supported deployment backends, the unchanged full30 panel, and measured warm/cold installation demonstrations. Reusable motif mining must retain full comparison identity and eligibility; new operators need ablation and compatibility tests. GPU inference portability and a broad claim of CPU/native-policy numerical equivalence remain future work.

3 / Product operations and maintainer runbook

Scope and evidence status

Algorithm Synthesis Model (ASM) means **“learns to generate and optimize executable algorithms from data.”** The product is the private `veox-ai/JopeDime` repository. On this host its canonical checkout is `/home/ubuntu/JopeDime`; the similarly named research repository at `~/jain-split/jope-dime` is a different upstream.

This chapter is pinned to source `68f56404c3e199e06413bfd645d3fe0803d7b96f`. Sixteen actual version/help invocations against its immutable CLI completed successfully. The executable reports `jope 0.1.0`; its SHA256 is `eab2d5b1b05629020aab3d540d51eb24ba0db18ff0917da188f89a34b6dc5e2b`. The machine-readable evidence is `agent-maintainer-manual/cli-help-68.json`. A version string alone cannot distinguish later builds; use the full source, executable digest and build receipt.

The source inventory, a help page, an executed fixture, exact-head CI and a live operational qualification establish different things. This chapter records interfaces and operational requirements. It does not turn an unexecuted command example into measured capability. The maintainer-private examples were subsequently executed: all fourteen commands passed. Their separate receipt and scope limits appear in the language chapter and `data/example-acceptance/`.

Surface	Present in the immutable 68 CLI	Remaining distinction
`doctor`	Yes	Static capability inventory; it does not inspect every runtime dependency or qualify a live campaign.
`benchmark verify-manifest`, `verify-inputs`, `methods`, `plan`, `run`, `report`	Yes	Planning performs zero fits. Run/report require exact admitted resources, inputs and journal identities.
`optimize capabilities`, `proof-plan`, `plan`, `run`, `report`	Yes, execution feature-gated	A compiled ASM feature is separate from qualified encoding, genuine neural-origin fits and successful evolution.
`fit`, `emit`, `search`, `pipeline run`	Yes	Legacy numerical regression/recipe paths; their outputs are not complete frozen-panel fitted inference exports.
`encode`, `champions`, `train-step`, `route`	Yes	`train-step` trains a local centroid router, not the bounded neural policy campaign.
`datasets`, `models`	Yes	Existing catalog/model tools; a fetchable file is not automatically admitted as an ASM input.
`train prepare-inputs`, `plan`, `run`, `resume`, `export`, `verify`	Pending combined training overlay	Includes native admission, actual updates/reload/continuation checks and experimental exports. See the training chapter.

Surface	Present in the immutable 68 CLI	Remaining distinction
`demo` and private CLI installer	Pending product overlay	Includes generic fitted demos and authenticated paired ASM demos; actual speed and parity must be measured.
`benchmark export-winner`, `qualify-export`	Pending winner add-on	Preserves the selected native state and ranking; backend qualification needs actual supervised execution.
`motifs mine`, `validate` under the umbrella CLI	Not present at 68	Separate preparation exists; production eligible-State reading, mine command and umbrella integration were identified as remaining work.

The stopped integration order is combined **training64** (including regression consumption, accounting and coverage), then **product28**, then unchanged **warm-target** and **winner-export** add-ons. The add-on refresh is pinned under `agent-delivery-refresh-68f56404`; two strict replays produced identical nine-file postimages, with 95 canonical guards. These are custody patches, not installed behavior. Never apply the training coverage/accounting constituents a second time or copy an older loose installer postimage over its corrected form.

Physical layout and identities

Keep source, bulk inputs and evidence in separate locations. The examples below use these names; resolve them from actual receipts before execution:

```
R=/home/ubuntu/JopeDime
C=/home/ubuntu/jope-rescue-20260921/asm-integration-20260923
D=/home/ubuntu/.cache/jope/asm-private-30-v1
M="$R/benchmarks/panel.json"
# S: full reviewed source SHA; J: physical immutable CLI path.
# A: new durable private acceptance directory, outside Git and /tmp.
```

`JOPE_HOME` chooses the source/assets root; `JOPE_STATE` chooses mutable local state; `JOPE_REGISTRY` chooses the dataset registry. Those roots are resolved at call time. Pass explicit paths when producing reproducible evidence. Do not repurpose `HOME` or globally override `CARGO_TARGET_DIR`. A managed installation symlink is convenient for interactive use; a campaign pins the physical executable path and checksum so changing the symlink cannot silently replace its worker.

There are two different hashes in the CLI contracts. An artifact SHA256 hashes exact file bytes. A campaign's `plan_sha256` is its validated semantic identity, printed by planning. Do not substitute `sha256sum plan.json` for that semantic ID. Worker/build/source inventories have their own schemas and pins; preserve each rather than renaming one receipt into another type.

Campaign journals require absolute durable paths outside Git working trees and `/tmp`. Private directories are restricted by the front doors. Do not publish entire run directories: they can include raw development features, labels, recipes, private context, model state and provenance.

Build and install

The current development build path builds the Rust CLI from the existing checkout:

```
cd "$R"
CARGO_INCREMENTAL=0 CARGO_PROFILE_DEV_DEBUG=line-tables-only \
  cargo build --locked --release -p jope-cli --features asm --bin jope
```

Allocate build/test resources through the repository's existing coordination before executing it. Use one reused `target/` per claimed checkout. The CLI's `asm` feature exposes the private Linux execution path; a default-feature build may show the command while reporting ASM execution unavailable. The ordinary CLI should not require the monolithic engine, Torch or CUDA. Separate context and policy workers and their authenticated model inputs are still necessary for a live neural run.

After the pending installer is integrated, prefer its measured source-build and checksum-install front doors:

```
python3 "$R/deploy/install-private.py" source \
  --checkout "$R" --source-sha "$S" --output "$A/cli-build" --jobs 4
python3 "$R/deploy/install-private.py" binary \
  --binary "$PRIVATE_BINARY" --sha256 "$BINARY_SHA256" \
  --source-sha "$S" --prefix "$A/install"
```

Read the binary path and digest from the actual source-build receipt. The source command checks the checkout before and after compilation, uses the same `asm` selection for dependency inventory and build, rejects unwanted engine/Torch/CUDA dependencies, and selects the executable from Cargo artifact messages. Its receipt includes the toolchain, dependency graph, features, source identity, compiler output, executable digest and measured cold duration. Compilation failures retain failure receipts.

The binary installer requires Linux x86_64, Python 3.10 or later and `readelf`. It validates the supplied checksum before executing the binary, checks architecture/glibc/dynamic dependencies, probes version and command help, and activates a digest-addressed private binary through a managed symlink. It does not download models or datasets. A caller-supplied checksum proves byte identity; retain independent source/review/build provenance as well. Cold build time, cold installation time, data preparation and warm demonstration time are separate measurements.

The installer's `jope.private-cli-source-build/v1` receipt does not replace a `jope.asm-cargo-build/v1` worker receipt or `jope.asm-worker-build-inventory/v1`. Build and retain the context/policy workers, source manifest, accepted assets and native context qualification through their owning workflow. The native canonicalizer used by experimental proposal admission must match the final source/build and physical executable bytes.

Prepare the panel and external runtimes

The approved panel contains ten regression, ten binary and ten multiclass datasets. Its exact manifest SHA256 is `ba7dace4cd535687002cc95538dbc7f7aaf77ec973177c7159b3c6d22471eee0`. Reformatting the JSON changes its bytes and is refused even when its apparent meaning is unchanged. Do not replace a failed dataset with an easier fixture.

```
uv run --project "$R/python" --extra benchmark python "$R/benchmarks/prepare.py" \
  --manifest "$M" --output "$D"
"$J" benchmark verify-manifest --manifest "$M"
"$J" benchmark verify-inputs --manifest "$M" --datasets "$D"
"$J" benchmark methods
```

Preparation must preserve actual frozen row assignments, original labels, category strings, nulls and groups. `verify-inputs` verifies development bytes and schema, not just file existence. The benchmark does not open the sealed evaluation partition. An operational proof of one regression family retains the original thirty-dataset manifest but selects a separately declared proof scope; it does not modify the official panel.

The native roster has 17 regression recipes, five binary recipes and four multiclass recipes. Aliases do not create extra screening slots. The second GPR configuration is distinct because it has a distinct fitted implementation. Every task also includes XGBoost, LightGBM, CatBoost and scikit-learn default adapters. External library versions, entry points, environment and package bytes must be authenticated before admission.

```
python3 "$R/deploy/prepare-panel-runtime.py" \
  --binary "$J" --native-command benchmark \
  --build-receipt "$CLI_BUILD_RECEIPT" --source-sha "$S" \
  --python "$EXTERNAL_PYTHON" --site-packages "$EXTERNAL_SITE_PACKAGES" \
  --adapters "$R/crates/jope-bench/adapters" --output "$A/runtimes.json"
"$J" benchmark plan --manifest "$M" --datasets "$D" --source-sha "$S" \
  --runtimes "$A/runtimes.json" --output "$A/base.json"
```

The preparation helper checks actual dependencies/constructors and pins the runtime. It is not a completed fit comparison. Retain unavailable-dependency reasons rather than dropping methods. Prior inventories observed XGBoost 3.4.1, LightGBM 4.7.0, CatBoost 1.2.10 and scikit-learn 1.7.2 on this host; these historical metadata observations do not authenticate a later environment. A new package installation invalidates old package-byte inventory. Keep the external CPU environment separate from the dedicated CUDA training environment.

Omitting `--runtimes` from `benchmark plan` produces a verified Rust screening inventory with zero fits; it is not the complete admitted ASM/comparator campaign. The runtime-aware plan is the base for `optimize plan`.

Plan, execute, resume and report

Prepare the qualified configuration before opening an execution window. It binds the actual context producer, source/worker/model inputs, numerical qualification and policy configuration. The standard comparison preregisters 64 neural proposals plus 16 exploration candidates against 80 generic candidates, with equal arm budgets. Generation-zero neural slots must actually fit; a proposal file or generic-only result cannot demonstrate neural consumption.

```
"$J" optimize plan \
  --base-plan "$A/base.json" --base-plan-sha256 "$BASE_PLAN_ID" \
  --configuration "$ASM_CONFIGURATION" \
  --configuration-sha256 "$ASM_CONFIGURATION_FILE_SHA256" \
  --output "$A/panel.json"

"$J" benchmark run --plan "$A/panel.json" --plan-sha256 "$PANEL_PLAN_ID" \
  --manifest "$M" --datasets "$D" --journal "$A/panel-journal"

"$J" benchmark report --plan "$A/panel.json" --plan-sha256 "$PANEL_PLAN_ID" \
  --manifest "$M" --datasets "$D" --journal "$A/panel-journal"
```

`optimize run` and `optimize report` delegate to the same campaign contracts. Repeating run with the same plan and journal resumes the original authenticated state and absolute deadline. A restart does not grant another two hours. Do not repair journals by editing receipt fields or copying successes from another experiment. Read-only report recovery authenticates retained artifacts and recomputes scores; it does not fit another predictor.

Constraint	Official panel policy
Worker/CPU count	Twenty isolated workers shared by search and benchmarking; at most twenty CPUs.
Thread count	One thread per worker, including external libraries.
Memory and swap	64 GiB total; zero swap. The shared policy reserves 4 GiB for the controller and caps each fit at 3 GiB.
GPU capacity	At most one GPU task; the approved training hardware is one RTX 3090. GPU training has its own separate total hour.
Fit and search time	At most 30 seconds per fit, normally 120 seconds per search arm.
Panel wall time	7,200 seconds total, including 300 seconds reserved for reporting. No new fit/qualification starts after the original execution cutoff at 6,900 seconds.
Coexistence	Do not overlap the experiment with exclusive full-CPU CI.

The CLI uses the shared Linux systemd/cgroup-v2 supervisor and process-group cleanup. Missing host capability is unavailable execution. Do not put an additional outer allocation lock around a command that already acquires the shared allocation. A child crash, timeout, cancellation, launch failure and unavailable dependency have different recorded states; unknown work/cost is not zero.

For each dataset, applicable methods screen on development fold zero. The top five recipes freeze before confirmation across all five development folds. ASM is also confirmed when it misses that top five. Report every original outcome. Selection can use adaptive development folds; these are not untouched external generalization estimates. Preserve the explicit selection-exposure metadata, and keep sealed evaluation untouched.

Regression is scored by RMSE. Binary classification uses AUROC, with the frozen second class as positive and loss `1 − AUROC`. Multiclass uses the full probability matrix and original class order for log loss. Predictions must be finite; class order and row count must match; probability rows are validated without silently renormalizing. The pinned log-loss epsilon is `1e-15`, and probability row-sum tolerance is `1e-6`.

Warm fitted demonstration after integration

The pending generic demo fits one task-compatible registered recipe, saves complete private state, reloads it, verifies unchanged predictions and compares against a train-mean or train-prior baseline. It uses one CPU/worker, 2 GiB total, zero swap/GPU, 30 seconds per fit and 60 seconds total with five seconds reserved for reporting. Its synthetic mode is explicitly labelled; it is not evidence from a real panel dataset.

```
"$J" demo --dataset "$DATASET_ID" --manifest "$M" --datasets "$D" \
  --source-sha "$S" --output "$A/demo-real"
```

```
"$J" demo --synthetic multiclass --source-sha "$S" \
  --output "$A/demo-synthetic" --export python
```

The authenticated ASM route consumes a separately prepared single-regression proof plan:

```
"$J" demo --asm-plan "$WARM_PLAN" --asm-plan-sha256 "$WARM_PLAN_ID" \
  --manifest "$M" --datasets "$D" --output "$A/demo-asm" \
  --export python --export rust
```

The warm-target add-on treats under 60 seconds as a measured target, rather than a cap that prevents a required real fitted demonstration. Both complete equal arm allowances must fit within the declared original execution window. Unequal, missing or globally clipped arms fail the matched-comparison condition. An explicitly admitted longer allowance can produce a successful complete execution with `target_met: false`; it cannot claim under-60 acceptance. It does not change the official panel's 120-second arm defaults.

The previously retained source-3fd proof started the policy about 22.1 seconds into its original window, spent about 20.8 seconds in policy production and started the first fit around 50.4 seconds. Those are historical measurements, not a forecast or a proof that equal 20-second warm arms work. A suggested longer configuration is a sizing proposal until explicitly admitted. Neither a generic substitute nor clipping the second arm solves this requirement.

Artifact inventory, cost and privacy

Evidence	What to retain and inspect
Source/build/install	Full commit, source manifest, Cargo messages, exact features/toolchain, dependency graph, binary and worker hashes, physical paths, cold timings and exits.
Plan and configuration	Frozen panel identity, all runtime/model/encoder pins, resource limits, semantic plan ID, seeds, numerical policy and original window.
Worker execution	Started/finished union, exact job/input identity, output inventory, exits/signals, timeouts, process-group drainage, memory/CPU counters where known.
Scientific results	All screen/confirmation losses, frozen recipe set, original row/class order, baseline, selection exposure and unsupported/unavailable outcomes.
Fitted predictor	Complete private state, fitted preprocessing, recipe, native reload proof, prediction hashes and exact artifact identity.
Export qualification	Exact winner/fold/state, public bundle, toolchain/input/class/policy hashes, actual commands and parity result. See the language chapter.

Count a unique fitted predictor by authenticated complete fitted-artifact identity, and a reusable recipe by the declared recipe identity. This does not establish behavioral uniqueness. Fit failures, policy/context work and preparation still incur cost. A worker CPU subtotal is not automatically total creation CPU. Validated systemd unit CPU can include setup, controller and children; it overlaps worker CPU and must not be added to it. Unknown counters stay null with a reason.

In the pending ASM export accounting, `fitted_export_commands` is a subset of global worker commands. Comparator-only accounting requires its separate export cost to be added. Export preparation is measured separately; final receipt publication and other unmeasured overhead remain explicit. Retain the original panel cutoff: backend qualification after a campaign expires is unattempted through the existing front door. A distinct post-panel bounded validation stage is not implemented and cannot be claimed inside the two-hour panel allowance.

Allowlisted public fitted bundles contain inference code, necessary fitted input dictionaries, class/schema metadata and their hashes. They exclude the private recipe, discovery history and scoring references. The private prepared plan and worker directory remain private. Canary/text-policy checks supplement the explicit file allowlist; passing static checks does not replace actual complete-prediction parity.

Troubleshooting and safe recovery

Symptom	Meaning and next action
`compiled: false` or ASM feature unavailable	Use the reviewed Linux CLI built with `asm`; then separately qualify its model inputs and workers.
Running binary differs from pinned native runtime	Resolve the installed physical binary, verify its hash and prepare a new pre-execution inventory/plan; do not alter a running plan.
Frozen manifest or development checksum differs	Stop admission, preserve both files and find the preparation/source mismatch. Do not regenerate folds or substitute datasets to make a check pass.
Missing external dependency/compiler	Retain unavailable status and inventory. Install/qualify before opening a new authorized experiment; never drop the method or replace a winner.
Context/reference/worker identity mismatch	Rebuild or requalify through the owning source-bound workflow; a copied success flag is insufficient.
Deadline expired	Recover/report the original window. A repeated command cannot renew it. Any new experiment needs its own explicit scope and resource declaration.
Started job lacks a finished receipt	Treat execution cost/status as unknown; let the shared pool perform terminal recovery/drainage. Do not blindly retry the child or delete its artifacts.
No complete five-fold winner	Keep incomplete/failed confirmations visible. Do not select a different recipe merely because it has an emitter.
Warm result exceeds 60 seconds	Preserve fitted artifacts, comparisons and measured target miss. Do not relabel elapsed time or omit preparation performed inside the warm invocation.
S3 access stalls at instance metadata	On the host, select the configured credential-process profile such as `AWS_PROFILE=veox-jepson-prod`; never source the bare-token credential note files. Runners have no AWS credentials.
No eligible training families/no-signal admission	Preserve refusal; obtain real eligible data under the declared protocol before GPU training. Do not use development/test gain as a substitute for the train-side OOF admission rule.

Repository operation remains serialized: read the current queue/ledger, claim paths under `AGENT_CHAT.md.lock`, keep one writer per checkout, and preserve pending work. Changes require exact-head CI, independent eligible review/native approval, guarded rebase merge and verification of remote `main` plus `main-guard`. A changed head invalidates an earlier head's approval. Preserve evidence before deleting closed lanes or build directories; `target/` can contain irreplaceable receipts.

Source anchors

Current interface/semantics: `crates/jope-cli/src/main.rs`, `crates/jope-cli/src/kernel.rs`, `crates/jope-pipeline/src/cli.rs`, `crates/jope-pipeline/src/front_door.rs`, `crates/jope-paths/src/lib.rs`, `crates/jope-bench/src/panel/{cli,campaign_cli,campaign,optimize_cli,methods,score,receipt_timing}.rs`, `crates/jope-pipeline/src/{experiment,execution,supervisor,neural_search}.rs`, `deploy/prepare-panel-runtime.py`, `benchmarks/{panel.json,prepare.py}` and `AGENTS.md`.

Pending interfaces: final product postimages of `deploy/install-private.py`, `docs/private-install-demo.md` and panel `{demo,demo_asm,inference}.rs`; stopped add-on postimages of panel `{fitted_export,fitted_export_run}.rs` and `docs/{private-warm-target,private-fitted-qualification}.md`. Exact source and receipt hashes accompany this chapter's custody handoff.

4 / Training operations, eligibility and checkpoint consumption

What can be claimed now

At source `68f56404c3e199e06413bfd645d3fe0803d7b96f`, the latest bounded regression origin proof has actual neural proposals, fitted generation-zero candidates, fitted descendants and a successfully authenticated recovered report. The newly authored training path is still a stopped follow-on. Its guarded 64-file patch is `595824ef6eb8e15872be058668503cedad1dd9a5ff98ac4700c6f713251d384c`; manifest `b86b550d7ee39fd93d5202accd5f77c975b7bcb4fd8f8e7706c69c5f279294a2` binds every original packet, overlap resolution and current source/API guard.

The follow-on passed custody replay guards, source syntax, pure metadata checks and feature-graph inspection. That evidence does not establish Rust type checking, a GPU optimizer run, a real newly trained checkpoint, or a successful live trained-checkpoint consumption loop. The current admissible-teacher count remains zero in the inspected evidence. Source support, fixture tests, exact-head CI, live qualification and serving promotion must remain separate status fields.

All commands below involving `jope train` or experimental consumer configuration describe the **pending overlay's implemented interface**. They are a maintainer runbook for integration and authorized acceptance, not claims that these commands were executed at the current PR83 source. Preserve this status notice when extracting examples.

Evidence and identity before execution

A training run has at least four separate identities:

Identity	What it binds	Why it is separate
Learning state	Eligible HWM records and teaching evidence	Accumulated data authority is distinct from a single training attempt
Training plan	Task/category inputs, source, runtime, numerics, seed, passes and resource policy	Changing it invalidates an existing run's compatibility
Execution window	Actual first admission and immutable deadline	Resume must use elapsed budget rather than restart the clock
Checkpoint/export	Parameters, architecture, vocabulary, task, optimizer ancestry and completed proof	A file's existence is not evidence of an update or eligibility

Use durable directories outside Git working trees and outside `/tmp`. Keep corpus and model assets out of Git. The native front door inventories the actual Python interpreter, preserves its virtual-environment invocation path, captures environment and installed runtime identity, and supervises workers rather than treating a bare Python subprocess as resource isolation.

The training runtime inventory hashes installed dependency bytes, excluding generated bytecode, and records all installed distributions rather than only a short library allowlist. Version strings alone do not protect against modified installed files. The full source commit and source-package hashes are part of compatibility. An incompatible environment should be rebuilt/replanned before admission, not silently accepted during resume.

Sources: pending `crates/jope-cli/src/train.rs:1`; `python/src/jope/policy/training_campaign.py:120`.

Resource policy and clocks

Operation	Required policy
Official frozen30 panel	20 workers/CPUs, 64GiB, zero swap, at most one GPU task, 30 seconds per fit, 7200 seconds total including300-second reporting reserve
Paired search	64 neural+16 exploration versus80 generic; default120 seconds per arm; same declared budget and grammar/variation identity
Separate origin/teacher qualification	Explicitly preregistered resource manifest and separate journal; no inferred extra allowance and no reset of another campaign
Training	One visible RTX3090, one campaign of at most3600 seconds, at most1200 seconds per category, one active category worker, CPU allowance20,64GiB,zero swap
Training numerical runtime	One Torch thread, FP32 storage, CUDA TF32 enabled by declared policy, AMP disabled; explicit experimental lineage
CPU training	Explicit synthetic qualification only; never an automatic fallback from requested CUDA
Full-CPU CI	Exclusive; no overlapping campaign/training resource allocation

The training device check requires exactly one visible CUDA device, an RTX3090 name and a usable GPU UUID. It records requested/resolved device identity. Resume requires the same identity. A missing GPU is an explicit failure; it must not become a successful CPU run under a CUDA claim.

Native planning does not open the training clock. First supervised admission creates the global window. Each category receives at most20 minutes, capped by the remaining global deadline minus its five-second reporting reserve. All passes, validation, reloads and checkpoint work must fit the original window. Two passes are required for continuation evidence; they do not receive two category budgets. A fresh directory or modified plan is a new campaign, not a legitimate way to finish an expired approved campaign.

Sources: current `crates/jope-bench/src/panel/campaign_cli.rs:51`; pending `python/src/jope/policy/training_campaign.py:164`, `:704`, `:720`, `:1043`; `crates/jope-cli/src/train.rs:31`.

How a fitted recipe becomes an eligible teacher

The pending eligibility path freezes independent facts before optimization: original dataset and prepared-data identity, ordered rows and groups, target role, exact inner-fold assignment, evaluator and vocabulary identities, numerical policy, baseline and worker/resource pins. This freeze cannot be created retrospectively from a winning candidate's own output. A historical successful fit is not equivalent to preregistered teaching evidence.

The train-side inner split uses five whole-group folds. Groups are assigned deterministically by largest size first, with lexical tie-breaking and balancing. No group's rows are split simply to fill a fold. The outer development hold rows remain outside the search's inner fitting/selection closure.

Each candidate must produce complete authenticated OOF predictions for all five inner folds. Scores are recomputed from retained predictions, exact targets and ordered class probabilities. The trivial baseline is fitted from each fold's training rows: target mean for regression, class frequencies for classification. The guard uses mean fold loss and admits only if:

- all five fold losses and baseline losses are finite;
- mean candidate loss improves over mean baseline loss;
- at least four of five fold losses improve;
- each counted improvement exceeds `1e-12 + 1e-6 * abs(baseline_loss)`;
- actual worker requests, process outcomes, output hashes, fitted artifacts, seeds, source, resources and deadlines match the original freeze.

The criteria are intentionally stronger than “a run completed” or “the score field is positive.” Constant/no-signal predictors, missing folds, reordered rows, changed class order, stale scores, unauthenticated prediction blobs and unknown execution accounting cannot provide teaching authority. A failed admission is retained as evidence rather than discarded.

Optimization-visible labels initially called `test` become development labels, with a durable record of that exposure. `sealed` and `meta_test` inputs are refused for optimization/teaching. The HWM identity includes the full comparison context, so an improvement under a different evaluator, split, vocabulary, preprocessing or numerical protocol cannot displace an unrelated incumbent. These details belong to the pending eligibility/HWM correction, not a claim that every historical HWM record already meets this rule.

Sources: pending `crates/jope-learning/src/eligibility.rs:170`, `:253`, `:277`, `:427`, `:700`, `:1023`; `crates/jope-learning/src/eligibility/inner.rs`; `crates/jope-learning/src/record.rs`; `crates/jope-pipeline/src/search_plan.rs`.

Present teacher readiness and the smallest real path

The metadata readiness audit verified28 source-family evidence files and the retained original process/context receipts. The inspected historical55e9 and3fd origin runs had no preregistered teaching freeze or vocabulary pins; they cannot be upgraded after the fact. The new68 operational origin proof also was not a teaching campaign. There is therefore no currently demonstrated eligible real corpus to train from.

The prospective small pair selected from frozen provenance/size, before observing losses, is:

Dataset	Evidenced family	Fold0 fitting rows
`230_machine_cpu`	UCI computer-hardware relative-performance source	135

Dataset	Evidenced family	Fold0 fitting rows
712_chscase_geyser1	Chatterjee casebook geyser source	140

Both must independently produce eligible recipes. The minimum of ten completed inner fits is only the five-fold×two-family floor for one successful recipe per family. It is not a replacement for the specified matched search and does not promise that either family will improve.

If exactly these two families qualify, seed7 assigns machineCPU to development and geyser to training under the current partition rule. If more families qualify, use the entire admitted set and let the native family partition determine membership. Do not omit inconvenient eligible comparisons, force the two-family allocation, or reuse the official30 panel's deadline for a separate trial. Historical checkpoint exposure remains a separately reported unknown; family provenance alone does not prove zero prior exposure.

Source evidence: custody `agent-training-readiness/readiness-v3.json`, SHA256 `9197900d4b696811909c4ba4e779e6c5d488569dd9c360165cc55fd43bf8c735`. Its metadata inspection does not itself admit training or re-run numerical qualification.

Prepare a teaching campaign before it starts

The final reviewed build must supply its CLI, worker inventory, source manifest, runtime manifests, component assets and qualification receipts. Record the actual built source SHA; do not substitute the latest checkout hash for the binary's source. Recheck that no exclusive CPU CI is running before opening an authorized CPU campaign.

The `teaching` member of the complete ASM configuration has this shape:

```
{
  "state_root": "/durable/private/learning-state",
  "prepared_root": "/durable/private/prepared-panel",
  "panel": {"path": "/source/benchmarks/panel.json", "sha256": "EXACT_PANEL_FILE_SHA256"},
  "original_label_role": "development"
}
```

The capitalized value is a placeholder, not a valid digest. The complete configuration also requires exact source/build/runtime/checkpoint/vocabulary/context fields; the small object above cannot be passed as the whole configuration. Planning preregisters the teaching freezes before final experiment identity, so worker evidence can bind them.

The following shell variables must already name authenticated artifacts. Commands are interface examples and allocate no budget merely by appearing here:

```
"$JOPE_FINAL_CLI" benchmark plan \
  --manifest "$JOPE_PANEL" --datasets "$JOPE_PREPARED" \
  --source-sha "$JOPE_FINAL_SOURCE" --runtimes "$JOPE_RUNTIMES" \
  --output "$JOPE_CAMPAIGN/comparator-plan.json"

"$JOPE_FINAL_CLI" optimize plan \
  --base-plan "$JOPE_CAMPAIGN/comparator-plan.json" \
  --base-plan-sha256 "$JOPE_BASE_PLAN_ID" \
  --configuration "$JOPE_TEACHING_CONFIGURATION" \
  --configuration-sha256 "$JOPE_CONFIGURATION_FILE_SHA" \
  --output "$JOPE_CAMPAIGN/asm-plan.json"
```

Use the semantic plan identity emitted by native planning for `--base-plan-sha256` and `--plan-sha256`. A JSON file's byte hash and its semantic identity need not match. Configuration/resource arguments explicitly refer to their required file hashes.

For a separately admitted small proof, substitute this planning command. `proof-plan` selects one regression dataset from the verified frozen base panel, replaces the resources with the explicitly pinned proof resource manifest, and preregisters teaching only for that selected dataset:

```
"$JOPE_FINAL_CLI" optimize proof-plan \
  --base-plan "$JOPE_CAMPAIGN/comparator-plan.json" \
  --base-plan-sha256 "$JOPE_BASE_PLAN_ID" \
  --configuration "$JOPE_TEACHING_CONFIGURATION" \
  --configuration-sha256 "$JOPE_CONFIGURATION_FILE_SHA" \
  --dataset "$JOPE_SELECTED_DATASET" \
  --resources "$JOPE_PROOF_RESOURCES" \
  --resources-sha256 "$JOPE_PROOF_RESOURCES_SHA" \
  --output "$JOPE_CAMPAIGN/asm-plan.json"
```

A second family requires its own actual preregistered proof plan and journal, with its separately declared approved resource allocation. Both may point at the same learning state and frozen30 metadata. Neither proof changes or spends an unstarted official panel's immutable window. Proposed example durations in earlier queue documents were not resource admissions.

Run and recover using the same immutable plan and original journal:

```
"$JOPE_FINAL_CLI" optimize run \
  --plan "$JOPE_CAMPAIGN/asm-plan.json" --plan-sha256 "$JOPE_ASM_PLAN_ID" \
  --manifest "$JOPE_PANEL" --datasets "$JOPE_PREPARED" \
  --journal "$JOPE_CAMPAIGN/journal"

"$JOPE_FINAL_CLI" optimize report \
  --plan "$JOPE_CAMPAIGN/asm-plan.json" --plan-sha256 "$JOPE_ASM_PLAN_ID" \
  --manifest "$JOPE_PANEL" --datasets "$JOPE_PREPARED" \
  --journal "$JOPE_CAMPAIGN/journal"
```

Sources: pending `crates/jope-bench/src/panel/optimize_cli.rs:13`, `crates/jope-bench/src/panel/origin_proof.rs`; current `crates/jope-bench/src/panel/campaign_cli.rs:18`.

Build the real corpus from admitted evidence

`train prepare-inputs` takes a request, not hand-written training examples. The request schema binds requested tasks, seed, family registry, vocabulary, encoder and optional initialization checkpoint. Its `contexts` map has one independently pinned proof for every eligible comparison. Each proof includes producer/source, original input, qualification/reference manifest, vector/artifact, frozen transforms, component bundle, actual context execution and eight case executions.

The exporter reopens the original exact fitting input, verifies feature schema/values/labels and row order, verifies complete context admission, and checks each recipe against exact executable vocabulary bins. It refuses parameter snapping, wrong-task heads and fabricated vectors. It then writes corpus rows with1536 dataset plus512 target dimensions, family-disjoint train/development metadata and retained source/context identities.

```
"$JOPE_FINAL_CLI" train prepare-inputs \
  --learning-state "$JOPE_LEARNING_STATE" \
  --request "$JOPE_CORPUS_REQUEST" --request-sha256 "$JOPE_CORPUS_REQUEST_SHA" \
  --output "$JOPE_CORPUS_OUTPUT"
```

Do not author a request using invented eligible comparison IDs: these must come from actual successful admission. The exported request requires all eligible requested comparisons. At least two independent families are necessary for a real train/development partition. Source: pending `crates/jope-cli/src/train/corpus.rs:43`, `:195`, `:362`, `:485`.

Plan, run, resume, verify and export

The intended real all-category path preserves real admitted regression while allowing explicitly labeled synthetic fallback only for classification categories whose real admission fails:

```
"$JOPE_FINAL_CLI" train plan \
  --state "$JOPE_TRAINING_STATE" --python "$JOPE_TRAINING_PYTHON" \
  --inputs "$JOPE_CORPUS_OUTPUT/inputs.json" \
  --task all-admitted --device cuda:0 --source-sha "$JOPE_FINAL_SOURCE" \
  --learning-state "$JOPE_LEARNING_STATE" --seed 7 --passes 2

"$JOPE_FINAL_CLI" train run --state "$JOPE_TRAINING_STATE"
"$JOPE_FINAL_CLI" train verify --state "$JOPE_TRAINING_STATE"
"$JOPE_FINAL_CLI" train export \
  --state "$JOPE_TRAINING_STATE" --output "$JOPE_TRAINING_EXPORT"
```

If interrupted before the original deadlines and while all plan/runtime/input pins still agree, continue with:

```
"$JOPE_FINAL_CLI" train resume --state "$JOPE_TRAINING_STATE"
```

Do not use the top-level `--synthetic-fixture` switch in place of missing real regression evidence. For deliberately small mechanical CPU qualification only, the explicit interface is `train plan --synthetic-fixture --device cpu`, with the same source/runtime/state requirements. Such a result stays synthetic and does not satisfy the real-data task.

The planner records requested, admitted, unavailable and fallback categories. With `all-admitted`, a missing real regression category can remain explicitly unavailable while classifier fixtures proceed; that is partial coverage and cannot be reported as completed real-regression training. Current real-classification corpus admission is not qualified. Binary/multiclass fixture fallback retains its own synthetic provenance and task labels.

Verification must establish positive optimizer updates, changed parameter digests, exact reload, an actual resumed optimizer state and increasing step counters, at least two committed passes, complete planned category coverage where claimed, and compatible device/source/runtime/vocabulary/input identities. The worker reloads both `policy.pt` and `resume.pt` and requires equal parameter proof. Each pass belongs to the original category/global windows and contributes an immutable execution/receipt chain. A zero-update checkpoint or interrupted partial output cannot become a successful pass.

Export first performs verification, then writes the exact plan, verification and category artifacts into a private output bundle with a file inventory. It creates experimental training artifacts; it does not update the accepted serving pointer. Sources: pending `python/src/jope/policy/training_campaign.py:744`, `:780`, `:1043`, `:1203`, `:1303`.

Consume the new trained checkpoint

The pending final regression consumer is selected by the `experimental.regression` entry of the complete `jope.asm-configuration/v1` object. This entry is an object with:

Field	Required meaning
`schema`	`jope.experimental-python-configuration/v1`
`task`	`regression`, `binary`, or `multiclass`, matching its configuration key
`runtime`	Full pinned PythonRuntime object, including program, manifest, source/package files and runtime digest
`export_manifest`	Exact exported training manifest artifact
`checkpoint`	Exact task's exported `policy.pt` artifact
`vocabulary`	Exact frozen action vocabulary artifact
`canonicalizer` / `canonicalizer_build`	Native executable and actual successful build evidence
`canonicalizer_features`	Exact feature set from Cargo artifact messages
`sampling`	Bounded count, temperature, top-p and maximum sequence length
`generic_draws`	Declared bounded generic bank allowance

Artifacts use `{path, sha256}` with absolute paths and real file hashes. Runtime is not merely a Python executable/version: source files and installed package files must be inventoried. The native canonicalizer must have a successful build receipt whose source and source-manifest pins agree with the consumer. Do not copy an accepted checkpoint receipt into this export field.

Regression additionally requires the export to identify real admitted regression training, the same source commit, and the exact native encoder identity. The consumer admits the complete native context and independently verifies architecture, task, vocabulary, checkpoint and proof. Classification uses its separate context lineage. Missing architecture fields, missing/extra model state, altered vocabulary, source changes, changed encoder assets or a different training task fail closed. No explicit experimental selection silently falls back to the accepted model.

After building a new configuration around the verified export, preregister a separately authorized operational proof using the same `optimize proof-plan` front door. Its report must bind the new checkpoint hash to actual generation-zero neural fits, retain fitted artifacts/predictions and, when evolution is claimed, show completed descendants with neural ancestry. Recover the report from the same original journal and deadline. Only after those facts are demonstrated should the final official panel configuration be frozen around that checkpoint.

Source: pending `crates/jope-bench/src/panel/experimental_proposal.rs:36`, `crates/jope-pipeline/src/experimental_producer.rs:36`, `python/src/jope/policy/experimental_proposal.py`; overlay documentation `docs/experimental-regression-consumption.md`.

Resume, failures and proof interpretation

The current pool treats started-only attempts as uncertain until process/cgroup recovery authenticates their outcome. It does not launch a replacement child merely because the parent process disappeared. Terminal records are persisted before potentially slow drainage. The search uses immutable observed journals and refuses changed completed evidence. At cutoff, only proven never-admitted inventory may normalize; the per-fold replay digest preserves ordered real attempts and output identities. Both live PoolBackend feedback and recovered FitAdapter scoring compute the same shared digest.

A regression caught during acceptance illustrates the distinction: the d0 source completed964 commands, but its recovered report failed because the two feedback constructors disagreed about replay evidence. Those fit counts did not justify declaring the report passed. The68 correction restored one shared digest computation, followed by a fresh bounded proof and recovered report. The d0 records were retained unchanged.

Never extend an expired original deadline, mutate a failed result into a success, reinterpret an unknown cost as zero, or cite syntax parsing as model execution. A report can honestly say “source implementation prepared; CPU fixture passed; live training pending.” That is more useful than collapsing all three into one completion flag.

Remaining acceptance queue and maintenance risks

The required next integration/acceptance work is sequential: apply guarded training64 after PR83 lifecycle completion; compile and run all required tests at the final head; admit fresh genuine teachers; materialize and verify the corpus; execute the one approved GPU hour; verify/export; consume the new checkpoint in actual fits; qualify fitted portability and warm/cold delivery; execute the unchanged30 panel and report all outcomes. The product chapters cover the remaining deployment stages in detail.

Maintain explicit checks for changed dependency bytes, an unsupported GPU/device, absent families, zero updates, wrong task heads, changed vocabulary bins, missing continuation, missing native component receipts, stale code hashes, counterfeit fitted outputs, incomplete process-group cleanup and partially written journals. The experimental route is intentionally separate from accepted serving. Any future promotion mechanism must add its own independent evaluation and pointer discipline rather than rename an experimental export.

The current source-only regression follow-on has an independent review with no confirmed production/API blocker. Seven pure Python checks passed; the three actual-model regression/binary/multiclass cases were explicitly deselected because custody work was barred from model execution. The next required CI/runtime acceptance must execute those cases and verify their collected inventory. This chapter records the gap rather than claiming it closed.

5 / Executable algorithms, language support and private GP examples

Four different artifacts

The private program `(pipe X (standardize) (ridge))` is a small executable algorithm description. It takes a feature table, fits standardization from training rows, then fits a ridge-regression head. It is an ordered typed recipe, rather than an already fitted set of coefficients. A different transform or head changes the algorithm. GP search works within its admitted grammar and task constraints; the presence of a code emitter does not grant arbitrary-language or arbitrary-operator coverage.

Keep these artifact types distinct:

1. **Private recipe:** the lossless canonical program and its configuration. It can be fitted again on new training data and can carry private discovery provenance.
2. **Native fitted predictor:** preprocessing state, category dictionaries, imputation statistics, transforms, calibration and head parameters learned from one exact fit. It predicts without training again.
3. **Emitted recipe:** a program in another language with `fit` and `predict` behavior. Running it on training data fits a new instance of the algorithm. Agreement between these refits tests recipe/runtime equivalence.
4. **Fitted inference bundle:** an allowlisted implementation of one exact already fitted predictor. It contains the selected inference state and an input boundary; qualification compares its predictions against the original saved native artifact.

At source `68f56404c3e199e06413bfd645d3fe0803d7b96f`, the legacy `jope emit` and `jope pipeline run --emit` paths produce type 3. `jope fit` computes a fit and reports scores but does not provide a saved fitted-artifact output flag. The pending product/winner packets expose type 4 from complete benchmark/demo fitted state. Do not advertise a legacy recipe file as the exact fitted winner.

This chapter and the supplied tiny fixture are **maintainer-private**. The DSL, private recipe manifest, discovery metadata and qualification references do not belong in a public inference bundle. The public allowlist described below is deliberately narrower than the manual's working directory.

Language interfaces at the frozen source

Interface	Input and output	Coverage and operational meaning
Native Rust `feat_math`/advanced-v3	Typed recipe or authenticated `BoundModel`; numeric or raw-table prediction boundary	The benchmark uses the raw-table boundary and preserves complete private fitted state. The native roster is task-specific.
Legacy Python emitter	Generated single-file Python with `Model` fit/predict and CSV CLI	Numerical recipe execution; host `libm`. Supports the emitter's legacy program set, not every advanced-v3 state.

Interface	Input and output	Coverage and operational meaning
Legacy Rust emitter	Generated standard-library Rust with fit/predict and CSV CLI	Numerical recipe execution; source emitted with Python as a pair. Compiler/runtime and fixture qualification remain required.
Legacy C++ emitter	One C++ source, C ABI, numeric input, fit/predict; external CSV driver for examples	Ridge/ElasticNet heads and the emitter's admitted transform subset. C++17 and C++20 use this same source with different standard flags.
Legacy JavaScript emitter	ES2020 numeric fit/predict API; external CSV driver	Ridge/ElasticNet heads and its admitted transform subset. `Math` may differ from native platform libm.
Existing private fitted C++ library	`supports_fitted_cpp17`, `emit_fitted_cpp17` over `BoundModel`	Broader scalar-state emitter exists at 68. Its source gate is not the pending public bundle gate or proof of winner parity. Multiclass probability matrices are refused by its scalar ABI.
Pending public fitted Python/Rust	`predict_raw` over ordered raw rows with string/null cells	Complete selected-state lowering for the supported subset below; no model refit.
Pending public fitted C++17/C++20	Same generated C++ inference core plus `codec.json` and Python `ctypes` raw-table boundary	Public gate currently restricts this broader emitter to regression ridge and binary logit ridge. It is not a self-contained C++ raw-table parser.
Pending public fitted JavaScript	`predictRaw(rows)` over raw rows; Node/CommonJS qualification harness	Linear and multiclass subset below; a declared ECMAScript numerical policy and tolerance are used.
CUDA	No qualified public fitted-export route in this delivery sequence	Deferred until Python/Rust, the one C++ emitter under both standards and JavaScript are qualified. A training GPU does not establish CUDA inference parity.

Source anchors: `crates/jope-cli/src/kernel.rs`, `crates/jope-pipeline/src/emit.rs`, `crates/feat-math/src/{transpile,emit_py,emit_rs,emit_cpp,emit_js}.rs`, `crates/feat-math/src/advanced_v3/fitted_cpp.rs`, and pending `advanced_v3/fitted_portable.rs` plus `crates/jope-bench/src/panel/inference.rs`.

Complete registered-recipe matrix

The table covers all 26 task-compatible native benchmark recipes. **Yes means the source admits the recipe/head under the additional state restrictions; it is not a measured parity result.** Every row has native Rust fit/save/reload support in the benchmark contract. Public fitted columns are pending source packets and require actual selected-state qualification. The private C++ column describes its existing library head gate only. External adapters follow the next subsection.

Task	Registered method ID	Private C++ scalar head gate	Public Python/ Rust fitted	Public C++17/20 fitted	Public JavaScript fitted
Regression	`rust.ridge`	Yes	Yes	Yes	Yes
Regression	`rust.ridge_gcv`	Yes	Unsupported	Unsupported	Unsupported

Task	Registered method ID	Private C++ scalar head gate	Public Python/ Rust fitted	Public C++17/20 fitted	Public JavaScript fitted
Regression	`rust.elasticnet`	Yes	Unsupported	Unsupported	Unsupported
Regression	`rust.histgbm`	Yes	Unsupported	Unsupported	Unsupported
Regression	`rust.histgbm2`	Yes	Unsupported	Unsupported	Unsupported
Regression	`rust.ksmooth`	Yes	Unsupported	Unsupported	Unsupported
Regression	`rust.ensemble`	Yes	Unsupported	Unsupported	Unsupported
Regression	`rust.krr`	Yes	Unsupported	Unsupported	Unsupported
Regression	`rust.gpr`	Yes	Unsupported	Unsupported	Unsupported
Regression	`rust.hurdle2`	Yes	Unsupported	Unsupported	Unsupported
Regression	`rust.stack`	Yes	Unsupported	Unsupported	Unsupported
Regression	`rust.tab_boost`	Yes	Yes	Unsupported	Unsupported
Regression	`rust.extra_trees`	Yes	Yes	Unsupported	Unsupported
Regression	`rust.gam_boost`	Yes	Yes	Unsupported	Unsupported
Regression	`rust.pls`	Yes	Yes	Unsupported	Unsupported
Regression	`rust.portfolio (core_v1, preset=core)`	Yes, members checked	Unsupported	Unsupported	Unsupported
Regression	`rust.gpr.rbf_linear`	Yes	Unsupported	Unsupported	Unsupported
Binary	`rust.logit_ridge`	Yes	Yes	Yes	Yes
Binary	`rust.tab_boost`	Yes	Yes	Unsupported	Unsupported
Binary	`rust.extra_trees`	Yes	Yes	Unsupported	Unsupported
Binary	`rust.gam_boost`	Yes	Yes	Unsupported	Unsupported
Binary	`rust.portfolio`	Yes, members checked	Yes	Unsupported	Unsupported
Multiclass	`rust.multiclass_ridge`	Unsupported matrix ABI	Yes	Unsupported	Yes
Multiclass	`rust.multiclass_ovr_logistic`	Unsupported matrix ABI	Yes	Unsupported	Yes
Multiclass	`rust.multiclass_diagonal_lda`	Unsupported matrix ABI	Yes	Unsupported	Yes
Multiclass	`rust.multiclass_nearest_centroid`	Unsupported matrix ABI	Yes	Unsupported	Yes

This yields static public coverage of 14/26 recipes for Python/Rust, 2/26 for each C++ standard and 6/26 for JavaScript. Those counts are not percentages of datasets, winners or actual successful parity runs. Python/Rust's twelve unsupported registered recipes are legacy regression heads outside the public fitted lowering. Generated programs can introduce additional unsupported state even when their final head appears in a supported row.

The registry records aliases separately: regression ridge has the historical alias `logistic`, histogram boosting has `gbm`, the ensemble has `ensemble_ridge_gbm`, and binary logit ridge has `logistic`. Aliases do not consume extra fit slots. `gpr.rbf_linear` remains a distinct configuration. Do not infer task compatibility from an ambiguous alias; use task plus exact registered recipe identity.

Source anchors: `crates/jope-bench/src/panel/methods.rs::rust_methods`, `advanced_v3/fitted_cpp.rs::{supports_fitted_cpp17,check_head}`, pending `advanced_v3/fitted_portable.rs::{fitted_portable_unsupported,fitted_javascript_unsupported,fitted_public_cpp_unsupported}` and `advanced_v3/fitted_portable/scalar.rs::supports`.

State restrictions beyond the head table

The pending public lowering validates the full model binding and fitted input dimensions. It admits IID rows and fitted IID row order only. Historical state, other model forms, target mappings, nonempty general transform lists and calibration are currently unsupported. Categorical encoding must be the admitted fitted one-hot mode. Missing-aware state is limited to the supported tabular-boosting and extra-tree implementations. Invalid coefficients, schema bindings or dimensions are errors, not legitimate unsupported fallbacks.

The benchmark's registered recipes use numeric median imputation, categorical one-hot encoding, missing indicators, `min_count=1` and `max_onehot=256`. Their fitted dictionaries and imputation statistics are part of the state. A deployment must submit columns in the recorded order and use the recorded raw string/null representation. An unseen category must follow the fitted input policy; neither the host nor caller should invent a new category mapping. Binary probability outputs include the two original class names and order; multiclass outputs include every original class column.

The existing private C++ library has broader checks for target mappings, transforms, calibration and ordered/raw input than the public fitted gate. Its broader admission must not be used to skip public privacy, raw-boundary or numerical qualification. The selected public subset is a deliberate current boundary, not evidence that every other native model is intrinsically unexportable.

External winners

XGBoost, LightGBM, CatBoost and scikit-learn are executable benchmark adapters for the admitted tasks. Their native fitted artifact retains the estimator pickle and preprocessing/reload evidence. The pending public winner-export helper reports standalone Python, Rust, C++17, C++20 and JavaScript export of that external library state as **typed unsupported**. It preserves the external winner, all losses and native fitted artifact. Installing a library in Python does not make its pickle a dependency-free Python inference bundle.

Example A: a tiny private recipe, fit and predictions

The supplied `examples/gp-synthetic.json` contains 24 explicitly synthetic rows with two integer, non-null features and target `3*x0 − 2*x1 + 5`: 16 training rows, four validation rows and four test rows. It is a maintainer example, not one of the frozen thirty datasets or the official ten generated fixtures. `train.csv` contains the same sixteen training rows as headerless `x0,x1,target`; `predict.csv` contains the eight validation/test feature rows with targets removed.

One training row is `-12,-5,-21`; the corresponding typed JSON row is:

```
{"id":0,"features":
[-12,-5],"target":-21,"split":"train","group":null,"timestamp":null,"series":null}
```

Set `R` to the repository root, `E` to `$R/docs/internal/asm-manual/examples`, `J` to a checksum-verified immutable CLI and `A` to a new durable private output directory. Execute numerical work only after its allocation is available. The following commands are reproducible instructions. The separate measured execution below records all fourteen commands passing; the commands themselves are not receipts:

```
mkdir "$A"
"$J" fit --input "$E/gp-synthetic.json" \
  --genome '(pipe X (standardize) (ridge))' --json > "$A/fit-report.json"
"$J" emit --genome '(pipe X (standardize) (ridge))' \
  --lang all --output "$A/recipe" --json > "$A/emit-report.json"
```

`fit-report.json` records training/holdout R^2 for this numeric legacy path. The holdout prefers validation rows; the fallback is test rows if validation cannot be loaded. Do not use this convenience path to establish the frozen-panel leakage or classification protocol. It does not persist a complete fitted artifact. `emit-report.json` lists language source sizes, paths and hashes, and each emitted program below fits its own model from `train.csv`.

```
python3 "$A/recipe/model.py" "$E/train.csv" "$E/predict.csv" "$A/recipe/pred-python.txt"
rustc -O "$A/recipe/model.rs" -o "$A/recipe/model-rs"
"$A/recipe/model-rs" "$E/train.csv" "$E/predict.csv" "$A/recipe/pred-rust.txt"

cat "$A/recipe/model.cpp" "$R/deploy/ci/emitters/driver.cpp" > "$A/recipe/program.cpp"
g++ -std=c++17 -O3 -ffp-contract=off -fno-fast-math \
  "$A/recipe/program.cpp" -o "$A/recipe/model-cpp17"
"$A/recipe/model-cpp17" "$E/train.csv" "$E/predict.csv" "$A/recipe/pred-cpp17.txt"
g++ -std=c++20 -O3 -ffp-contract=off -fno-fast-math \
  "$A/recipe/program.cpp" -o "$A/recipe/model-cpp20"
"$A/recipe/model-cpp20" "$E/train.csv" "$E/predict.csv" "$A/recipe/pred-cpp20.txt"

cat "$A/recipe/model.js" "$R/deploy/ci/emitters/driver.js" > "$A/recipe/program.js"
node "$A/recipe/program.js" "$E/train.csv" "$E/predict.csv" "$A/recipe/pred-js.txt"
python3 "$R/deploy/ci/emitters/compare-bits.py" "$A/recipe" > "$A/predictions.bits"
python3 "$E/validate-outputs.py" --recipe-output "$A/recipe" > "$A/recipe-parity.json"
```

The supplied `validate-outputs.py` requires all five prediction files to exist, each with eight finite values, and checks their complete bit patterns before producing a success receipt. The helper compares every existing `pred-*.txt`; by itself it does not enforce that the required five files were produced. Record each command's nonzero exit as failure and retain its stderr. A comparison of only one surviving file cannot qualify the language matrix. The observed exact bit agreement on this ridge fixture does not imply exact JavaScript agreement for arbitrary transcendental operations.

`examples/commands.json` supplies the exact argument arrays and placeholders for a maintainer runner. Compiler and interpreter versions, source hashes, fixture hashes, commands, exit codes and wall/CPU measurements belong in the execution receipt. Preparation and execution receipts are retained separately. The measured execution below supersedes the original static-only status.

Example B: actual legacy pipeline/search shape

The existing pipeline takes a locally cached catalog dataset directory with `metadata.json` or `manifest.json` and verified files. A bare CSV and the synthetic JSON above are not that catalog format. Use the committed regression fixture for a reproducible offline command:

```
"$J" pipeline run \
  --dataset "$R/datasets/test/pmlb-229-pwlinear" \
  --budget evaluations=4,seconds=60 --seed 7 --rng-seed 7 --cuts 1 \
  --emit rust,python,cpp17,cpp20,js --policy skip \
```

```
  --output "$A/pipeline" --json > "$A/pipeline-result.json"
"$J" pipeline verify-receipt --receipt "$A/pipeline/receipt.json" --json
```

This performs numerical GP candidate measurements, selects a recipe, emits supported recipe sources and writes a pipeline receipt. `--policy skip` explicitly records the absent neural proposal stage; `auto` also records a skip and `require` fails closed in this legacy route. It is not the qualified neural ASM arm.

The legacy budget is per cut, combines exploration/comparison attempts and active seconds, and is nonpreemptive for an in-flight fit. Multiple cuts can consume multiple per-cut allowances. It is not the twenty-worker panel's hard isolated 30-second fit and original global deadline contract. Preserve the recorded `deadline_overrun` and stage counts rather than applying official panel claims to this command.

The emitted winner may not be covered by C++ or JavaScript. The receipt retains exclusions and their reasons; it never replaces the selected recipe solely to make all requested languages emit. Reusing the same C++ source under `-std=c++17` and `-std=c++20` establishes the intended single-emitter structure. The maintained `deploy/ci/emitters.sh` and `deploy/ci/e2e-smoke.sh` provide larger qualification workflows; running them builds and executes models and requires its own resource allocation.

Example C: exact fitted winner export after pending integration

Use the final integrated installed CLI, original admitted plan and journal, unchanged panel/development inputs and exact selected recipe. The pending front door accepts a dataset and an existing fitted fold, rather than a replacement recipe:

```
"$J" benchmark export-winner \
  --plan "$PLAN" --plan-sha256 "$PLAN_ID" --manifest "$M" --datasets "$D" \
  --journal "$JOURNAL" --dataset "$DATASET_ID" --fold 0
"$J" benchmark qualify-export \
  --prepared "$PREPARED_EXPORT" --prepared-sha256 "$PREPARED_EXPORT_ID" \
  --tools "$TOOLS" --tools-sha256 "$TOOLS_FILE_SHA256"
"$J" benchmark report \
  --plan "$PLAN" --plan-sha256 "$PLAN_ID" --manifest "$M" --datasets "$D" \
  --journal "$JOURNAL"
```

The first command returns the semantic prepared identity, private `prepared.json` and public bundle root. It chooses the smallest complete five-fold mean loss, breaking ties by method identity. Incomplete confirmation has no final winner. Choose another existing fold only when that exact fitted state is the intended qualification target; fold-zero parity does not prove the other four states.

The tools file uses the existing private `Runtime` schema keyed by command name, normally `python3`, `rustc`, `g++` and `node`. Each entry pins the physical executable and SHA256, exact version, empty argument prefix, required toolchain/package artifacts, admitted environment and unavailable reason or null. Use the real compiler executable, not an unpinned selector such as a mutable rustup shim. Tool inventory must be prepared from actual installed bytes; an illustrative JSON snippet with invented digests is not admissible.

Qualification order is Python, Rust, the same C++ emitter under C++17 then C++20, then JavaScript. Unsupported settles that backend's inventory without replacing the winner. A compiler absence, crash, timeout or parity failure prevents silently proceeding as though that backend qualified. The complete probability matrix, row order and class order are compared, not just the winning class or a rounded score.

The child receives generated inference files and unlabeled raw prediction rows. Expected predictions, original scoring labels and dataset-root access stay with the controller. Each job uses the existing bounded worker supervisor. Report recovery checks source/tool/input/artifact/command/receipt identities and runs no compiler or predictor. Started work without a finished receipt remains unknown until shared terminal recovery; it is not a successful retry.

Backend	Public files	Pinned numerical comparison
Python	`model.py`	Exact IEEE-754 bits, host platform libm.
Rust	`model.rs`	Exact IEEE-754 bits, host platform libm.
C++17 / C++20	`model.cpp`, `codec.json`, `boundary.py`	Exact IEEE-754 bits through the typed Python/ctypes boundary; no fast-math/FMA contraction drift.
JavaScript	`model.js`	Absolute/relative `1e-12` policy with explicit ECMAScript Math identity.

Necessary public metadata records the task, ordered columns/classes, fitted raw-cell/input policy, emitter/template hashes, runtime, numerical policy and file checksums. The initial bundle is `emitted-unqualified`. `all_backends_qualified` becomes true only after five actual successful qualifications. An operation can complete with explicit unsupported outcomes while that all-backends flag remains false.

These jobs use the original campaign's remaining execution allowance; they do not open a new export clock. For the official two-hour panel, no new export job may start after original `t0 + 6900 seconds`, and the last five minutes remain reporting reserve. If confirmation consumes the window, preserve unattempted export statuses. A future separately bounded post-panel validation stage would need its own explicit identity and cost accounting; the current front door does not implement one.

Future work and acceptance boundaries

Priority gaps are actual full-state parity for supported winners, broader fitted-state lowering, a typed multiclass C++ boundary, qualified external-library export where feasible, and only later CUDA. Each extension needs exact native-state replay, missing/unknown-category tests, numerical policy fixed before scoring, stale-evidence refusal and public-bundle canaries. New operators additionally need ablation and compatibility evidence; adding an emitter branch alone does not establish reusable learning value.

Retain native winners even when every requested language is unsupported. Report the full loss/ranking and export gap together. Performance claims need measured creation/prediction time on the actual artifact and hardware; binary size, source-line count, emission throughput or a passed static support function cannot substitute for those measurements.

Measured execution of the private examples

On 23 September 2026, all **14 unchanged example commands passed**, with zero failed, unavailable, timed-out or unstarted commands. This supersedes the examples' earlier “prepared, not executed” status. The original command packet and static-only preparation receipts remain unchanged as historical inputs; this execution is separate evidence.

The immutable CLI came from reviewed source `68f56404c3e199e06413bfd645d3fe0803d7b96f`; merged main `a75e543d2166017cb0139e0a251d51d9a0c98e08` has the identical source tree `176b75bc36d76c7ce4eaeb7234677ec013468889`. The executable SHA256 was `eab2d5b1b05629020aab3d540d51eb24ba0db18ff0917da188f89a34b6dc5e2b`. The original fourteen-command packet SHA256 was `f34771938934d1360cbe054689393198a53929479ba119f5a0faf2b0da7a170a`.

The private synthetic fixture contains **24 rows: 16 training, four validation and four test**. The recipe `(pipe X (standardize) (ridge))` achieved training R^2 **0.9995230651** and validation R^2 **0.9918641493**. Its emitted Python, Rust, C++17, C++20 and JavaScript programs each fitted from the same training CSV and produced **eight finite predictions with identical IEEE-754 bits**. A separate validator required all five output files and all eight rows, so a missing backend could not pass by omission.

The packet also ran the separately identified committed **PMLB 229 pwlinear** legacy pipeline fixture (`1cb97b1d7314d43d`). It completed four successful candidate measurements with no failed measurements, selected `(pipe X (ridge))`, reported validation RMSE **0.0775685574** and R^2 **0.8151398564**, and passed receipt verification. Its policy stage was explicitly skipped. This fixture is not the synthetic twenty-four-row example and this invocation is not the official thirty-dataset campaign.

Measurement	Recorded value
Controller elapsed time	10.407138 seconds
Launcher elapsed time	10.506140 seconds
Exact systemd unit CPU	9.382049 CPU-seconds
Fourteen-command child CPU subtotal	2.952963 CPU-seconds
Python	3.12.3
Rust	1.97.1
G++	13.3.0
Node.js	26.1.0

The unit CPU includes overlapping command/controller work; **do not add it to those subtotals**. The separately granted allocation declared one CPU, 4 GiB memory, zero swap/GPU, network denial and 300 seconds total, with 30-second numerical-command and 60-second compiler limits. The shared exclusive CPU lock was held and released; all command process groups drained. CPU affinity `[0]` and the dedicated unit cgroup were observed. A live property readback raced unit collection and returned unloaded defaults, so independent readback of memory/swap/quota/network/device limits is **not established**. The unit reported 5,332,992 bytes peak memory and zero swap, while cumulative reaped-child maximum RSS was 168,808 KiB; retain these different accounting domains without claiming a reconciled total peak.

This result qualifies the documented **recipe refits and legacy pipeline example**. It does not qualify exact saved-fitted-state export, neural ASM consumption, an ASM under-sixty-second warm demonstration, GPU training or the full panel. No Cargo build or download ran. Full command outputs, tool/input/source/output hashes and the original unit journal are retained in the private `agent-maintainer-manual/examples-run/` evidence directory.

Evidence pins: `HANDOFF.json` SHA256 `a05e8a0e9f9430b0c71ff23c57e5e29532a61bbe118daac3fe358b45f2423346`; `audit.json` SHA256 `363fbe62a706b44030205da78411243ba206ad2366d546a277babe966b802c91`; `result.json` SHA256 `71395a038feeef768bd6c69bb43dcabbfced20333d989cd1161c07213f4f68d6`.

6 / Benchmark protocol, rankings and interpretation

Three evidence collections with different purposes

The frozen ASM panel, the historical same-cut benchmark and the imported SOTArena ladders have different inputs and protocols. Keep their figures and identities separate. A strong historical Elo is not a result for the current thirty-dataset ASM campaign. A rank computed on development labels is not a sealed-test estimate.

Collection	Inputs and purpose	Current interpretation
Frozen `asm-private-30-v1`	Ten regression, ten binary and ten multiclass datasets; original frozen row assignments	Current implementation target; one regression operational proof, full panel unrun
Historical same-cut v1/v2	Ten committed regression datasets; all methods share each cut; screening and deepening	Reproducible historical comparisons; v1 and v2 are separate snapshots
Historical tabular-regression-v2	Imported score inventory and corrected benchmark-valid/public-development views	Historical ladder with its own coverage and validity rules; not a current ASM qualification

The plot source tables retain method, dataset, snapshot, coverage and source hashes. Missing cells stay missing. The book does not manufacture thirty current outcomes from the historical collections. Historical method names are identifiers; the word “jope” in an identifier does not prove the method is this exact product checkpoint or executable.

Frozen-panel execution contract

The panel is immutable once comparison begins. Its identity covers dataset metadata, original labels and types, row IDs, group membership, original development folds, sealed membership and prepared data hashes. Do not replace difficult datasets, reshuffle rows, map categories to arbitrary numeric order, drop missing rows or substitute a different target to make a task executable. Missing source subject or time identifiers remain limitations even when duplicate-feature groups are protected.

For each dataset, screen every applicable native method, configured external adapter and admitted search arm on development fold zero. Freeze the top five recipe identities and confirm those recipes across all five original development folds. Additionally confirm ASM if it missed the top five. A configuration such as `gpr.rbf_linear` is distinct; a spelling alias does not justify an extra independent fit. The recipe is frozen, while the fitted state for each fold is learned from that fold's permitted training rows.

ASM compares 64 neural plus 16 exploration generation-zero slots with 80 generic slots. Both arms receive the same declared search and fit limits. A producer refusal, duplicate or failed candidate is an outcome to record. It cannot silently become another neural success or a free replacement arm. Actual inner-fold fits determine selection and offspring ancestry. A model's predicted reward never substitutes for measured loss.

This screen/freeze/confirm design uses development data adaptively. The selected recipe can already have learned from rows that occur in a later confirmation fold. Fold-zero labels influence screening. The implementation records these intersections; the five-fold summary remains adaptive development evidence. Sealed rows and sealed meta-test families stay outside learning, optimization and this documentation audit. The word “test” in an optimization-visible file does not make it a valid test set.

Metrics and class order

Task	Primary reported metric	Smaller-is-better search loss	Required prediction shape
Regression	RMSE	RMSE	One finite real prediction per row
Binary classification	AUROC	`1 − AUROC`	Probabilities with original two-class order
Multiclass classification	Log loss	Log loss	One probability for every original class on every row

RMSE is the square root of the mean squared prediction error. It has the target's units, so a raw RMSE on machine CPU cannot be averaged meaningfully with an unrelated dataset's RMSE without an explicitly declared normalization. AUROC measures score ordering between positive and negative examples; the positive class and class order must be fixed by the task contract. A one-class evaluation fold cannot produce a meaningful AUROC. Multiclass log loss averages the negative logarithm of the probability assigned to the true class. Integer class predictions alone cannot supply that probability-matrix metric.

Probability shape, finiteness, class names and order belong to the authenticated prediction boundary. A backend that produces the same numbers in a different class order has failed parity. A backend with only scalar output cannot be advertised as supporting a multiclass winner. Any numerical clipping, normalization, accumulation, tie or tolerance rule must be fixed before scoring and recorded with the evaluator; changing the rule afterward changes comparison identity.

The historical same-cut command has an older contract. It constructs its own hashed cuts, uses regression R^2, binary AUROC and multiclass macro-F1 for screening, and applies its recorded robust-fitness summary during deepening. Its regression ladder uses RMSE; its classification ladder can maximize AUROC or macro-F1. Those historical definitions must not be silently copied into the frozen ASM panel's multiclass log-loss contract. Source: `docs/same-cut-bench.md` and `crates/jope-bench/src/eval.rs`; current panel code lives under `crates/jope-bench/src/panel/`.

What Elo means here

The arena implementation fits a regularized Bradley–Terry model on an Elo scale. For two ratings A and B, its modeled pairwise probability is:

```
P(A beats B) = 1 / (1 + 10 ** ((rating_B − rating_A) / 400)).
```

Each dataset with both eligible scores supplies one equally weighted comparison for that method pair. The observed outcome is 1 for a win, 0 for a loss and 0.5 for a tie. Metric direction is explicit. The relative tie condition is `abs(a − b) <= 1e-12 * max(abs(a), abs(b), 1)`. A missing or ineligible score supplies no invented numerical loss.

The default implementation starts and recenters ratings at 1500, runs at most 250 diagonal-Newton iterations, applies L2 regularization of `1e-4`, clips each iteration's update to ±40 Elo points and stops when the largest update is below `1e-8`. The ±40 setting is an iteration clamp, not an online match-by-match chess update. Source: `crates/jope-arena/src/elo.rs`.

Two source paths preserve historical arithmetic: one accumulates individual sorted dataset/pair matches in the original Python order, and the reporter aggregates each pair's sufficient statistics. They are equivalent in exact arithmetic but can differ slightly in floating-point accumulation. Reproduction must identify the path and snapshot, not merely say “Elo.”

The rating is relative to the included method universe, evidence eligibility and overlap graph. A high rating on a handful of datasets has different coverage from the same rating on every dataset. Report dataset coverage, full-coverage rank, average rank, wins, pending cells and protocol beside Elo. Adding methods, removing invalid rows or changing overlap can change ratings without any predictor changing. Elo does not measure the magnitude of an RMSE improvement, runtime, memory, statistical certainty or future unseen-data performance. No confidence interval or significance claim is implied by a plotted point.

Reading the performance figures

Historical ladders retain their original snapshot names. Benchmark-valid and public-development views are separated. Same-cut screening includes all admitted methods; deepening includes only finalists, so empty cells have a deliberate meaning. A rank heatmap compares methods within a dataset and stage; it is not a plot of absolute loss across datasets. Any chart with only the top methods states that limit, while its CSV preserves the full underlying inventory.

The current operational figure contains only the two development RMSE values from the preselected `230_machine_cpu` proof: neural 166.3181717454848 and generic 155.25909998168862. Lower is better. Generic wins that comparison. It would be misleading to compute a new thirty-dataset Elo from these two values or describe the 832 fitted predictors as 832 independent datasets.

The corpus-size figures are metadata/development statistics, not performance. The language coverage figure counts admitted source recipes under additional state restrictions; it is not a backend success rate. The architecture diagram is explanatory, and the completion matrix is an evidence inventory; neither is an experimental result.

Resource limits and accounting

The default admitted panel allocation is twenty CPUs/workers, 64 GiB memory, zero swap, one RTX 3090 when GPU work is explicitly allocated, thirty seconds per fit and 120 seconds per search arm. The full panel receives two hours total, including a five-minute reporting reserve. Training receives one GPU-hour total and twenty minutes per category. Do not overlap numerical campaigns with exclusive full-CPU CI. Resume continues the original campaign deadline and resource envelope.

Report elapsed wall time, summed worker wall time, worker CPU time, aggregate supervisor CPU, GPU time if measured, launched commands, successful fits, failures, unique fitted predictors and unique reusable recipes separately. Aggregate supervisor CPU contains worker CPU; adding them double-counts. Parallel worker durations do not equal elapsed wall time. Deduplicated/cache reuse is useful but is not another physical fitted predictor.

The current one-dataset proof completed 964 worker commands: 834 completed and 130 failed. It counted 832 unique fitted predictors and 164 recipes, plus two final refits with 34 development predictions each. Its run took 239.143121970 seconds; independent report recovery took 18.718997325 seconds. These values are measured at the qualified source and protocol, not projected full-panel throughput. The status chapter gives CPU totals and amortized denominators.

The supplied one-hour/ten-million-token scenario is a contextual scenario only. Its workload, model, hardware, predictor inventory and token/cost traces are unavailable. This proof used approximately 6.64% of an hour for its run, but that fraction is not a matched speedup or token-efficiency result. Dollar cost and token usage remain unmeasured.

From optimization to reusable knowledge

The pending strict HWM key includes task, dataset hash, cut identity, evaluator, vocabulary and protocol. Scores from different identities cannot overwrite each other or supply training feedback. A recipe must show finite train-side out-of-fold improvement over a trivial baseline and improve on four of five folds before teaching admission. No-signal data should be refused by this gate. An optimization-visible test label is development evidence even if its filename says otherwise.

Typed motifs can reuse eligible program fragments and macros while retaining provenance. The prepared motif library does not yet establish the production eligible-state reader, `motifs mine` integration or operational reuse. A new operator requires compatibility tests and an ablation under the unchanged comparison protocol. Repeatedly selecting the best development result can improve development performance without proving improved sealed generalization.

Minimum complete future campaign report

For every one of the thirty immutable datasets, retain the input and cut identities; applicable, skipped and unavailable methods with reasons; screen outcomes; frozen finalists; all confirmation-fold outcomes; ASM's extra confirmation when needed; original and fitted recipe identities; class order; selected loss and exposure labels; resource limits and original deadlines; failures and cancellations; actual fit/predict counts; and retained winner/backend outcomes. Report all losses and gaps, not only wins. A deadline-exhausted dataset remains in the denominator with an explicit incomplete outcome.

Ten wins per category is a measured target. The current evidence does not promise that result. The full-panel report, newly trained policy consumption, selected fitted-winner parity and serving promotion remain separate acceptance milestones.

7 / Data estate, statistics and measured evidence

Observation date: 2026-09-23T16:41:48.371336+00:00. Frozen source: `68f56404c3e199e06413bfd645d3fe0803d7b96f`. These are observations at that source and time. They do not turn stored metadata, a source packet, or historical results into a current operational qualification.

What is measured and what remains pending

The current source has a successful operational proof on `230_machine_cpu`: the verified encoder produced context; the neural producer supplied actual generation-zero fits; a fitted descendant followed; two fitted refits each returned 34 development predictions. The neural refit did not beat the matched generic arm. This proof uses the accepted checkpoint, not a newly trained experimental export. Training, final installed warm demonstrations, and the unchanged thirty-dataset panel remain unstarted at this observation. No current thirty-dataset ranking or Elo exists.

Every number in this chapter comes from a pinned manifest, prepared development table, immutable report, committed historical snapshot, filesystem metadata, or authenticated AWS metadata request. The audit did not run a fit, load a model, open sealed target bodies, or execute a report command.

The machine-readable companions are `dataset-inventory.json` (all thirty datasets, 522 feature records, complete available upstream metadata and detailed development statistics), `performance-evidence.json` (measured current origin plus historical score cells, timing and ladders), `catalog-summary.json`, `local-estate.json`, `s3-estate.json`, and `metadata-receipts/`. The catalog JSONL retains every available per-dataset catalog field for all 3,859 corpus datasets. No claim is made to enumerate every mathematically possible statistic.

Frozen thirty-dataset panel

Panel `asm-private-30-v1` uses ten regression, ten binary, and ten multiclass datasets from PMLB revision `7c1f4bdc00136dc2e55c87fa6b8ba6e8af6d1a68`. Manifest SHA-256: `ba7dace4cd535687002cc95538dbc7f7aaf77ec973177c7159b3c6d22471eee0`. It was selected before measuring ASM outcomes, excludes the historical favorable test-10 collection, and cannot replace a dataset because a method failed or lost.

Local preparation: `/home/ubuntu/.cache/jope/asm-private-30-v1/`. `development/` contains the permitted development tables; `splits/` contains original row/group/fold assignments; `metadata/` contains pinned upstream YAML; `downloads/` contains the original compressed archives; `sealed/` contains withheld partitions. The audit opened development, assignment, and metadata files only. It inspected sizes and expected hashes of raw and sealed files without opening their bodies.

There are **55,029 total rows**, **44,048 development rows**, and **10,981 sealed rows** according to the frozen assignments. Feature counts sum to 522 across datasets; this is not a union of unique semantic variables. Development data contains 1,612 missing feature cells in 302 rows. All thirty development, split-assignment and metadata file SHA-256 values match the frozen manifest. Every row ID is unique within its dataset, and every group stays wholly within one original development fold or sealed partition.

Dataset	Task	Total rows	Development	Sealed metadata	Numeric features	Categorical features	Missing development cells	Fold sizes
197_cpu_act	regression	8192	6553	1639	21	0	0	1311,1311,1311,1310,1310
225_puma8NH	regression	8192	6553	1639	8	0	0	1311,1311,1311,1310,1310
230_machine_cpu	regression	209	169	40	6	0	0	34,32,37,33,33
522_pm10	regression	500	402	98	7	0	0	81,81,91,67,82
529_pollen	regression	3848	3078	770	4	0	0	616,616,616,615,615
579_fri_c0_250_5	regression	250	200	50	5	0	0	40,40,40,40,40
auto_insurance_price	regression	201	162	39	13	10	10	35,32,31,31,33
690_visualizing_galaxy	regression	323	258	65	4	0	0	52,52,52,51,51
712_chscase_geyser1	regression	222	172	50	2	0	0	32,34,38,35,33
503_wind	regression	6574	5238	1336	14	0	0	1066,1074,1030,1032,1036
breast_cancer_wisconsin_diagnostic	binary	569	455	114	30	0	0	91,91,91,91,91
credit_approval_germany	binary	1000	800	200	3	17	0	160,160,160,160,160
congressional_voting_records	binary	435	347	88	0	16	0	77,57,64,84,65
haberman	binary	306	245	61	3	0	0	49,49,46,50,51
ionosphere	binary	351	281	70	32	2	0	55,57,56,57,56
sonar	binary	208	166	42	60	0	0	34,33,33,33,33
spambase	binary	4601	3723	878	57	0	0	696,781,710,789,747
phoneme	binary	5404	4330	1074	5	0	0	866,867,866,870,861
tic_tac_toe	binary	958	766	192	0	9	0	154,153,153,153,153
horse_colic_surgery	binary	368	292	76	7	13	1583	58,61,58,58,57

Dataset	Task	Total rows	Development	Sealed metadata	Numeric features	Categorical features	Missing development cells	Fold sizes
iris	multiclass	150	122	28	4	0	0	25,23,25,25,24
wine_recognition	multiclass	178	142	36	13	0	0	29,29,28,28,28
car_evaluation	multiclass	1728	1382	346	0	6	0	277,277,276,276,276
balance_scale	multiclass	625	500	125	0	4	0	100,100,100,100,100
dermatology	multiclass	366	292	74	1	33	0	59,59,58,58,58
segmentation	multiclass	2310	1848	462	18	1	0	364,372,369,373,370
vehicle	multiclass	846	677	169	18	0	0	135,136,135,136,135
optdigits	multiclass	5620	4496	1124	51	13	0	900,899,899,899,899
penguins	multiclass	344	275	69	5	2	19	55,55,55,55,55
tae	multiclass	151	124	27	3	2	0	20,25,27,30,22

Regression uses RMSE; binary uses AUROC with minimization loss `1−AUROC`; multiclass uses probability-matrix log loss with the frozen class ordering. Features retain numeric/categorical type and missing cells; original labels are not converted to manufacture a task. Screen all applicable methods on original fold zero, freeze five recipes, confirm across all five development folds, and confirm ASM additionally if it missed the five. These optimization-visible folds remain development evidence. Sealed evaluation is separate.

Known date columns and identical-feature groups are retained by the prepared split. Missing original subject/time identifiers remain a limitation. A grouped split cannot establish independence when the source never supplied those identifiers. Checkpoint exposure is unknown for these datasets; an unseen-family generalization claim is not supported.

Panel storage

File class	Logical bytes	Content read in this audit
development	17,923,164	Yes; metadata/development statistics and SHA-256
metadata	82,681	Yes; metadata/development statistics and SHA-256
sealed	4,518,613	No; size and expected identity only
source_archive	2,075,982	No; size and expected identity only
split_assignments	9,812,165	Yes; metadata/development statistics and SHA-256

File sizes are logical bytes, not compressed-memory use or total disk allocation. The raw archive sizes sum to 2,075,982 B, matching the frozen declared download byte counts; content hashes of those raw archives were not revalidated here because the archives also contain sealed targets. The prepared sealed file hashes are declared frozen identities, not freshly verified content checks in this audit.

Configured product stores and local roots

AWS metadata access used profile `veox-jepson-prod` and verified account `714044927654` with STS. No credentials are included in the receipts. Current-object listings exclude object versions and deleted objects. ETags are retained as concurrency guards for metadata downloads; they are not assumed to be SHA-256 content hashes. Catalog/collection/bundle manifest downloads use the exact listed ETag and verify size; their bytes receive local SHA-256 pins. No S3 dataset blobs, sealed tables, or model weight bodies were fetched.

`jope-datasets` defaults to `s3://veox-jopedime-datasets-use2`; `jope-models` defaults to `s3://veox-jopedime-models-use2`. The known learning store is `s3://veox-jopedime-learning-use2`; the current `jope-learning` mirror resolver uses `--store`, then `JOPE_LEARNING_URL`, and otherwise no remote mirror. It must not be described as an automatic remote default. No JOPE path/store overrides were set in this audit process.

There are two source-level path conventions to name explicitly: dataset cache resolution defaults to `${XDG_CACHE_HOME:-~/.cache}/jope`, while general `jope_paths::jope_home()` resolves a marked checkout. General mutable state defaults to `${XDG_STATE_HOME:-~/.local/state}/jope`. Use an explicit root or the owning command's resolver; do not assume the shared variable name makes the defaults identical.

S3 bucket / prefix	Current listed objects	Current listed logical bytes
`veox-jopedime-datasets-use2`	15,442	3,234,922,515
↳ `blobs/`	11,577	3,224,104,137
↳ `catalog/`	2	5,363,942
↳ `collections/`	4	94,356
↳ `datasets/`	3,859	5,360,080
`veox-jopedime-learning-use2`	645,281	430,645,459,168
↳ `staging/`	645,281	430,645,459,168
`veox-jopedime-models-use2`	967	3,929,105,273
↳ `blobs/`	21	1,461,921,570
↳ `bundles/`	3	32,045
↳ `staging/`	943	2,467,151,658

The three bucket listings completed with zero payload-body reads. The learning bucket contains current `staging/lineage-a/`, `staging/lineage-b/` and `staging/mirror-weights/` prefixes; per-lineage byte counts were not separately aggregated by this listing. Current listed learning bytes are 430,645,459,168. A separate CloudWatch daily statistic at 2026-09-23T00:00:00Z reports 645,281 objects and 430,732,116,688 StandardStorage bytes. It is a dated metric with a different scope/capture time; no exact reconciliation with the current listing is claimed. Paginated S3 listing is an observation over an interval, not an atomic bucket snapshot.

These are the three product stores inspected. Known archive and alternate-region locations documented in the host contract were not re-enumerated in this task; their current object/version counts and bytes remain unknown here. Counts of stored objects do not establish recovery completeness, teacher admission or serving eligibility.

Local root	Present	Regular files	Logical bytes	Unique-inode logical bytes
`/home/ubuntu/.cache/jope`	True	170	36,754,881	35,585,244
`/home/ubuntu/.local/state/jope`	False	0	0	0
`/home/ubuntu/JopeDime/datasets/test`	True	40	1,096,413	1,096,413
`/home/ubuntu/JopeDime/datasets/by-id`	False	0	0	0
`/home/ubuntu/JopeDime/models`	True	5	33,687	33,687
`/home/ubuntu/JopeDime/state`	False	0	0	0

The local inventory uses `lstat`, does not follow symlinks, and counts hardlinked payload bytes once in the unique-inode column. It covers the named product roots, not every archive, research checkout or build directory on the host. The standard local state roots were absent. That does not imply the host has no learning evidence: campaign and recovery receipts live in explicit custody paths outside those defaults.

The local dataset cache has two cataloged regression datasets, `0014b9dad5725a1e` (2,000 rows, 16 features) and `006bca140f93e411` (2,000 rows, 32 features), each with 1,000 declared training and 1,000 declared test rows. Their payloads were not reopened for this inventory. The committed historical `datasets/test/` collection is 10 datasets/40 files/1,096,413 B; it is distinct from the frozen thirty-dataset cache.

S3 corpus catalog

The pinned current catalog is `ab9fda8d2dea46067b70a42812e9d3c1d9dc6ba025780100df34377e81aa1120`: 3,859 distinct dataset hashes, all marked real, 6,208,051 declared rows. Shape metadata sums to 265,105 features and 4,628 categorical features; sums across datasets are not deduplicated semantic quantities. The catalog contains no task explicitly named multiclass; 322 entries are multi-label, which is a different task and must not be silently relabeled.

Catalog task	Datasets	Total declared rows	Training rows	Test rows metadata only	Feature-count range	Row-count range
binary	843	1,490,965	892,307	598,658	[1, 2000]	[10, 69624]
multi_label	322	786,436	555,756	230,680	[1, 20531]	[23, 50000]
regression	2,611	3,025,922	1,898,974	1,126,948	[0, 1688]	[10, 50000]

Catalog task	Datasets	Total declared rows	Training rows	Test rows metadata only	Feature-count range	Row-count range
timeseries	83	904,728	775,519	129,209	[2, 86]	[81, 49999]

Catalog discrepancy: the stored registry collection description says 1,416 unknown licenses. Counting the current catalog's actual license-status fields gives **3,755 unknown and 104 recorded**. This chapter uses the counted fields and retains the stale description as provenance. All 2 collections remain private: `registry-real-v1` has 3,859 members and `test-10` has 10; every listed member occurs in the catalog. No payload or training eligibility validation was performed simply by counting them.

Seven regression catalog entries declare zero features. These are retained anomalies, not automatically usable supervised tasks: `5ffbe1543f683e4d`, `66fea7ec7dffa02a`, `72235646c52be632`, `7d1497edfe70495d`, `d12486ba35015a09`, `e1c1ff8cdf4ee36f`, `f773853db5b3dc9c`. Corpus metadata lacks feature distributions, missingness, actual class balance and row-level dependence evidence for many entries; those statistics remain unknown here.

The three current model bundle manifests are `hyperion-v7`, `jope-prime-gen31-v2`, and `policy-step-60000`. Their existence and object sizes do not establish new training, inference parity or serving promotion. The model/training chapter describes their roles and acceptance boundaries. Historical lineage-A material in learning storage cannot replace its missing canonical training-state chain.

Measured current-source operational proof

Quantity	Measured result
Source	`68f56404c3e199e06413bfd645d3fe0803d7b96f`
Dataset	`230_machine_cpu`, regression
Immutable build wall seconds (preparation; not a cold-install measurement)	170.4488760890672
Eight-case native context qualification wall seconds (separate preparation)	29.248627813067287
Run wall seconds	239.14312197000254
Commands started / finished	964 /964
Completed / failed command outcomes	834 /130
Train OOF / diagnostic / refit-screen commands	960 /2 /2
Unique complete fitted predictor artifacts	832
Unique reusable recipes	164
Successful final refits	2, each 34 predictions
Neural development RMSE	166.3181717454848
Generic development RMSE	155.25909998168862
Generic minus neural loss	−11.059071763796169; neural did not win
Known worker CPU seconds	181.220773
Sum of worker wall seconds	202.266164543

Quantity	Measured result
Exact supervisor aggregate CPU seconds	270.889403
Unknown worker CPU / wall counters	0 /0
Teachers admitted from this proof	0

The supervisor figure is the exact `CPU_USAGE_NSEC=270889403000` event from local PID 1 systemd, linked to the precise supervisor unit, source, launch receipt and original experiment window. It includes controller/setup threads and child workers and excludes the outer launcher, immutable build and external preparation. It overlaps the worker CPU total; **never add them**. The rounded systemd stderr message is only diagnostic. No exact controller/setup overhead is derived by subtracting a rounded total. Isolated controller CPU, GPU seconds, token usage and currency cost remain unknown in the report.

Denominator	Worker CPU seconds	Supervisor aggregate CPU seconds	Amortized run wall seconds
One unique fitted predictor	0.2178134291	0.3255882248	0.2874316370
One unique recipe	1.1050047134	1.6517646524	1.4581897681

These are accounting ratios for one retained run, not extrapolated throughput or equal-quality claims. A recipe and a fitted predictor differ: the same recipe can create distinct fitted artifacts on separate folds. Unique artifact bytes do not prove behavioral uniqueness. Every failed fit remains in the inventory; failed or unavailable commands are not silently counted as zero cost.

The supplied one-hour/ten-million-token scenario has 3,600 seconds and 10,000,000 tokens by stipulation. This run used 239.143 seconds of wall time but has no measured token or monetary receipt. A dollar comparison, token savings claim, or one-hour quality extrapolation would require actual token usage, contemporaneous input/output/cache rates and an equivalent task/quality protocol. None is invented here.

The preceding d0 source run failed after 194.789382994 seconds at scorer replay. All 964 commands had terminal receipts, with 834 completed and 130 failed. Its exact supervisorCPU was 252.614611 seconds and workerCPU 181.863951 seconds. It remains a failed reproducer; the old journal and deadline were not repaired or reopened.

Existing historical rankings and Elo

Six committed snapshots are available and their ladder and score SHA-256 values were verified against retained bytes. No new Elo computation occurred in this audit. The snapshots use their own method pools and dataset protocols, so their Elo numbers cannot be compared across snapshots as a universal score.

Snapshot	Rankable rows	Methods	Datasets	Interpretation
same-cut-v1-screen	110	11	10	historical_same_cut_reproducible_snapshot_not_current_ASM 30
same-cut-v1-deep	50	11	10	historical_same_cut_reproducible_snapshot_not_current_ASM 30

Snapshot	Rankable rows	Methods	Datasets	Interpretation
same-cut-v2-screen	150	15	10	historical_same_cut_reproducible_snapshot_not_current_ASM 30
same-cut-v2-deep	50	15	10	historical_same_cut_reproducible_snapshot_not_current_ASM 30
historical-sotarena-published	8763	305	45	historical_published_best_known_test_selected_not_benchmark_valid
historical-sotarena-benchmark-valid	8718	304	45	historical_corrected_public_development

Same-cut v1 covers native heads plus resampled search; v2 adds XGBoost 2.1.4, LightGBM 4.6.0, CatBoost 1.2.10 and scikit-learn 1.7.2 random forest under Python 3.12.3. Both use the favorable historical test-10 regression selection, seed 20260907, five folds, top-five deepening and 32×8 search. Deep rows include only screened finalists; coverage must be printed with rankings. Their timing fields are then-recorded wall times under uncontrolled host load, not the present isolated-worker timing protocol.

The published SOTArena snapshot puts historical `jope-dime` at rank 1/Elo 2259.8 with 11 wins and 45/45 coverage. All 45 rows are test-selected: 42 best-test evolution-envelope rows plus 3 test-adaptive accepted replays. The corrected benchmark-valid snapshot excludes those 45; **that rank 1 is not a held-out ASM claim**. The corrected snapshot has 304 methods/8,718 rankable rows. Its fixed-seed historical `jope-dime-regressor` entry is rank 31/Elo 1687.0, 1 win, 42/45 coverage; TabFM Ensemble is rank 1/Elo 1934.9. It remains historical public-development evidence.

For plots, use `performance-evidence.json`: 6 complete ladder row sets; 520 same-cut score cells, including missing/rejected statuses; 20 per-dataset reports with exact cut identities, source heads and screen/deep timings; 8,763 historical SOTArena score rows; and the 2 current-origin refit loss rows. Do not blend these into one chart without explicit separate protocol panels. The current frozen 30 matrix has no completed cells yet; represent it as pending, not a fabricated ranking.

How the retained Elo scale is constructed

`crates/jope-arena/src/elo.rs` fits regularized Bradley–Terry ratings. Each dataset contributes one equal-weight comparison for each pair of methods with available scores. The preferred direction is taken from the metric; RMSE comparisons minimize. A tie has outcome 0.5 when `|a−b| ≤ 1e−12 × max(|a|,|b|,1)`; otherwise the better score receives 1 and the other 0. Missing cells create no comparison and must not become invented defeats.

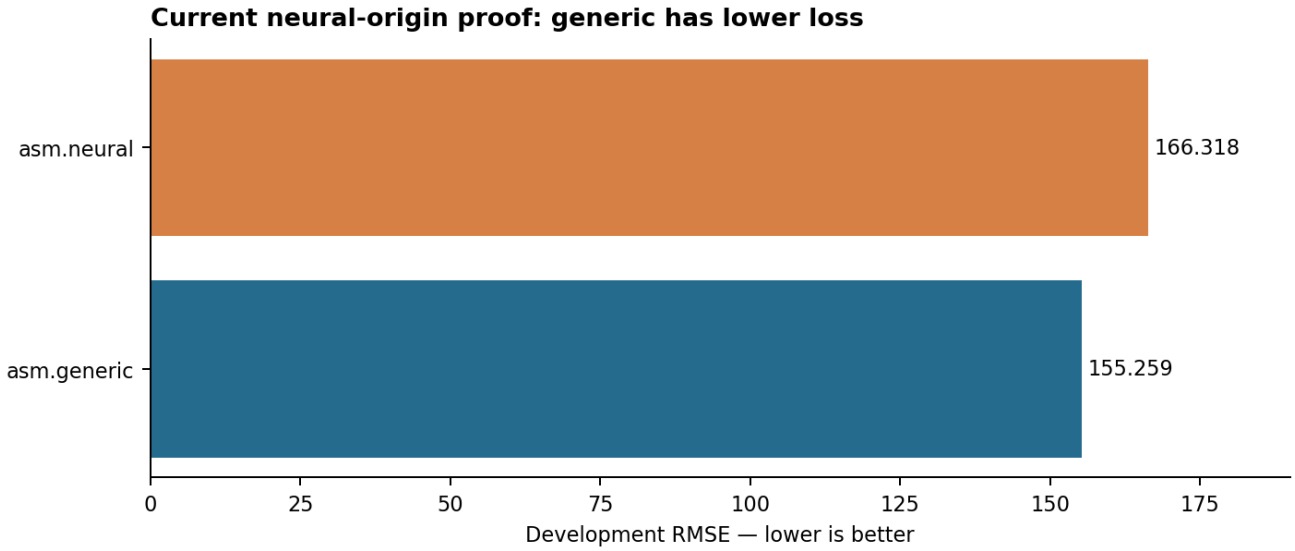
The expected pairwise outcome is `P(A beats B)=1/(1+10^((rating_B−rating_A)/400))`. The committed implementation centers at 1500, uses an L2 regularization coefficient of 1e−4, caps diagonal-Newton iterations at 250, clamps each iteration's rating update to ±40, and stops when the largest update falls below 1e−8. It preserves sorted dataset/method-pair accumulation and the historical numerical summation policy for reproduction. This describes the source algorithm; the audit did not rerun it or infer convergence where the snapshot does not report it.

A rating difference summarizes outcomes within one comparison pool. It does not encode an absolute RMSE difference, probability of unseen-dataset superiority, confidence interval, or universal ranking across tasks. A method observed only on its screened favorable subset needs its coverage displayed beside its rating. The historical `afm.lite-race.v1.resampled` row describes that search procedure; it does not prove the current neural ASM model.

8 / Visual evidence: data, measured performance and historical rankings

Current measured ASM outcome

This is the current bounded operational proof, with one preselected regression dataset and matched search-arm allowances. It demonstrates authenticated neural-origin fits and fitted descendants. The generic arm has lower development RMSE. The complete thirty-dataset ASM panel has not run, so this figure cannot establish its wins, ranking or Elo.

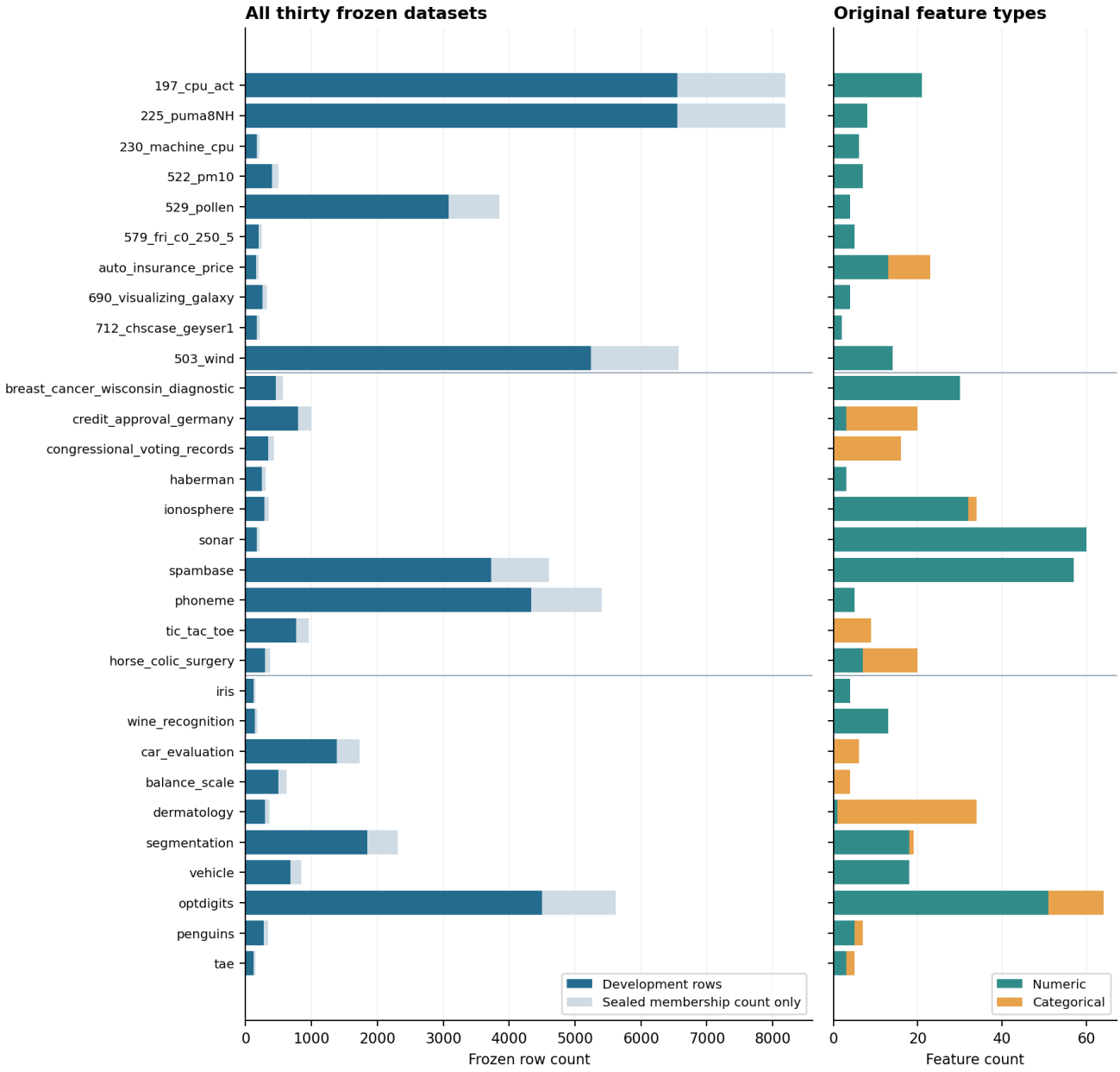


230_machine_cpu · 34 predictions per final refit · reviewed source 68f56404
One operational proof; no full-panel Elo, sealed-test result or teaching admission.

The two underlying records are in `data/plot-current-origin.csv`; exact evidence and denominators are in `data/performance-evidence.json`. There are 34 predictions per final refit. The proof's run wall time is 239.143 seconds; its independent recovered report takes another 18.719 seconds. Numerical measurements, worker counts and costs are detailed in the status and benchmark chapters.

Frozen dataset coverage

The following chart covers all thirty immutable datasets in task order: regression, binary, multiclass. It preserves the original feature-type metadata and row assignments. Sealed counts come from membership metadata; sealed target bodies were not opened. The complete statistics appendix gives readable numeric tables, class/category frequencies and every feature record.

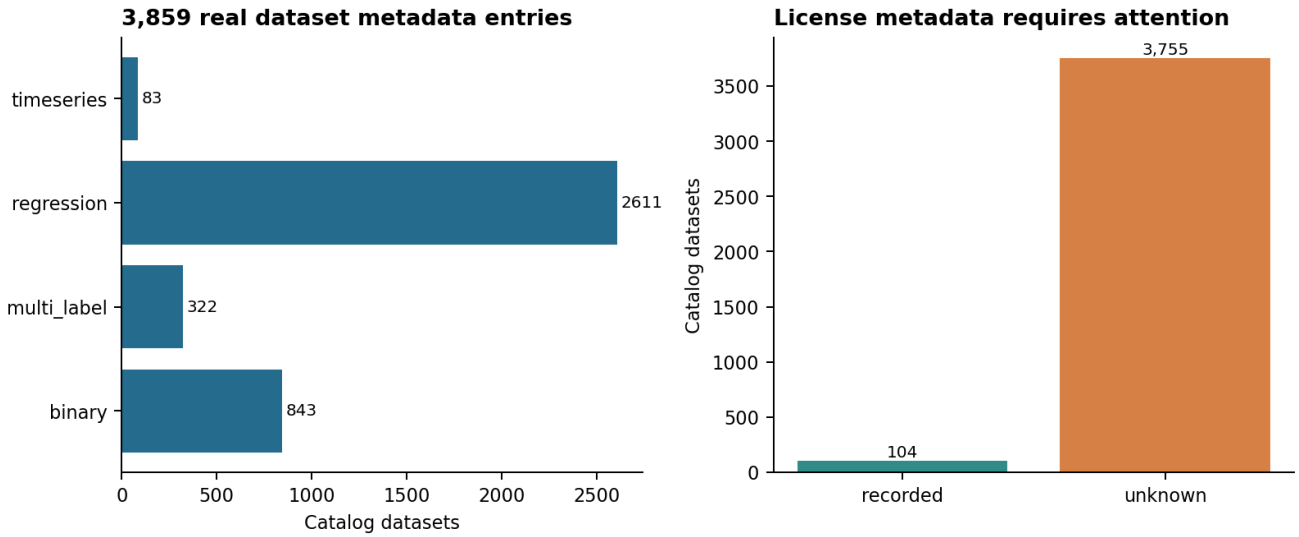


Regression / binary / multiclass: ten each. Sealed target bodies were not opened.
Source: pinned plot-dataset-sizes.csv and dataset-inventory.json; 23 September 2026.

Totals are 55,029 rows, 44,048 development rows, 10,981 sealed membership records and 522 summed feature records. Feature counts are not a union of semantic variables across datasets. The chart and the detailed tables share the same pinned inventory.

Broader corpus and metadata concerns

The broader catalog has 3,859 real datasets and 6,208,051 metadata-reported rows. It is a different collection from the frozen panel. Its `multi_label` task is not interchangeable with admitted multiclass classification. Per-payload statistics for the entire corpus were not recomputed.



Metadata statistics only. multi_label is a catalog task, not multiclass admission.
3,755 licenses unknown; the collection's older prose count is stale. Source: catalog-summary.json.

The catalog has 3,755 entries with unknown license metadata and 104 with a recorded license. The collection's older prose count of 1,416 unknown licenses is inconsistent with this current pinned catalog. This is a metadata quality gap; neither the chart nor a repository-level license resolves each dataset's provenance. See the estate chapter for exact storage bytes, prefixes and observation scope.

Ranking figures: interpretation contract

The historical ranking plots that follow use their separately pinned snapshots. They are not new model runs and are not the current ASM30 campaign. Benchmark-valid and public-development views retain their original eligibility distinctions. Same-cut deep-stage coverage varies because only finalists were deepened; absent cells are not zero losses. Refer to the benchmark-methodology chapter for the exact Elo model, tie rule, coverage limits and adaptive-development interpretation.

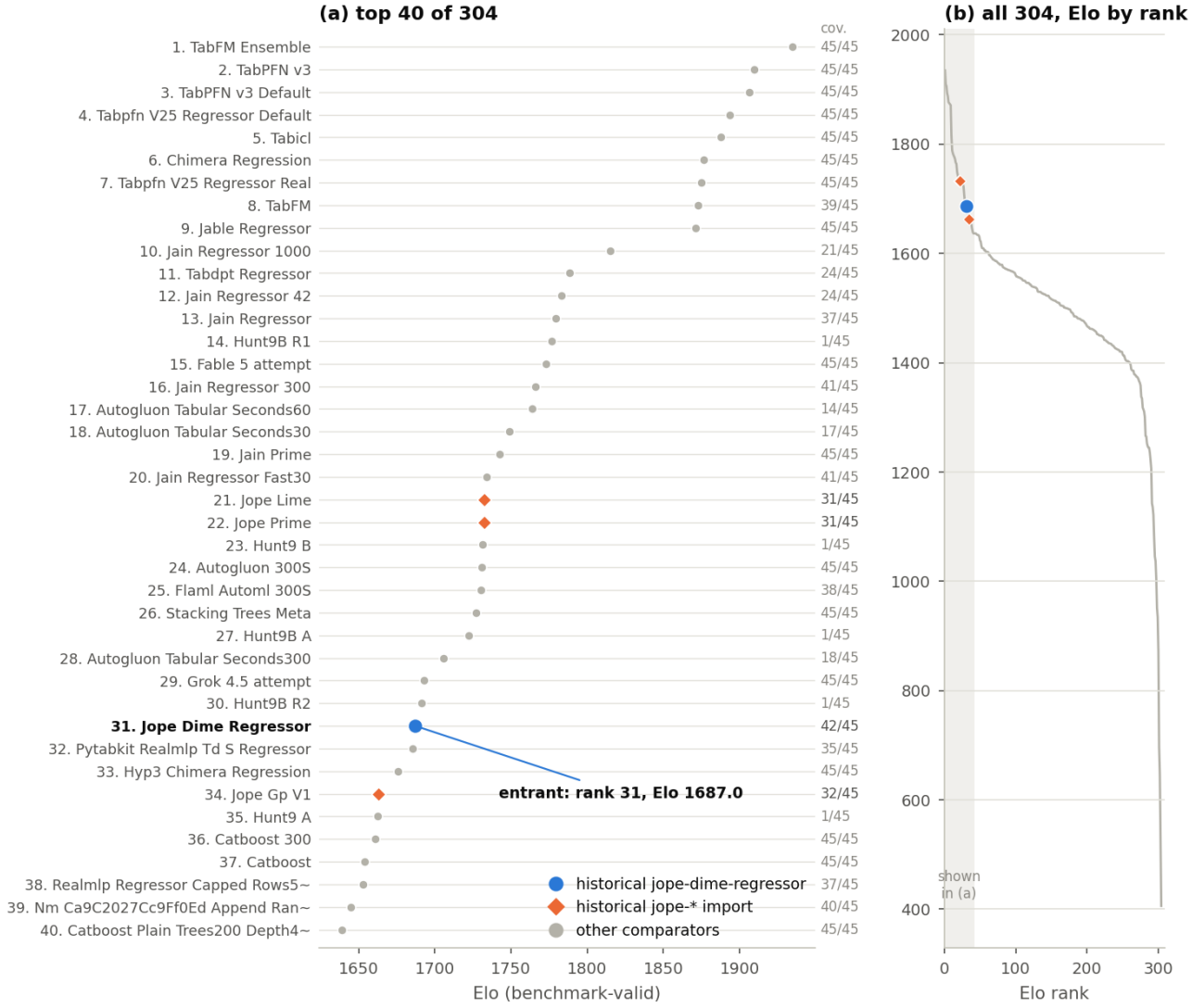
All machine-readable rows are retained in `data/performance-evidence.json` and `data/plot-historical-elo.csv`. A displayed top-method subset does not remove the other methods from those inventories. Dataset selection and historical protocol caveats travel with each figure.

Historical benchmark-valid Elo

This view retains 304 methods and the historical corrected eligibility filter. The historical `jope-dime-regressor` entrant is rank 31 with Elo 1687.0 in this snapshot. “Benchmark-valid” means no retained artifact contradicts a row's flag; comparator protocols were not independently verified. It does not certify that every comparator obeyed today's frozen ASM protocol.

Historical benchmark-valid Elo: Golden45 tabular regression

304 methods, 8,718 rankable scores, 45 published rows excluded as test-selected. Historical entrant: jope-dime-regressor at rank 31, Elo 1687.0, 1 dataset win, 42/45 coverage.



Source: assets/regression/leaderboard-tabular-regression-v2.benchmark-valid.csv (sha256 4c39f51c2e269114), cross-checked against assets/regression/elo-tabular-regression-v2.benchmark-valid.json; the published scores.csv with assets/regression/protocol-corrections-tabular-regression-v2.json applied. Committed arena snapshot; nothing re-run. Elo: regularized Bradley-Terry on the Elo scale, protocols ranked together, partial coverage allowed.
Caveat: 'benchmark-valid' means no artefact in this repository contradicts the row's published flag; comparator protocols were not verified. Entrant = the fixed-seed reproducible run (public-development). Orange = other historical jope-* imported run reports, not claimed by this repository. Coverage n/45 at right. As of 2026-09-23.

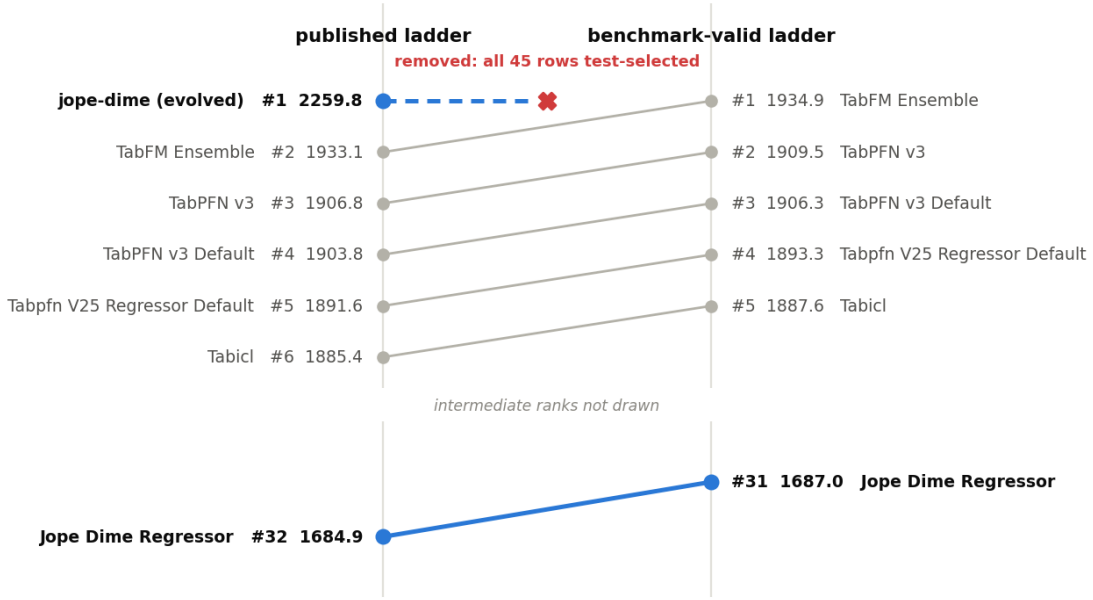
The top-method view is paired with the distribution of the full method universe. Dataset coverage and the identity of the historical entrant remain visible. These numbers describe the retained historical ladder, not the current accepted step-60000 model's thirty-dataset performance.

Why the two historical ladders differ

The published view includes development-selected rows; the corrected benchmark-valid view removes 45 test-selected rows. The historical `jope-dime` rank-one Elo of 2259.8 belongs to the former view. It is a best-known development result, not a held-out claim. An eligibility change alters the overlap graph and the fitted ladder, so these ratings must not be combined into a single current score.

Historical ladders: published vs benchmark-valid

Same scores.csv, two Elo fits. Published keeps every row; benchmark-valid drops the 45 jope-dime rows this repository's own code labels test-selected, then refits.



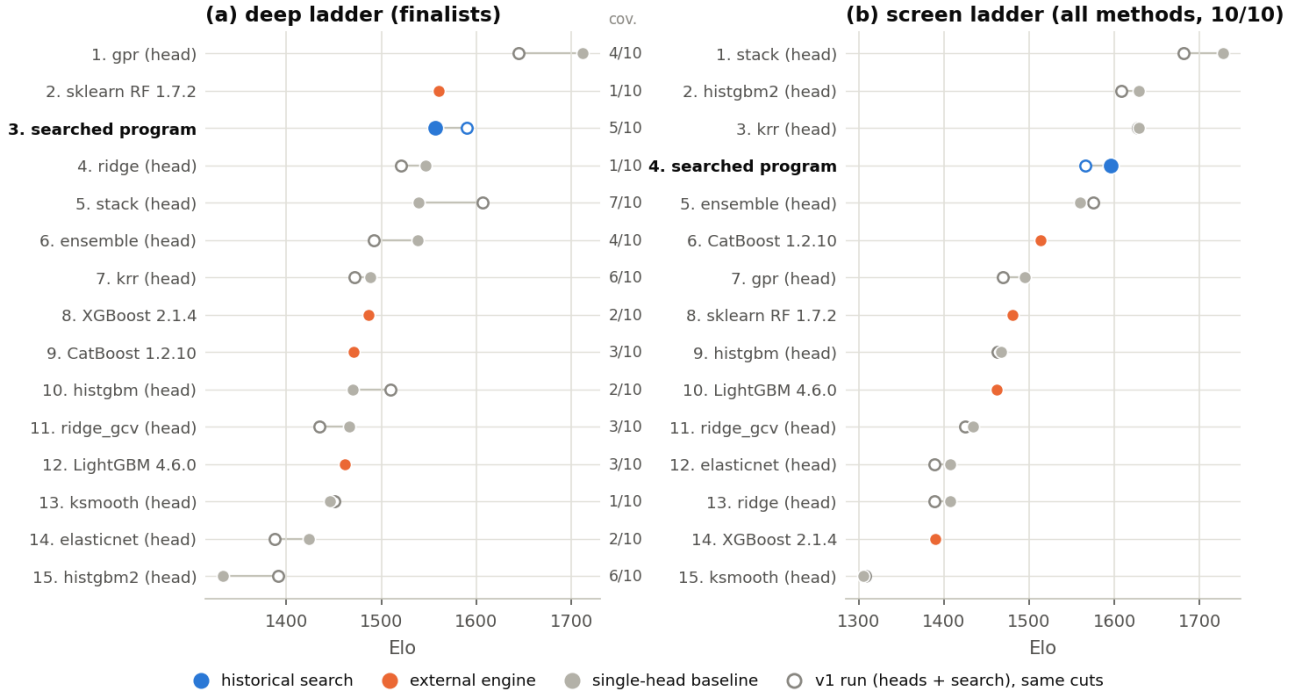
Sources: published assets/regression/sotarena-tabular-regression-v2/results/tabular-regression-v2/leaderboard.csv (byte-exact copy of the benchmark's ladder); benchmark-valid assets/regression/leaderboard-tabular-regression-v2.benchmark-valid.csv; corrections assets/regression/protocol-corrections-tabular-regression-v2.json: 42 Golden42 rows are each the per-dataset minimum RMSE over 63,108 test-scored evaluations (assets/regression/SOURCES.md:62); 3 Golden46 rows come from run golden46-new3-20260723-v4, which the campaign code writes as test_adaptive: true. Caveat: the published jope-dime position is a best-known result per dataset, not a held-out result. Only the top 5 of each ladder and this repository's two entries are drawn; the ranks in between are elided. As of 2026-09-23.

Historical same-cut screening and deepening

Same-cut v2 compares all fifteen methods during screening on ten regression datasets. Only the five finalists per dataset are deepened across all five cuts, so the deep-stage denominator differs by method. Hollow markers preserve the earlier v1 snapshot; they are a separate run, not confidence intervals. The historical searched program is distinct from the current qualified neural proposal path.

Historical same-cut v2: Elo by method on test-10

Every method fitted on identical training rows and scored on identical held-out rows by one evaluator.
Deep: finalists re-scored on all 5 cuts (coverage n/10 = datasets where the method reached the deep stage). Screen: every method, one cut.



Source: assets/same-cut/v2/leaderboard-jope-same-cut-v1-{deep,screen}.csv with their elo-*.json snapshots (hash-checked); hollow markers: assets/same-cut/v1 (heads + search only, same cuts). Recorded run, nothing re-run; search budget 32 programs x 8 rounds, re-run on every cut; engines run with library defaults, seed 0, one thread.
Caveat: FAVOURABLE SELECTION. test-10 was chosen where the historical engine ranked first on the published ladder (docs/datasets.md); nothing here generalises to a broader panel. The deep ladder ranks finalists only, so coverage differs by method. As of 2026-09-23.

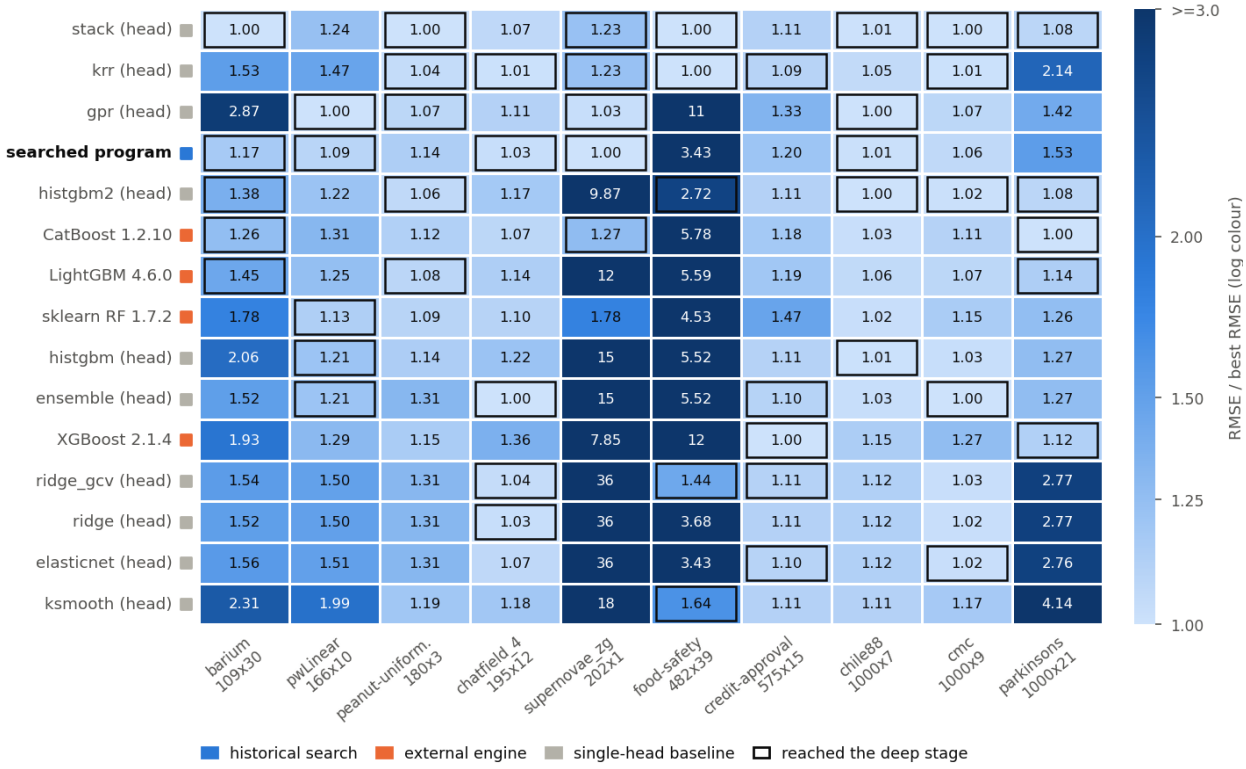
The ten historical datasets were a favorable selection for the earlier engine. Their ranking cannot establish performance on a broader corpus or substitute for the unchanged frozen thirty. External engine versions printed in these plots are the historical run versions, not a claim about currently installed adapter versions.

Historical relative loss and per-dataset rank

Each heatmap cell is a method's screen RMSE divided by the best screen RMSE on the same dataset. A ratio of one is best; the color saturates at three while displayed values are rounded. This normalization supports within-dataset visual comparison without averaging incompatible raw target units. Outlines show which cells reached the deep stage.

Historical same-cut v2: relative error per dataset (screen cut)

Cell = method RMSE / best RMSE on that dataset (1.00 = best). Rows sorted by median ratio; columns are datasets (training rows x features), sorted by rows.
Outlined cells reached the deep stage (top 5 on the screen cut). Colour saturates at 3.0; labels are rounded ratios.

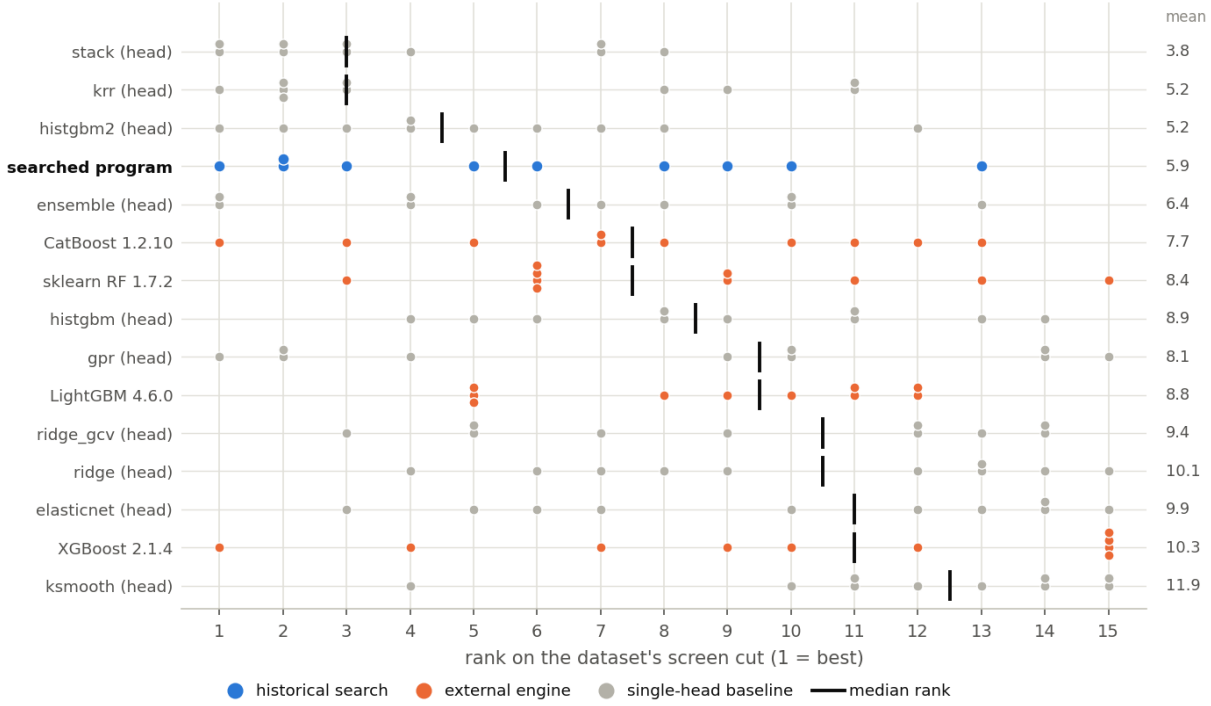


Source: assets/same-cut/v2/reports/*.bench-report.json (screen metrics per method; the screen cut is fold 0 of fold_of('<dataset_hash>:20260907', row, 5)), extracted to figures/data/fig04_samecut_v2_per_dataset.csv. Recorded run, nothing re-run.
Caveat: FAVOURABLE SELECTION (test-10, docs/datasets.md); one cut per cell, so a single split's noise is included; the deep ladder (fig03) is the pooled 5-cut view. As of 2026-09-23.

The rank-distribution plot shows all ten per-dataset screening ranks for each method, with mean and median. It reveals variability hidden by a single aggregate ladder. A rank difference conveys ordering, not the magnitude of a loss improvement or a confidence interval.

Historical same-cut v2: per-dataset rank by method (screen cut)

Each dot is one dataset (n = 10); rank 1 = lowest RMSE among the 15 methods on that dataset's screen cut. Vertical bar = median; number at right = mean rank.



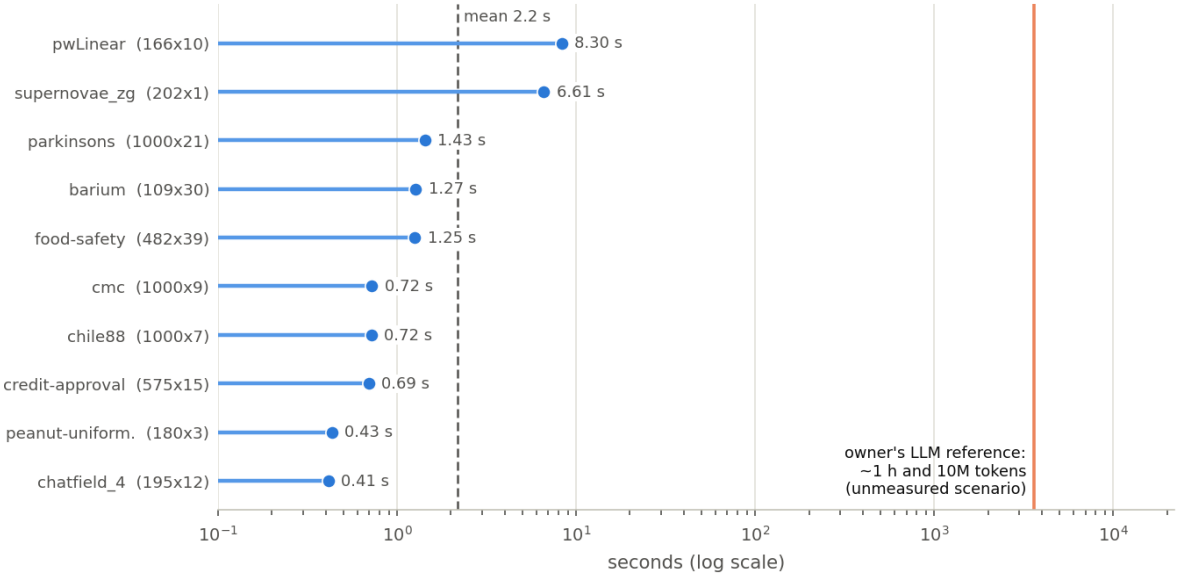
Source: assets/same-cut/v2/reports/*.bench-report.json field screen_rank (ranked by the screen score on one shared cut; on a shared cut R2 and RMSE order identically), extracted to figures/data/fig04_samecut_v2_per_dataset.csv. Recorded run, nothing re-run.
Caveat: FAVOURABLE SELECTION (test-10, docs/datasets.md); one cut per dataset; ranks ignore effect sizes (see fig04 for the ratios). As of 2026-09-23.

Historical production time and the supplied reference scenario

The retained same-cut reports record mean program-production wall time 2.1837062093 seconds, median 0.9870389355 seconds and maximum 8.299870126 seconds across ten datasets. These are recorded timings under the host's contemporary load, not isolated controlled latency measurements and not the current paired ASM proof's elapsed time.

Historical seconds to write each program (same-cut v2, test-10)

Wall seconds the search took to produce the program on the screen cut's training rows: mean 2.2 s, median 0.99 s, longest 8.3 s over 10 datasets.



Source: assets/same-cut/v2/reports/*.bench-report.json, method afm.lite-race.v1.resampled field produce_seconds (budget 32 programs x 8 rounds, 16 workers on a 128-core host), extracted to figures/data/fig06_samecut_v2_search_seconds.csv. Timings are recorded wall time under whatever else the host ran, not controlled.
Reference line: the owner's stated scenario for a large language model writing a program for the same task, about 1 hour and 10,000,000 tokens (deploy/ci/readme-benchmarks.py:73-77). This is a contextual duration reference, not a measured comparison. Workload, hardware, tokens and currency were not matched; no speedup is established. FAVOURABLE SELECTION (test-10). As of 2026-09-23.

The one-hour/ten-million-token marker is the user's contextual scenario. Workload, hardware, token trace, currency and result quality were not measured under matching conditions. The chart therefore establishes no matched speedup, cost saving, token-efficiency ratio or quality equivalence. The measured current proof and its per-predictor/per-recipe accounting remain separately reported.

9 / Maintaining, reviewing and recovering the product

Repository authority

The canonical product checkout is `/home/ubuntu/JopeDime`, upstream `veox-ai/JopeDime`. `/home/ubuntu/jain-split/jope-dime` is a different research repository; `/home/ubuntu/jope-dime` may be its symlink. Similar names are not interchangeable source, checkpoint or promotion authorities.

Before source changes, inspect the open PR queue, worktrees, branch/full HEAD, dirty state and latest `AGENT_CHAT.md`. Under `flock` on `AGENT_CHAT.md.lock`, reread overlapping claims and append checkout, branch, full HEAD, owner, paths and verification. The append lock does not permit two writers in one checkout. An idle session or old claim is not a stopped-head handoff.

Keep one writer per checkout and one implementation PR queue. Preserve pending work and authorship. A resumed campaign retains its resource limits and deadline.

Worktrees and builds

Extra copies belong under `<repo>/.agent/worktrees/`, at most four lanes. `.claude/worktrees` stays a symlink into that shared root. Verify it before worktree operations and keep `.agent/` locally excluded from Git status.

Each lane reuses its own `target/`. Before Cargo:

```
export CARGO_INCREMENTAL=0
export CARGO_PROFILE_DEV_DEBUG=line-tables-only
export CARGO_BUILD_JOBS=4
```

Do not set `CARGO_TARGET_DIR` globally or create a new target per task. CI selects persistent PR and trusted-main storage separately. Resolve binaries from Cargo artifact messages: guessing a hashed test executable can select an old binary built with different features.

Never put source, worktrees or builds in `/tmp`. A target directory can contain evidence. Cargo markers do not prove that every descendant is disposable.

Evidence checklist

Retain source revision and input manifest; exact command, directory, exit and timestamps; executable/runtime/features/environment; data/split/evaluator/vocabulary/protocol identities; original limits/start/deadline/cutoff; collected/executed/failed/skipped/excluded/unavailable inventories; artifact hashes and worker receipts; report replay; limitations and next owner.

Verify a source manifest against the committed blobs it claims. The current bytes at `target/release/jope` need not be the bytes used by an earlier result. Preserve immutable executable copies with their build receipts before another build replaces the usual path.

Current ASM custody is outside Git at `/home/ubuntu/jope-rescue-20260921/asm-integration-20260923/`. This host location is not a portable runtime default.

Coverage and reviewed lifecycle

Read `deploy/ci/coverage.md`, mapping and generated inventory together. Discovery counts packages, features and families; execution receipts count what one invocation actually collected and ran. Invocations overlap. Path exclusions, ignored tests and unavailable dependencies do not become passing tests.

Fast CI has eight lanes plus aggregate. Full-main diagnostics cover additional families and have recorded failures/unavailable capabilities. A fast gate does not establish all-feature, GPU or live-S3 qualification. Regenerate inventories and scores through their owning proof scripts; never hand-edit generated evidence.

The established writer is `jepsontaylor`; the designated independent Claude reviewer uses `neverhuman`. Verify them for every PR. The reviewer must differ from the opener and every associated commit author and committer. Use the full paginated inventory and record unassociated commits.

```
/home/ubuntu/.local/bin/gh-role writer api user --jq .login
/home/ubuntu/.local/bin/gh-role eligible reviewer veox-ai/JopeDime PR_NUMBER
```

The helper prevents identity mistakes but cannot prove independence. Changing credentials does not turn a writer into a reviewer. Never print tokens or credential-bearing environments.

Freeze the full head; dispatch fresh manual CI; retain source SHA, run ID, attempt and dispatch time. Verify current-attempt jobs/aggregate and nonzero meaningful coverage. Supply the independent reviewer with diff, acceptance commands and evidence. A native approval must explicitly bind the final head and remain undismissed. Changed source repeats checks and review.

```
/home/ubuntu/.local/bin/gh-role writer pr merge PR_NUMBER \
  --rebase --match-head-commit FULL_REVIEWED_SHA
```

Immediately recheck head, eligibility, CI and approval before merging. Afterwards verify `merged_at`, fetch remote main, compare the reviewed tree and verify main-guard. Preserve any ledger tail or other dirty work before restoring clean verified main. Protection enforcement has been procedural; a protection API 403 does not prove a preventive branch lock. Main-guard detects unreviewed pushes after they occur.

Shared compute

Fast CI runs on xbabe1. The canonical Ubuntu machine is local xbabe2 for full CPU work. The user's SSH alias `xbabe2` points elsewhere; use the runbook's local form for this machine.

The exclusive full runner and numerical campaigns share a host lock. Drain owned listeners and verify no active full job before numerical allocation. Do not seize the host by stopping another job. Record allocation ownership and restore the runner after numerical work releases it. Keep full CPU CI separate from the one-GPU-hour training campaign and the frozen panel.

Missing cgroups or dependencies cause explicit failures/unavailable outcomes; they do not authorize an unbounded fallback. Worker control readbacks, not prose, establish the limits actually enforced.

Recovery phases

Identify whether failure occurred before admission, in setup, after the gate, during cleanup, in controller scoring or during report generation.

- A pre-admission dependency refusal may be retried only inside the original window.
- Setup can consume CPU without starting a fit. Unknown setup cost stays unknown.
- A started identity is durable. Controller death does not authorize silently repeating it.
- Completed folds inside a failed candidate still require artifact authentication.
- Original-cutoff inventory settles without extending the clock.
- A report can fail after successful fits. Retain both; do not call the whole operation successful.
- Saving a terminal outcome does not release cleanup authority before descendants drain.

Resume with the same plan and journal. A changed source, dataset, budget or protocol requires a separately identified experiment with the old evidence retained, not a relabelled continuation.

Preservation and lane closure

Before deletion inspect `git status`, `git stash list`, `git log --all --not --remotes`, and local versus remote tags. A branch-only log misses commits elsewhere.

Preserve unique source/evidence for the tree being removed, restore independently into the reusable bare verification directory without alternates, and verify the exact commit, tree and connectivity. A valid bundle checksum does not prove the desired detached HEAD was included.

After preservation and a fresh consumer check, remove a closed lane with `git worktree remove`. Indefinite redundant copies are not an alternative to the required reclamation step. Unknown contents remain protected. Follow `deploy/maintenance/README.md`; its reporting timer proposes candidates and never authorizes deletion.

Updating the book

Refresh inventories and receipts, then section Markdown, figures, checklist and PDF. Keep a rendering input manifest. A checkpoint does not update performance figures by implication; merged training source does not prove actual learning. Historical Elo remains historical after regeneration.

Before delivery check local links, source identities, totals, metric directions, captions, capability status and rendered pages. Extract PDF text and inspect dense tables, examples and figures. Preserve unavailable statistics and measurements explicitly rather than filling them with zero.

10 / Prioritized acceptance checklist and roadmap

Purpose and scope

This checklist is the requested property-list of remaining work. `roadmap.json` is its machine-readable source, with stable IDs, priorities, evidence levels, dependencies, commands, required receipts, budgets and source references. It describes source68f56404 and the stopped training64/product28/warm/winner follow-ons. PR83 is merged: remote main `a75e543d2166017cb0139e0a251d51d9a0c98e08` has the reviewed source tree, and main-guard35889731910 attempt1 succeeded. Clean canonical main was verified before the documentation branch. LIFE-01 records the completed readback; later source work still needs its own lifecycle.

P0 blocks the specified downstream proof or integration; it does not mean every item is a newly discovered defect. P1 is required for the full requested demonstration or manual. P2 is explicit subsequent work. Commands are unexecuted acceptance templates unless their named evidence says otherwise. Variables must resolve to actual final-source artifacts and emitted identities. An empty command list records a missing truthful interface rather than inventing one.

Critical order

Verified reviewed main → guarded source integration → actual tests, host qualification and final packaging → fresh qualified contexts and preregistered real teachers → all-eligible family-disjoint corpus → one GPU-hour training with continuation → explicit new-checkpoint consumption → real fitted warm demonstration → unchanged thirty-dataset campaign, selected-state backend qualification inside its remaining window, and report → eligible motif reuse.

Library parity fixtures precede the full panel. Actual selected-winner exports require complete confirmations and must still fit the original execution window. They cannot consume its reporting reserve or silently run after expiry. Documentation can proceed alongside source preparation; actual numerical examples need an allocated execution window.

Do not spend the GPU hour until the regression consumer and its actual fixture tests are ready. Do not freeze the final official panel configuration until its checkpoint is chosen and consumed. Separate proof allowances suggested in older plans are proposals, not new budget authorization.

Compact task index

ID	Priority	Stage	Workstream	Current state	Dependencies
LIFE-01	P0	0	coordination	complete with retained readback	None
DOC-01	P1	0	manual	canonical private manual assembled	LIFE-01
DOC-02	P1	0	manual examples	complete at named recipe/refit scope	DOC-01

ID	Priority	Stage	Workstream	Current state	Dependencies
DOC-03	P1	0	manual figures	figures rendered; final PDF inspection pending	DOC-01
DOC-04	P1	0	manual quality	rendered; final publication gates pending	DOC-01, DOC-02, DOC-03
INT-01	P0	0	serial integration	stopped unmerged packets	LIFE-01
INT-02	P0	0	integration qualification	pending	INT-01
BENCH-01	P1	1	frozen benchmark foundation	implemented; retain during integration	INT-02
BENCH-02	P1	1	external comparators	refresh at final environment	INT-02
EXEC-01	P1	2	bounded workers	implemented; CI coverage gap retained	INT-02
EXEC-02	P1	2	fault regression tests	test gap: drain-and-fixtures-F2	INT-01
EXEC-03	P1	2	receipt replay regression tests	test gap: resume-fix-F2	INT-01
EXEC-04	P1	2	cutoff semantics regression	test gap: regressions-F2	INT-01
EXEC-05	P1	2	accounting and unknown outcomes	pending execution	INT-02
NEURAL-01	P0	3	context qualification	requires final-source binding	INT-02
TEACH-01	P0	6	HWM comparison identity	not runtime qualified	INT-02
TEACH-02	P0	6	label exposure readers	residual eligibility-B6	TEACH-01
TEACH-03	P0	3	real teacher collection	blocked on fresh admitted evidence	BENCH-01, BENCH-02, NEURAL-01, TEACH-01, TEACH-02
TEACH-04	P0	3	real corpus materialization	blocked on two real families	TEACH-03
TRAIN-01	P0	3	actual trainer tests	residual training-F3	INT-02
TRAIN-02	P0	3	native supervised training	host proof pending	TRAIN-01

ID	Priority	Stage	Workstream	Current state	Dependencies
TRAIN-03	P0	3	bounded real GPU training	blocked on admitted data and consumption readiness	TEACH-04, TRAIN-02, NEURAL-02
TRAIN-04	P0	3	checkpoint continuation and export	requires actual campaign	TRAIN-03
NEURAL-02	P0	3	trained-checkpoint consumer readiness	compile/runtime fixture pending	INT-02, TRAIN-01
NEURAL-03	P0	3	live newly trained consumption	newly trained consumption pending	TRAIN-04, NEURAL-01
INFER-01	P0	4	fitted state and privacy	qualification pending	INT-02
INFER-02	P1	4	exact selected-winner portability	actual export/ parity unrun	INFER-01, PANEL-01
INSTALL-01	P1	5	private installation	cold/live path pending	INT-02
INSTALL-02	P1	5	final input/ runtime preparation	pending final source	INSTALL-01, BENCH-01, BENCH-02
DEMO-01	P1	5	measured warm fitted demonstration	realtrained warm proof pending	NEURAL-03, INSTALL-02
PANEL-01	P1	5	unchanged thirty-dataset campaign	pending finaltrained configuration	BENCH-01, BENCH-02, NEURAL-03, DEMO-01, INFER-01, INSTALL-02
REPORT-01	P1	5	complete measured report	fullcampaign report pending	PANEL-01, EXEC-05
MOTIF-01	P1	6	eligible reusable fragments	minefront door incomplete	TEACH-01, TEACH-02, TEACH-03
MOTIF-02	P1	6	reuse validation	live reuse pending	MOTIF-01
MOTIF-03	P2	6	new operators	future conditional work	MOTIF-02
IP-01	P1	0	public privacy boundaries	residual D-IP4 and nextPRtextwork	INT-01
MAINT-01	P2	0	broad diagnostic qualification	not closedbyfastCI	LIFE-01

ID	Priority	Stage	Workstream	Current state	Dependencies
FUTURE-01	P2	4	CUDA inference and serving promotion	deferred	INFER-02, REPORT-01

Detailed acceptance

LIFE-01 — coordination

Priority: P0. **Current evidence:** reviewed merge, exact-tree remote main and successful main-guard. **Status:** complete with retained readback.

Action: Retain the verified lifecycle receipt; future source changes repeat exact-head CI and eligible independent native review.

Dependencies: None.

Acceptance commands:

```
/home/ubuntu/.local/bin/gh-role writer pr view 83 --json mergedAt,mergeCommit,headRefOid,state
/home/ubuntu/.local/bin/gh-role writer run view 35889731910 --json
headSha,attempt,status,conclusion,jobs
```

Required evidence: PR83 merged2026-09-23T16:34:47Z; remote main a75e543d2166017cb0139e0a251d51d9a0c98e08 has tree176b75bc36d76c7ce4eaeb7234677ec013468889, equal to reviewed68. Main-guard35889731910 attempt1 succeeded; job107278860508 finished16:40:11Z. Clean canonical main verified before docs/asm-maintainer-manual branch claim.

Resources: Read-only/static custody work; no fitting, models, GPU or campaign admission.

Source/evidence references: `AGENTS.md:Pull requests`, `status.md:Scope`, `maintenance.md:Coverage and reviewed lifecycle`, `neural-integration/lifecycle-68f56404c3e199e06413bfd645d3fe0803d7b96f/remote-main-verification.json`.

Qualification limit: This task is complete; subsequent implementation remains pending. Keep the measured68 source identity distinct from the rebase-merge commit identity.

DOC-01 — manual

Priority: P1. **Current evidence:** assembled thirteen chapters and complete data companions. **Status:** canonical private manual assembled.

Action: Maintain the dated evidence inventory and current source/pending distinctions.

Dependencies: LIFE-01.

Acceptance commands:

```
python3 docs/internal/asm-manual/tools/build_manual.py
```

Required evidence: Input manifest pins every chapter, figure dataset, source receipt and render tool; duplicates resolved; current/pending/historical/unknown labels retained.

Resources: Read-only/static custody work; no fitting, models, GPU or campaign admission.

Source/evidence references: `architecture-training-evidence.json`, `product-language-source-pins.json`, `dataset-inventory.json`, `performance-evidence.json`.

Qualification limit: Root integrated the book and renderer. Claude historical plot inputs were copied under stable before/after hashes; no stopped external renderer handoff is required.

DOC-02 — manual examples

Priority: P1. **Current evidence:** fourteen actual commands passed; five backend predictions bit-identical. **Status:** complete at named recipe/refit scope.

Action: Retain the completed fourteen-command receipt and rerun after meaningful recipe/runtime changes; qualify saved fitted winners separately.

Dependencies: DOC-01.

Acceptance commands:

```
python3 "$EXAMPLES/validate-outputs.py" --recipe-output "$EXAMPLE_RUN/recipe"
"$J" pipeline verify-receipt --receipt "$EXAMPLE_RUN/pipeline/receipt.json" --json
```

Required evidence: examples/commands.json instantiated with actual paths and pinned source/tool versions; five outputs with8finite values and complete-bit comparison; pipeline receipt verification. Stopped audit SHA256 363fbe62a706b44030205da78411243ba206ad2366d546a277babe966b802c91.

Resources: Completed separate 300-second allocation: one CPU, 4 GiB, zero swap/GPU, network denial declared; 30-second numerical and 60-second compiler caps. Actual receipt retained; independent live property readback limitation is disclosed. No Cargo or frozen-panel campaign; no deadline reset.

Source/evidence references: `examples/commands.json`, `examples/validate-outputs.py`, `languages-examples.md:Example A`, `languages-examples.md:Example B`.

Qualification limit: Actual acceptance: data/example-acceptance/audit.json. Five recipe refits each predict eight synthetic rows; separate legacy pmlb229 pipeline has four successful measurements. No fitted-export, ASM demo, training or full-panel qualification. Resource-readback race retained.

DOC-03 — manual figures

Priority: P1. **Current evidence:** ten local figure artifacts generated and source-checked. **Status:** figures rendered; final PDF inspection pending.

Action: Retain historical/current labels, complete source CSVs and final PDF visual checks.

Dependencies: DOC-01.

Acceptance commands:

```
OPENBLAS_NUM_THREADS=1 OMP_NUM_THREADS=1 python3 docs/internal/asm-manual/tools/
make_inventory_figures.py
OPENBLAS_NUM_THREADS=1 OMP_NUM_THREADS=1 python3 docs/internal/asm-manual/tools/make_figures.py
```

Required evidence: Plot source CSV/JSON, tool source, plot hash, precise captions, missing-stat flags and historical-vs-current labels; no invented sealed statistics or causal speedup.

Resources: Read-only/static custody work; no fitting, models, GPU or campaign admission.

Source/evidence references: `data-evidence.md`, `dataset-inventory.json`, `performance-evidence.json`.

Qualification limit: Historical Elo belongs to its original ladder/pool/protocol; the current one-dataset proof is not a new Elo ladder. Source and figure checks are retained; authored architecture SVG is a repository source artifact.

DOC-04 — manual quality

Priority: P1. **Current evidence:** thirteen chapters rendered; full-text and sampled-page checks available. **Status:** rendered; final publication gates pending.

Action: Render docs/*.pdf, extract PDF text, inspect all dense tables/code/figures, verify local links and filenames, and produce a readback checklist.

Dependencies: DOC-01, DOC-02, DOC-03.

Acceptance commands:

```
pdftotext -layout "$MANUAL_PDF" "$MANUAL_TEXT"
```

Required evidence: PDF SHA/page count; extracted-text receipt; visual inspections of title/contents, widest tables, commands, plots and final checklist; no clipping/omitted rows; Markdown counterparts present.

Resources: Read-only/static custody work; no fitting, models, GPU or campaign admission.

Source/evidence references: `maintenance.md:Updating the book`.

Qualification limit: A successful renderer exit is insufficient: the PDF must remain readable and complete.

INT-01 — serial integration

Priority: P0. **Current evidence:** two exact custody replays + guarded hashes. **Status:** stopped unmerged packets.

Action: Refresh guards only after verified main; integrate training64 then product28 then warm/winner add-ons without replacing newer source corrections.

Dependencies: LIFE-01.

Acceptance commands:

```
python3 "$TRAINING_PACKET/replay.py" --expect-head "$SOURCE_SHA" --manifest-sha256
"$TRAINING_MANIFEST_SHA" --output "$TRAINING_PACKET/replay-NEW"
python3 "$PRODUCT_PACKET/replay.py" --expect-head "$SOURCE_SHA" --manifest-sha256
"$PRODUCT_MANIFEST_SHA" --output "$PRODUCT_PACKET/replay-NEW"
```

Required evidence: Exact-source guard receipts; unchanged original packet hashes/skips/resolutions; new applied diff; root scorer digest/cutoff/terminal persistence and current ledger hooks preserved.

Resources: Read-only/static custody work; no fitting, models, GPU or campaign admission.

Source/evidence references: `agent-training-integration/refresh-68f56404/HANDOFF.json`, `agent-product-final-68f56404/HANDOFF.json`, `agent-delivery-refresh-68f56404/HANDOFF.json`.

Qualification limit: NEW is an unused custody directory name. Add-on replay invocation must follow its own pinned helper. Never force stale guards or apply constituent accounting/coverage twice.

INT-02 — integration qualification

Priority: P0. **Current evidence:** static syntax/feature checks; no integrated compilation yet. **Status:** pending.

Action: Compile/type-check the actual combined source, format, run meaningful suites and strict Clippy, regenerate coverage/inventories through owners, then obtain exact-head CI/native approval.

Dependencies: INT-01.

Acceptance commands:

```
CARGO_INCREMENTAL=0 CARGO_PROFILE_DEV_DEBUG=line-tables-only cargo test --locked -p jope-bench --
features asm
CARGO_INCREMENTAL=0 CARGO_PROFILE_DEV_DEBUG=line-tables-only cargo test --locked -p jope-cli --
features asm
CARGO_INCREMENTAL=0 CARGO_PROFILE_DEV_DEBUG=line-tables-only cargo test --locked -p jope-pipeline
--features qualified-context
cargo fmt --all --check
python3 deploy/ci/coverage.py discover
python3 deploy/ci/coverage.py validate
python3 deploy/ci/test-training-proof.py
TMPDIR="$ACCEPTANCE/tmp" JOPE_BUILD_JOBS=4 bash deploy/ci/test-inventory.sh --update
```

Required evidence: Collection/exit logs, actual tests passed/failed/ignored/deselected, refreshed inventory diff, feature/dependency graph, build executable hashes, current-attemptCI and independent exact-head approval.

Resources: Existing exclusive CPU CI allocation; one reused lane target, jobs4 unless lane policy sets otherwise; no concurrent numerical campaign.

Source/evidence references: `deploy/ci/coverage.json`, `deploy/ci/training-proof.py (pending)`, `deploy/ci/fast-lane.sh`, `deploy/ci/full-lane.sh`.

Qualification limit: Custody syntax tests are not this gate. Required18Python/27Rust/10policy names are proof minima, not total coverage. New code needs actual test collection.

BENCH-01 — frozen benchmark foundation

Priority: P1. **Current evidence:** current source + local/exact-head tests. **Status:** implemented; retain during integration.

Action: Reverify unchanged manifest and all prepared inputs, actual row/group assignments, original category/missing/label semantics, loss directions, full class probability ordering and task method inventory.

Dependencies: INT-02.

Acceptance commands:

```
"$J" benchmark verify-manifest --manifest "$M"
"$J" benchmark verify-inputs --manifest "$M" --datasets "$D"
"$J" benchmark methods
```

Required evidence: Original panel SHAba7dace4…,30datasets10/task, exact input-byte verification, native17/5/4recipes and4external families; alias/configuration inventory and unavailable reasons.

Resources: Read-only/static custody work; no fitting, models, GPU or campaign admission.

Source/evidence references: `crates/jope-bench/src/panel/data.rs`, `crates/jope-bench/src/panel/methods.rs`, `crates/jope-bench/src/panel/score.rs`, `benchmarks/panel.json`.

Qualification limit: Verification is metadata/data validation, no new fits. Do not replace a failed panel dataset.

BENCH-02 — external comparators

Priority: P1. **Current evidence:** four adapters implemented; earlier runtime inventories. **Status:** refresh at final environment.

Action: Pin all runtime/package/adapter bytes and actual constructors; execute relevant adapter contract tests and preserve unavailable dependencies.

Dependencies: INT-02.

Acceptance commands:

```
python3 deploy/prepare-panel-runtime.py --binary "$J" --native-command benchmark --build-receipt
"$CLI_BUILD_RECEIPT" --source-sha "$SOURCE_SHA" --python "$EXTERNAL_PYTHON" --site-packages
"$EXTERNAL_SITE_PACKAGES" --adapters crates/jope-bench/adapters --output "$ACCEPTANCE/
runtimes.json"
"$EXTERNAL_PYTHON" -m pytest crates/jope-bench/adapters/test_panel.py -q
```

Required evidence: Current4engine version/source/package hashes, constructors for3tasks, actual contract-test receipt and explicit unavailability inventory.

Resources: Preparation owns sharedCPUallocation; tests use allocatedCPU; no panel/training window begins.

Source/evidence references: `deploy/prepare-panel-runtime.py`, `crates/jope-bench/adapters/_panel.py`, `crates/jope-bench/adapters/test_panel.py`.

Qualification limit: Do not nest an outer allocation lock around a front door already taking it. Environment mutation invalidates old inventory.

EXEC-01 — bounded workers

Priority: P1. **Current evidence:** ten real synthetic host cases passed at 68. **Status:** implemented; CI coverage gap retained.

Action: Requalify final worker/controller bytes through ten current host cases and make required-lane coverage of critical async cleanup paths explicit.

Dependencies: INT-02.

Acceptance commands:

```
python3 deploy/ci/qualify-execution.py --test-binary "$PIPELINE_TEST_BINARY" --output
"$ACCEPTANCE/host-execution"
```

Required evidence: Ten nonzero actual cases,20concurrentworkers,limit readbacks, process-group drainage, setup failures/cancel gates, original-window equality and fixture code reachability.

Resources: Existing exclusive CPU CI allocation; one reused lane target, jobs4 unless lane policy sets otherwise; no concurrent numerical campaign.

Source/evidence references: `deploy/ci/qualify-execution.py`, `crates/jope-pipeline/src/execution.rs`.

Qualification limit: Current host qualification is not automatically covered by the ordinary required unit lane; missing privileges fail, not skip.

EXEC-02 — fault regression tests

Priority: P1. **Current evidence:** production terminal-persistence fix qualified indirectly. **Status:** test gap: drain-and-fixtures-F2.

Action: Add a real async-reap fixture with pre-gate BudgetExhausted and an undrained group. Prove terminal record survives before cleanup and admission remains blocked until drain.

Dependencies: INT-01.

Acceptance commands:

```
CARGO_INCREMENTAL=0 CARGO_PROFILE_DEV_DEBUG=line-tables-only cargo test --locked -p jope-pipeline
--features qualified-context execution::tests
```

Required evidence: Reproduction fails when persist is moved back under drained branch or admission_blocked handling is removed; corrected actual test passes with cleanup ordering evidence.

Resources: Existing exclusive CPU CI allocation; one reused lane target, jobs4 unless lane policy sets otherwise; no concurrent numerical campaign.

Source/evidence references: `crates/jope-pipeline/src/execution.rs:812`, `AGENT_CHAT.md: 2026-09-23T16:17 delta review`.

Qualification limit: Exact new test name is not present yet; existing module command is the truthful current gate. Ten old host cases do not refute this coverage gap.

EXEC-03 — receipt replay regression tests

Priority: P1. **Current evidence:** shared per-fold helper fixed and real68 recovered report passed. **Status:** test gap: resume-fix-F2.

Action: Add PoolBackend-shaped candidate receipt roundtrip asserting live Feedback equals recovered FitAdapter Feedback, including digest and partial completed-fold evidence.

Dependencies: INT-01.

Acceptance commands:

```
CARGO_INCREMENTAL=0 CARGO_PROFILE_DEV_DEBUG=line-tables-only cargo test --locked -p jope-bench --
features asm
CARGO_INCREMENTAL=0 CARGO_PROFILE_DEV_DEBUG=line-tables-only cargo test --locked -p jope-pipeline
--features qualified-context neural_search::tests
```

Required evidence: Regression reproduces d0 mismatch; exercises Some(shared fold digest), changed completed outputs refuse, never-admitted normalization cannot gain evidence.

Resources: Existing exclusive CPU CI allocation; one reused lane target, jobs4 unless lane policy sets otherwise; no concurrent numerical campaign.

Source/evidence references: `crates/jope-pipeline/src/neural_search.rs:1735`, `crates/jope-bench/src/panel/optimize_fit.rs`, `AGENT_CHAT.md:resume-fix-F2`.

Qualification limit: The two original FixtureBackend reproductions use absent digests and do not cover the same_folds path.

EXEC-04 — cutoff semantics regression

Priority: P1. **Current evidence:** production cutoff ordering corrected; existing test too weak. **Status:** test gap: regressions-F2.

Action: Strengthen cancellation-at-cutoff test to assert original_deadline stop reason and persisted receipt, not only is_ok.

Dependencies: INT-01.

Acceptance commands:

```
CARGO_INCREMENTAL=0 CARGO_PROFILE_DEV_DEBUG=line-tables-only cargo test --locked -p jope-pipeline
--features qualified-context cancellation_at_cutoff_preserves_observed_generation_on_resume
```

Required evidence: Test fails if cancellation check moves before deadline check; successful resume emits no new fit and leaves observed bytes unchanged.

Resources: Existing exclusive CPU CI allocation; one reused lane target, jobs4 unless lane policy sets otherwise; no concurrent numerical campaign.

Source/evidence references: `crates/jope-pipeline/src/neural_search.rs:1380`, `AGENT_CHAT.md:regressions-F2`.

EXEC-05 — accounting and unknown outcomes

Priority: P1. **Current evidence:** additive accounting union source + four authored Rust cases. **Status:** pending execution.

Action: Execute started/finished-union accounting tests; retain started-only unknowns separately from terminal refusals and qualify report replay without child retries.

Dependencies: INT-02.

Acceptance commands:

```
CARGO_INCREMENTAL=0 CARGO_PROFILE_DEV_DEBUG=line-tables-only cargo test --locked -p jope-bench --
features asm asm_campaign
```

Required evidence: Actual four new accounting tests collected; started-only attempts counted unknown, no inferredzero, exports counted once inASMworker totals, recovered reporting no launch.

Resources: Existing exclusive CPU CI allocation; one reused lane target, jobs4 unless lane policy sets otherwise; no concurrent numerical campaign.

Source/evidence references: `agent-accounting-union/training-bridge/HANDOFF.json`, `crates/jope-bench/src/panel/asm_campaign.rs (pending)`.

NEURAL-01 — context qualification

Priority: P0. **Current evidence:** eight fixed native cases passed at 68. **Status:** requires final-source binding.

Action: Build exact context/policy/proof workers; qualify unchanged native references for final source and preserve eight fixed-case plus actual dataset process evidence.

Dependencies: INT-02.

Acceptance commands:

```
"$CONTEXT_PROOF" prepare --sources "$REFERENCE_SOURCES" --sources-sha256 "$REFERENCE_SOURCES_SHA"
--encoder-files "$ENCODER_FILES" --encoder-files-sha256 "$ENCODER_FILES_SHA" --worker
"$CONTEXT_WORKER" --worker-sha256 "$CONTEXT_WORKER_SHA" --source-sha40 "$SOURCE_SHA" --resources
"$CONTEXT_RESOURCES" --resources-sha256 "$CONTEXT_RESOURCES_SHA" --output "$ACCEPTANCE/context-
proof"
"$CONTEXT_PROOF" run --plan "$CONTEXT_PLAN" --plan-sha256 "$CONTEXT_PLAN_SHA"
```

Required evidence: All eight fixed cases pass the unchanged component/context maximum absolute difference of0.0003; source, assets and program identities match. Preserve actual bounded execution receipts and the qualified construction token, original labels and row order.

Resources: Separate explicit resource manifest and approved allowance required before admission;20 CPUs/workers maximum,64 GiB maximum,zero swap,one thread,30 seconds per fit,120 seconds per arm maximum. No new allowance granted by this checklist.

Source/evidence references: `crates/jope-policy/src/context_producer/admission.rs:4`, `crates/jope-pipeline/src/cpu_context_bridge.rs`.

Qualification limit: Metadata/path equality or checksum alone is not numerical qualification; no changed tolerance or checkpoint substitution.

TEACH-01 — HWM comparison identity

Priority: P0. **Current evidence:** pending full-identity source + test plan. **Status:** not runtime qualified.

Action: Apply and execute full comparison identity enforcement before feedback; ensure stale evaluator/vocab/preprocessing/protocol/class order cannot collide with an incumbent.

Dependencies: INT-02.

Acceptance commands:

```
CARGO_INCREMENTAL=0 CARGO_PROFILE_DEV_DEBUG=line-tables-only cargo test --locked -p jope-learning
```

Required evidence: Changed identity negative cases, old record migration/refusal behavior, hash-chain integrity and ties/strict improvement semantics.

Resources: Existing exclusive CPU CI allocation; one reused lane target, jobs4 unless lane policy sets otherwise; no concurrent numerical campaign.

Source/evidence references: `crates/jope-learning/src/record.rs (pending)`, `crates/jope-learning/src/hwm.rs (pending)`, `crates/jope-learning/src/eligibility.rs (pending)`.

TEACH-02 — label exposure readers

Priority: P0. **Current evidence:** burn writer and effective-role mapping prepared. **Status:** residual eligibility-B6.

Action: Wire durable label-burn checks into every path that may claim an untouched test score, including HWM ingest/freeze and report publication, or explicitly prohibit that claim.

Dependencies: TEACH-01.

Acceptance commands:

```
CARGO_INCREMENTAL=0 CARGO_PROFILE_DEV_DEBUG=line-tables-only cargo test --locked -p jope-learning
CARGO_INCREMENTAL=0 CARGO_PROFILE_DEV_DEBUG=line-tables-only cargo test --locked -p jope-bench --
features asm
```

Required evidence: Production caller inventory for require_unburned_test_partition; an integration test proving a burned partition is refused by test-valid readers; attempted sealed teaching refused; adaptive development reporting remains truthful.

Resources: Existing exclusive CPU CI allocation; one reused lane target, jobs4 unless lane policy sets otherwise; no concurrent numerical campaign.

Source/evidence references: `crates/jope-learning/src/eligibility.rs:253 (pending)`, `AGENT_CHAT.md:eligibility-B6`, `crates/jope-bench/src/panel/selection_exposure.rs`.

Qualification limit: No current live leak was observed because no producer emits benchmark-valid panel records. A burn file writer alone is not downstream enforcement.

TEACH-03 — real teacher collection

Priority: P0. **Current evidence:** zero eligible current teachers; two prospective evidenced families. **Status:** blocked on fresh admitted evidence.

Action: Preregister teaching before two separately bounded proof plans for machineCPU and geyser. Obtain actual finite OOF improvement and improvements on four of five folds, or retain the rejection.

Dependencies: BENCH-01, BENCH-02, NEURAL-01, TEACH-01, TEACH-02.

Acceptance commands:

```
"$J" optimize proof-plan --base-plan "$BASE_PLAN" --base-plan-sha256 "$BASE_ID" --configuration
"$TEACHING_CONFIGURATION" --configuration-sha256 "$TEACHING_CONFIGURATION_SHA" --dataset
"$FAMILY_DATASET" --resources "$TEACHING_RESOURCES" --resources-sha256 "$TEACHING_RESOURCES_SHA"
--output "$FAMILY_PLAN"
"$J" optimize run --plan "$FAMILY_PLAN" --plan-sha256 "$FAMILY_PLAN_ID" --manifest "$M" --datasets
"$D" --journal "$FAMILY_JOURNAL"
"$J" optimize report --plan "$FAMILY_PLAN" --plan-sha256 "$FAMILY_PLAN_ID" --manifest "$M" --
datasets "$D" --journal "$FAMILY_JOURNAL"
```

Required evidence: Before-fit freeze, vocabulary and source pins; two independent families and original windows; nine context proofs per dataset; complete five-fold predictions and trivial baselines. Apply the1e-12 absolute plus1e-6 relative gain margin and four-of-five rule. Retain no-signal refusals.

Resources: Separate explicit resource manifest and approved allowance required before admission;20 CPUs/workers maximum,64 GiB maximum,zero swap,one thread,30 seconds per fit,120 seconds per arm maximum. No new allowance granted by this checklist.

Source/evidence references: `agent-training-readiness/readiness-v3.json`, `operations-training.md:Present teacher readiness`, `crates/jope-bench/src/panel/teaching.rs (pending)`.

Qualification limit: Execute separately for230_machine_cpu and712_chscase_geyser1; the prospective choice predates losses. Old55e9,3fd and68 operational receipts cannot be retrofitted. The older queue suggested300 secondseconds per family without admitting that allowance.

TEACH-04 — real corpus materialization

Priority: P0. **Current evidence:** native exporter prepared. **Status:** blocked on two real families.

Action: Build the request from every eligible requested comparison and its original native-context proof; freeze a family-disjoint corpus and the exact vocabulary.

Dependencies: TEACH-03.

Acceptance commands:

```
"$J" train prepare-inputs --learning-state "$LEARNING_STATE" --request "$CORPUS_REQUEST" --
request-sha256 "$CORPUS_REQUEST_SHA" --output "$CORPUS_OUTPUT"
```

Required evidence: At least two real families; every eligible requested comparison included; seed7 partition derived from source families; exact1536+512 context values. No invented context, parameter-bin snapping, label remapping or omission of difficult examples.

Resources: Read-only/static custody work; no fitting, models, GPU or campaign admission.

Source/evidence references: `crates/jope-cli/src/train/corpus.rs (pending)`, `operations-training.md:Build the real corpus`.

Qualification limit: Only if exactly two families qualify does the recorded machineCPU-development/geyser-training assignment apply. More eligible families must all remain admitted.

TRAIN-01 — actual trainer tests

Priority: P0. **Current evidence:** seven pure input tests passed; actual model cases deselected. **Status:** residual training-F3.

Action: Execute actual optimizer, reload and continuation tests across regression, binary and multiclass, plus task, vocabulary and runtime compatibility negatives.

Dependencies: INT-02.

Acceptance commands:

```
python/.venv/bin/python -m pytest python/tests/test_policy_training_contract.py python/tests/
test_training_campaign.py python/tests/test_training_corpus_contract.py python/tests/
test_experimental_proposal.py python/tests/test_experimental_runtime.py -q --
junitxml="$ACCEPTANCE/training-fixtures.xml"
python3 deploy/ci/test-training-proof.py
```

Required evidence: Collected test IDs include all three parameters of test_actual_trained_classifier_reload_sampling_uses_task_head. Prove positive updates, changed parameters, task reward routing, reload and continuation; reject zero updates and missing architecture. The proof gate must see actual execution.

Resources: Existing exclusive CPU CI allocation; one reused lane target, jobs4 unless lane policy sets otherwise; no concurrent numerical campaign.

Source/evidence references: `python/tests/test_experimental_proposal.py (pending)`, `python/src/jope/policy/train_v3.py (pending)`, `deploy/ci/training-proof.py (pending)`.

Qualification limit: Required-name inventories and deselected tests do not count as execution. A synthetic CPU checkpoint remains fixture evidence.

TRAIN-02 — native supervised training

Priority: P0. **Current evidence:** supervisor/cgroup orchestration prepared. **Status:** host proof pending.

Action: Exercise the native training front door with bounded synthetic normal and cancellation cases: subgroup entry, release, gate refusal, cleanup and original-window resume.

Dependencies: TRAIN-01.

Acceptance commands:

```
"$J" train plan --state "$FIXTURE_TRAIN_STATE" --python "$TRAINING_PYTHON" --synthetic-fixture --
task all-admitted --device cpu --source-sha "$SOURCE_SHA" --seed 7 --passes 2
"$J" train run --state "$FIXTURE_TRAIN_STATE"
"$J" train verify --state "$FIXTURE_TRAIN_STATE"
```

Required evidence: Actual normal and interrupted CPU workers; controller-owned subgroup joined before execution; original deadline, device and runtime unchanged; children drained and resumed state compatible. The native wrapper must enforce admission.

Resources: Explicitbounded CPU fixture allocation; same planned3600 seconds/1200scaps are maxima, not authorization to spend them twice; no GPU.

Source/evidence references: `crates/jope-cli/src/train.rs (pending)`, `python/src/jope/policy/training_campaign.py (pending)`.

Qualification limit: Author the cancellation harness against the actual front door and target only its owned processes. CPU mode requires explicit fixture provenance.

TRAIN-03 — bounded real GPU training

Priority: P0. **Current evidence:** implemented follow-on; no GPU campaign started. **Status:** blocked on admitted data and consumption readiness.

Action: Run one all-admitted CUDA campaign with real regression and two passes. Retain synthetic classifier fallback separately and report every unavailable category.

Dependencies: TEACH-04, TRAIN-02, NEURAL-02.

Acceptance commands:

```
"$J" train plan --state "$TRAIN_STATE" --python "$TRAINING_PYTHON" --inputs "$CORPUS_OUTPUT/
inputs.json" --task all-admitted --device cuda:0 --source-sha "$SOURCE_SHA" --learning-state
"$LEARNING_STATE" --seed 7 --passes 2
"$J" train run --state "$TRAIN_STATE"
```

Required evidence: Original global/category windows; one RTX3090 UUID; positive updates, changed parameter hashes and task coverage with data_kind/fallback reasons. Save and reload model plus optimizer; record CPU/GPU cost where measured. Requested CUDA must not silently fall back to CPU.

Resources: One original3600 seconds total window;≤1200 seconds per category including both passes;one RTX 3090,20CPU,64 GiB,zero swap,1Torch thread;no exclusiveCPU CI overlap. Resume never resets.

Source/evidence references: `python/src/jope/policy/training_campaign.py:780 (pending)`, `operations-training.md:Plan, run, resume, verify and export.`

Qualification limit: NEURAL-02 gates source and fixture readiness before the GPU hour. It does not depend on a newly trained live export, avoiding a circular prerequisite.

TRAIN-04 — checkpoint continuation and export

Priority: P0. **Current evidence:** verification/export source prepared. **Status:** requires actual campaign.

Action: Verify two committed passes, optimizer counters and the immutable receipt chain; export complete architecture, vocabulary, task, source, encoder and runtime bindings.

Dependencies: TRAIN-03.

Acceptance commands:

```
"$J" train resume --state "$TRAIN_STATE"
"$J" train verify --state "$TRAIN_STATE"
"$J" train export --state "$TRAIN_STATE" --output "$TRAIN_EXPORT"
```

Required evidence: Resume only when needed inside the original window. Prove increased optimizer steps, identical parameters after reload, changed parameters relative to the prior checkpoint, matching export/resume proof and completed task coverage. Keep experimental non-serving status.

Resources: One original3600 seconds total window;≤1200 seconds per category including both passes;one RTX 3090,20CPU,64 GiB,zero swap,1Torch thread;no exclusiveCPU CI overlap. Resume never resets.

Source/evidence references: `python/src/jope/policy/training_campaign.py:1203 (pending)`, `python/src/jope/policy/training_campaign.py:1303 (pending)`.

Qualification limit: Resume is conditional, not an instruction to start a third pass after completing the plan. A new state or plan cannot extend the approved hour.

NEURAL-02 — trained-checkpoint consumer readiness

Priority: P0. **Current evidence:** independent source review + purechecks; regression packet appended. **Status:** compile/runtime fixture pending.

Action: Qualify the explicit experimental regression consumer before GPU use. Preserve accepted-baseline hard pins and strict architecture, task, vocabulary, source and encoder compatibility.

Dependencies: INT-02, TRAIN-01.

Acceptance commands:

```
CARGO_INCREMENTAL=0 CARGO_PROFILE_DEV_DEBUG=line-tables-only cargo test --locked -p jope-pipeline
--features qualified-context experimental_regression_requires_matching_task_and_executable_head
python/.venv/bin/python -m pytest python/tests/test_experimental_proposal.py -q
```

Required evidence: Execute all three actual task-model cases. Test accepted-versus-experimental decoding and refusal of incomplete architecture, missing state, changed vocabulary, source or encoder. The operational regression route requires a real admitted export.

Resources: Existing exclusive CPU CI allocation; one reused lane target, jobs4 unless lane policy sets otherwise; no concurrent numerical campaign.

Source/evidence references: `agent-experimental-regression/HANDOFF.json`, `crates/jope-pipeline/src/experimental_producer.rs (pending)`, `python/src/jope/policy/experimental_proposal.py (pending)`.

Qualification limit: The new regression consumer is not serving promotion. Synthetic regression may test mechanics but cannot enter the real operational route.

NEURAL-03 — live newly trained consumption

Priority: P0. **Current evidence:** current accepted checkpoint origin/evolution proved onone dataset. **Status:** newly trained consumption pending.

Action: Pin the actual exported checkpoints and runtime/canonicalizer into the final configuration. Prove new-checkpoint proposals reach real generation-zero fits and completed descendants.

Dependencies: TRAIN-04, NEURAL-01.

Acceptance commands:

```
"$J" optimize proof-plan --base-plan "$BASE_PLAN" --base-plan-sha256 "$BASE_ID" --configuration
"$TRAINED_CONFIGURATION" --configuration-sha256 "$TRAINED_CONFIGURATION_SHA" --dataset
230_machine_cpu --resources "$CONSUMPTION_RESOURCES" --resources-sha256
"$CONSUMPTION_RESOURCES_SHA" --output "$CONSUMPTION_PLAN"
"$J" optimize run --plan "$CONSUMPTION_PLAN" --plan-sha256 "$CONSUMPTION_PLAN_ID" --manifest "$M"
--datasets "$D" --journal "$CONSUMPTION_JOURNAL"
"$J" optimize report --plan "$CONSUMPTION_PLAN" --plan-sha256 "$CONSUMPTION_PLAN_ID" --manifest
"$M" --datasets "$D" --journal "$CONSUMPTION_JOURNAL"
```

Required evidence: The exported checkpoint hash appears in actual proposal and fit evidence; 64+16 versus80 origins remain authenticated. Retain completed descendants where claimed, predictions and fitted states. Regression must be admitted and classifier fixture limits explicit; replay the original window successfully.

Resources: Separate explicit resource manifest and approved allowance required before admission;20 CPUs/workers maximum,64 GiB maximum,zero swap,one thread,30 seconds per fit,120 seconds per arm maximum. No new allowance granted by this checklist.

Source/evidence references: `crates/jope-bench/src/panel/experimental_proposal.rs (pending), crates/jope-bench/src/panel/origin_proof.rs (pending).`

Qualification limit: The current proof-plan scopes one regression dataset. Classification needs its implemented campaign/configuration route; this command alone cannot prove binary or multiclass live consumption.

INFER-01 — fitted state and privacy

Priority: P0. **Current evidence:** native fitted state exists; product28source prepared. **Status:** qualification pending.

Action: Execute the lossless private codec, complete fitted preprocessing/state reload, and public allowlist/canary tests before export. Unsupported states remain explicit.

Dependencies: INT-02.

Acceptance commands:

```
CARGO_INCREMENTAL=0 CARGO_PROFILE_DEV_DEBUG=line-tables-only cargo test --locked -p feat-math
fitted_portable
CARGO_INCREMENTAL=0 CARGO_PROFILE_DEV_DEBUG=line-tables-only cargo test --locked -p jope-bench --
features asm inference
bash deploy/ci/ip-scrub.sh
```

Required evidence: Complete prediction parity after reload; missing/unseen category, class order and full probability-matrix cases. Preserve the private codec identity; public bundles must omit recipe, discovery and provenance canaries.

Resources: Existing exclusive CPU CI allocation; one reused lane target, jobs4 unless lane policy sets otherwise; no concurrent numerical campaign.

Source/evidence references: `crates/feat-math/src/advanced_v3/fitted_portable.rs (pending)`, `crates/jope-bench/src/panel/inference.rs (pending)`, `languages-examples.md:State restrictions`.

Qualification limit: The private C++ head gate is broader than the public gate. Python/Rust14of26, C++2of26 and JavaScript6of26 are static head counts, not runtime qualification rates.

INFER-02 — exact selected-winner portability

Priority: P1. **Current evidence:** winner/add-on source + statictests. **Status:** actual export/parity unrun.

Action: Qualify the selected existing fold state in order: Python, Rust, the same C++ emitter under C++17 and C++20, then JavaScript. Preserve unsupported external winners and the complete ranking.

Dependencies: INFER-01, PANEL-01.

Acceptance commands:

```
"$J" benchmark export-winner --plan "$PLAN" --plan-sha256 "$PLAN_ID" --manifest "$M" --datasets
"$D" --journal "$JOURNAL" --dataset "$DATASET_ID" --fold "$FOLD"
"$J" benchmark qualify-export --prepared "$PREPARED_EXPORT" --prepared-sha256
"$PREPARED_EXPORT_ID" --tools "$TOOLS" --tools-sha256 "$TOOLS_SHA"
"$J" benchmark report --plan "$PLAN" --plan-sha256 "$PLAN_ID" --manifest "$M" --datasets "$D" --
journal "$JOURNAL"
```

Required evidence: Exact state, artifact, input, class, toolchain and numerical-policy hashes; actual five-backend outcomes. Python/Rust/C++ use bitwise comparison; JavaScript uses1e-12 absolute and relative tolerance. Recover interrupted work without retrying its child; read-only reports launch nothing. Unsupported never replaces the winner.

Resources: Same original campaign execution window and worker caps; no new job after t0 + 6900 seconds for official panel. No post-panel allowance is implemented.

Source/evidence references: `crates/jope-bench/src/panel/fitted_export.rs (pending)`, `crates/jope-bench/src/panel/fitted_export_run.rs (pending)`, `languages-examples.md:Example C`.

Qualification limit: Library fixtures precede the panel. Actual selected winners require complete confirmations from PANEL-01 and qualification before the original execution cutoff. No separately bounded post-panel qualification front door exists. One fitted fold does not qualify the other four states.

INSTALL-01 — private installation

Priority: P1. **Current evidence:** installer source and15metadata tests passed. **Status:** cold/live path pending.

Action: Build the CLI from final source, measure cold build separately, checksum-install the exact Linux binary, and verify actual command help plus dependency/ELF boundaries.

Dependencies: INT-02.

Acceptance commands:

```
python3 deploy/install-private.py source --checkout "$R" --source-sha "$SOURCE_SHA" --output
"$ACCEPTANCE/cli-build" --jobs 4
python3 deploy/install-private.py binary --binary "$PRIVATE_BINARY" --sha256 "$BINARY_SHA" --
```

```
source-sha "$SOURCE_SHA" --prefix "$ACCEPTANCE/install"
"$J" doctor
```

Required evidence: Cold build/install durations, cache and already_present flags; Cargo executable messages, features, toolchain, dependency graph and ELF checks. Retain checksum/help readbacks and failure receipts. CLI dependencies exclude the engine, Torch and CUDA.

Resources: Existing exclusive CPU CI allocation; one reused lane target, jobs4 unless lane policy sets otherwise; no concurrent numerical campaign.

Source/evidence references: `deploy/install-private.py (pending)`, `product.md:Build and install`.

Qualification limit: An installer receipt is not policy/context-worker build authority. Do not include training credentials or model bodies in the public inference bundle.

INSTALL-02 — final input/runtime preparation

Priority: P1. **Current evidence:** prepared panel existing; runtime refresh required. **Status:** pending final source.

Action: Install the pinned final Python wheel without changing CUDA dependencies; build worker inventories and verify prepared datasets before campaign planning.

Dependencies: INSTALL-01, BENCH-01, BENCH-02.

Acceptance commands:

```
"$J" benchmark verify-inputs --manifest "$M" --datasets "$D"
"$J" benchmark plan --manifest "$M" --datasets "$D" --source-sha "$SOURCE_SHA" --runtimes
"$RUNTIMES" --output "$BASE_PLAN"
```

Required evidence: Wheel, source, runtime and dependency inventory; context/policy workers identified from actual Cargo messages; semantic base-plan identity; zero fits and no execution clock started.

Resources: Read-only/static custody work; no fitting, models, GPU or campaign admission.

Source/evidence references: `product.md:Prepare the panel`, `agent-acceptance-queue/QUEUE.md`.

Qualification limit: Packaging commands depend on final reviewed artifacts and the actual virtual environment. Preserve versions and file identities; changing the environment invalidates a pinned campaign.

DEMO-01 — measured warm fitted demonstration

Priority: P1. **Current evidence:** warmtargetadd-onprepared;historicalfirstfit≈50.4s. **Status:** realtrained warm proof pending.

Action: Produce actual trained-neural and generic fitted artifacts, reloaded predictions, baseline/comparison and truthful target_met. Do not clip arm budgets to pass sixty seconds.

Dependencies: NEURAL-03, INSTALL-02.

Acceptance commands:

```
"$J" demo --asm-plan "$WARM_PLAN" --asm-plan-sha256 "$WARM_PLAN_ID" --manifest "$M" --datasets
"$D" --output "$WARM_OUTPUT" --export python --export rust
```

```
python3 deploy/ci/qualify-demo.py --binary "$J" --sha256 "$J_SHA" --source-sha "$SOURCE_SHA" --
output "$ACCEPTANCE/generic-three-task" --require-under-60
```

Required evidence: Actual fitted state, predictions and baseline; both full paired allowances; start/ end, warm elapsed, execution_complete and target_met. Keep generic synthetic qualification separate from trained-neural proof.

Resources: Separate explicit resource manifest and approved allowance required before admission;20 CPUs/workers maximum,64 GiB maximum,zero swap,one thread,30 seconds per fit,120 seconds per arm maximum. No new allowance granted by this checklist.

Source/evidence references: `crates/jope-bench/src/panel/demo_asm.rs (pending)`, `docs/private-warm-target.md (pending)`, `product.md:Warm fitted demonstration`.

Qualification limit: Under sixty seconds is a measured target. The existing twenty-second-arm example is unqualified. An explicitly admitted longer proof may satisfy execution while missing the target. This checklist grants no implicit300-second allowance.

PANEL-01 — unchanged thirty-dataset campaign

Priority: P1. **Current evidence:** protocol/currentfoundation qualified;full panel unstarted. **Status:** pending finaltrained configuration.

Action: Freeze the final roster/configuration and run the unchanged thirty datasets once. Screen fold0, freeze the top five, confirm five development folds, and add asm.neural diagnostically when absent. Retain every failure.

Dependencies: BENCH-01, BENCH-02, NEURAL-03, DEMO-01, INFER-01, INSTALL-02.

Acceptance commands:

```
"$J" optimize plan --base-plan "$BASE_PLAN" --base-plan-sha256 "$BASE_ID" --configuration
"$TRAINED_CONFIGURATION" --configuration-sha256 "$TRAINED_CONFIGURATION_SHA" --output
"$PANEL_PLAN"
"$J" benchmark run --plan "$PANEL_PLAN" --plan-sha256 "$PANEL_PLAN_ID" --manifest "$M" --datasets
"$D" --journal "$PANEL_JOURNAL"
"$J" benchmark report --plan "$PANEL_PLAN" --plan-sha256 "$PANEL_PLAN_ID" --manifest "$M" --
datasets "$D" --journal "$PANEL_JOURNAL"
```

Required evidence: All thirty dataset identities, ten per task; every applicable method slot; screen/ frozen/confirmation losses; selection-exposure row intersection hashes; no sealed reads; winner backend outcomes and actual win counts.

Resources: Exactly7200 seconds including300 seconds reporting reserve;20 CPUs/workers,64 GiB,zero swap,one thread per fit,30 seconds per fit,120 seconds per search arm; sealed labels unopened; existing window never reset.

Source/evidence references: `benchmarks/panel.json`, `crates/jope-bench/src/panel/protocol.rs`, `crates/jope-bench/src/panel/selection_exposure.rs`.

Qualification limit: Current code forces asm.neural into extra confirmation, not asm.generic. Selected five-fold summaries are adaptive development, and ten wins per category are not promised. Qualify exports before the original6900-second cutoff or preserve unattempted status.

REPORT-01 — complete measured report

Priority: P1. **Current evidence:** one-dataset832predictor/164recipe operational audit. **Status:** fullcampaign report pending.

Action: Publish all thirty losses/gaps, outcomes, time/cost per unique artifact and recipe, coverage and failure inventories. Compare the supplied one-hour/ten-million-token scenario only using supported quantities.

Dependencies: PANEL-01, EXEC-05.

Acceptance commands:

```
"$J" benchmark report --plan "$PANEL_PLAN" --plan-sha256 "$PANEL_PLAN_ID" --manifest "$M" --
datasets "$D" --journal "$PANEL_JOURNAL"
```

Required evidence: Report recovery without fits; union of started/finished records; worker CPU distinguished from overlapping service CPU; known and unknown costs; training, installation, preparation and export accounted separately. Tokens and money stay unknown unless measured; retain win/loss/gap tables.

Resources: Original300 secondsreportingreserve forpanel; laterreadonlyauditopensnofitbudget.

Source/evidence references: `status.md:Cost`, `data-evidence.md`, `crates/jope-bench/src/panel/asm_campaign.rs (pending)`.

Qualification limit: The one-hour comparison supplies wall-duration context, not a matched speedup. Unique artifact hashes do not establish behavioral uniqueness. Never add aggregate service CPU to worker CPU already included in it.

MOTIF-01 — eligible reusable fragments

Priority: P1. **Current evidence:** separate typed library branch exists. **Status:** minefront door incomplete.

Action: Integrate the production eligible-State reader, typed fragment/macro miner and umbrella CLI. Preserve full comparison identity, provenance and private/public separation.

Dependencies: TEACH-01, TEACH-02, TEACH-03.

Acceptance commands:

```
"$J" motifs mine --help
"$J" motifs validate --help
```

Required evidence: Actual mine CLI contract/help; eligible-input identity inventory; deterministic typed fragment output. Exclude rejected, no-signal, unbound and sealed examples.

Resources: Read-only/static custody work; no fitting, models, GPU or campaign admission.

Source/evidence references: `status.md:Pending acceptance`, `agent-acceptance-queue/QUEUE.md:motifs`.

Qualification limit: These help commands are currently unavailable at 68 and are acceptance targets. No truthful exact mine invocation exists yet; record it after implementation.

MOTIF-02 — reuse validation

Priority: P1. **Current evidence:** validationlibrary/preparation only. **Status:** live reuse pending.

Action: Validate fragments/macros on independent host contexts, lossless roundtrips, deterministic fit compatibility and privacy. Measure reuse benefit under the full comparison identity.

Dependencies: MOTIF-01.

Command boundary: no truthful executable acceptance command yet; implement or obtain missing interface first.

Required evidence: Exact implemented validate command and input table; at least three hosts per fragment where its existing contract requires that count; roundtrip and actual fit receipts, independent context identity and canary absence. Support frequency is separate from measured uplift.

Resources: Explicitnewboundedreusequalification allocation required; noresetofpanelortrainingwindows.

Source/evidence references: `agent-acceptance-queue/QUEUE.md:motifs`, `docs/adr/ADR-0005-canonical-learning-loop.md`.

Qualification limit: Do not invent mine flags. The historical validate argument shape alone does not establish umbrella wiring or production State admission.

MOTIF-03 — new operators

Priority: P2. **Current evidence:** useracceptance requirement. **Status:** future conditional work.

Action: Admit new operators only after measured ablation and grammar, checkpoint and backend compatibility tests. Preserve the old vocabulary identity.

Dependencies: MOTIF-02.

Command boundary: no truthful executable acceptance command yet; implement or obtain missing interface first.

Required evidence: Preregistered ablation recipes, budget and contexts; measured marginal gain; task/type/codec/emitter/vocabulary compatibility; new identity and migration policy.

Resources: Separatelypreregistered ablationbudget; noneadmitted here.

Source/evidence references: `user approval plan:Eligible optimization and reuse`, `languages-examples.md:Future work`.

Qualification limit: Typed reusable macros may stay within existing operators. Fragment frequency alone does not justify adding an operator.

IP-01 — public privacy boundaries

Priority: P1. **Current evidence:** allowlists/canaries prepared; historicallintblindspots. **Status:** residual D-IP4 and nextPRtextwork.

Action: Resolve the approved public vocabulary/naming scope and test actual CLI help, receipts and NOTICE. Preserve the explicitly requested internal manual and the strict public inference allowlist.

Dependencies: INT-01.

Acceptance commands:

```
bash deploy/ci/ip-scrub.sh
```

Required evidence: Actual shipped-file inventory and help/receipt/NOTICE canaries; private provenance absent; allowlist negative cases. Keep the internal requested manual distinct from public bundles.

Resources: Existing exclusive CPU CI allocation; one reused lane target, jobs4 unless lane policy sets otherwise; no concurrent numerical campaign.

Source/evidence references: `AGENT_CHAT.md:D-IP4`, `deploy/ci/ip-scrub.sh`, `crates/feat-math/src/advanced_v3/fitted_portable.rs (pending)`.

Qualification limit: Current lint success does not prove unscanned surfaces clean. The user explicitly requested the private GP explanation, so ordinary creation of this manual needs no additional approval.

MAINT-01 — broad diagnostic qualification

Priority: P2. **Current evidence:** historicalfull-default failures inventoried. **Status:** not closedbyfastCI.

Action: Reproduce and repair recorded full-default, scratch, policy and neural families through their owning full lanes. Retain unavailable capabilities and resource limits.

Dependencies: LIFE-01.

Acceptance commands:

```
bash deploy/ci/full-lane.sh full-default
bash deploy/ci/full-lane.sh full-scratch
bash deploy/ci/full-lane.sh full-policy
bash deploy/ci/full-lane.sh full-neural
```

Required evidence: Source-bound full-suite collection and failures, reproductions and corrected receipts. Keep fast-CI and full-diagnostic totals separate; zero tests or skips cannot count as passes.

Resources: Existing exclusive CPU CI allocation; one reused lane target, jobs4 unless lane policy sets otherwise; no concurrent numerical campaign.

Source/evidence references: `deploy/ci/full-lane.sh`, `maintenance.md:Coverage and reviewed lifecycle`, `AGENT_CHAT.md:full-default failures`.

Qualification limit: Check lane environment requirements before invocation. The owning qualification supplies allocation and budget; this manual does not admit them.

FUTURE-01 — CUDA inference and serving promotion

Priority: P2. **Current evidence:** not qualified in this delivery. **Status:** deferred.

Action: After portable CPU backends, design CUDA fitted-inference qualification and independent experimental-to-serving promotion. Preserve the accepted baseline until promotion passes.

Dependencies: INFER-02, REPORT-01.

Command boundary: no truthful executable acceptance command yet; implement or obtain missing interface first.

Required evidence: New backend contract, fixed numerical policy, exact device/toolchain/state parity and promotion evaluation; reviewed pointer compare-and-swap and rollback where used.

Resources: NoCUDAinference/promotionbudgetadmitted bythisroadmap.

Source/evidence references: `docs/adr/ADR-0005-canonical-learning-loop.md`, `architecture.md:Experimental trained models`.

Qualification limit: Training GPU availability does not prove CUDA inference parity. Export is not serving promotion.

Stopping and completion rules

An unsuccessful prerequisite remains a recorded refusal or failure. Do not substitute synthetic real-data teachers, another winner, easier datasets or a new deadline. Source completion does not close a runtime task. A plan, help output or checkpoint file is not the corresponding live receipt.

Full delivery requires all thirty outcomes, losses and gaps, backend limitations, measured unique-predictor/recipe costs, honest training category scope and actual newly trained consumption. Ten wins per category and an under-sixty-second warm result remain measured targets. The one-hour/ten-million-token comparison must retain unknown hardware, workload, token and currency quantities.

The document gate requires one readable PDF plus matching section Markdown, reproducible figure inputs, source/lifecycle references and this maintained checklist. The maintainer-private examples are authorized documentation; their existence does not expand public inference bundle contents.

11 / Evidence index, reviewed delivery and glossary

Source and lifecycle identity

This edition describes the implementation reviewed at `68f56404c3e199e06413bfd645d3fe0803d7b96f`. PR83 merged on 23 September 2026 at 16:34:47 UTC. The resulting remote main is `a75e543d2166017cb0139e0a251d51d9a0c98e08`. Both revisions have Git tree `176b75bc36d76c7ce4eaeb7234677ec013468889`. Rebase merge changed commit identity; it did not change the reviewed source tree.

The manual itself is a later documentation change. Its publication identity must be read from its own Git history and CI receipt, not confused with the implementation snapshot. Follow-on patches described as pending are outside the implementation snapshot even if they are preserved on this host.

Gate	Exact observation
PR	`veox-ai/JopeDime#83`, merged
Manual CI dispatch	Run 35887284772, attempt 1, source `68f56404…`, success
PR CI	Run 35887255898, attempt 1, source `68f56404…`, success
Required CI jobs	Eight lanes and aggregate succeeded in both runs
CI test inventory	937 Rust passed, 14 skipped; 221 Python passed per run
First independent native review	5293748183, `neverhuman`, APPROVED, exact full reviewed head, 16:29:01 UTC
Second independent native review	5293788772, same eligible account, APPROVED, same head, 16:32:43 UTC; includes residual disclosures
Author eligibility	All twelve commits associated with `jepsontaylor` as author and committer; reviewer differs
Merge	Guarded rebase with `--match-head-commit` at the reviewed head
Remote-main verification	Fetched main and compared complete Git tree; canonical clean main observed before documentation branch
Main guard	Run 35889731910, attempt 1, success at resulting main; guard job 107278860508

Two review processes overlapped during coordination. Both were independent of the source writer and both approvals are retained; they are not two different human reviewers. One process's supplementary disclosures identify outstanding work rather than silently treating it as fixed. The writer did not select another credential to approve its own source.

The new full-main diagnostic run 35889732032 was cancelled while queued to preserve the admitted local allocation and let main-guard run. It did not pass or execute. Earlier full-main run 35852022466 failed and remains historical diagnostic evidence requiring repair. Passing fast CI does not replace that outstanding broad qualification.

Important local receipts

The durable custody root is `/home/ubuntu/jope-rescue-20260921/asm-integration-20260923/`. These are host-local receipt locations, not promises that every reader has the same files. The structured documentation evidence identifies the source observations copied into this edition. Weight bodies, source dataset bodies and sealed labels are not embedded in the manual.

Relative custody path	Meaning
`neural-integration/lifecycle-68f56404c3e199e06413bfd645d3fe0803d7b96f/`	Full author/committer inventory, CI, native reviews, merge and main-guard readbacks
`neural-integration/pr83-final-review-68f56404c3e199e06413bfd645d3fe0803d7b96f/`	Independent review prompt, source/proof inspection and native approval receipt
`neural-integration/source-match-68f56404c3e199e06413bfd645d3fe0803d7b96f.json`	All 2,855 Git blobs match the immutable build input inventory
`neural-integration/native-context-68f56404c3e199e06413bfd645d3fe0803d7b96f/`	Eight native encoder qualification cases under unchanged numerical policy
`neural-integration/origin-run-68f56404.json`	Final operational run command, timing and exit status
`neural-integration/origin-report-68f56404.json`	Independent recovered-report command and timing
`agent-final-acceptance/measured-origin-68f56404/audit.json`	Source/report/worker inventory and independently bound supervisor accounting
`neural-integration/final-head-pipeline-bench-68f56404.json`	Local 53 benchmark and 106 pipeline test results
`neural-integration/final-head-host-ten-68f56404.json`	Ten actual privileged host cases and binary identity
`agent-final-acceptance/failed-origin-d0f5748b/audit.json`	Preserved failed prior run; not an accepted current score
`agent-maintainer-manual/`	Section drafts, structured data inventory, example preparation and documentation observations

Treat a receipt's command, source SHA, input hashes, runtime and original deadline as one evidence unit. Copying a number out of that unit can lose its scope. A local path alone proves neither continued availability nor integrity: verify the retained hash and the contents required for the claim.

Reproduction boundaries

Rendering this manual uses its committed section Markdown, structured data, figure inputs and Python rendering tools. It should not fetch a corpus, load a model, run a fit or contact the sealed store. Plotting historical evidence is distinct from rerunning historical experiments. The included figure data keeps missingness, eligibility and snapshot identities.

The private toy example has its own synthetic input and output receipt. Even a successful five-language recipe/refit comparison cannot qualify a different selected fitted predictor. The pending fitted-inference acceptance must authenticate the retained winner and compare complete saved-state predictions with its original numerical policy.

Operational replay needs more than the document: the exact source/build, immutable workers, model/checkpoint/vocabulary, runtime dependencies, allowed data inputs, original plan and journals, and an available resource allocation. Reproduction that resumes an expired campaign cannot start fresh jobs by assigning a new deadline. A new experiment needs a separately recorded plan; it does not repair the old receipt.

Glossary

Term	Meaning in this repository
ASM	Algorithm Synthesis Model: learns to generate and optimize executable algorithms from data
Algorithm / recipe / program	Executable typed procedure and configuration; reusable across fits
GP	Genetic programming: search over executable programs using variation and measured selection
Fitted predictor	One recipe plus preprocessing and model state learned from exact training inputs
Checkpoint	Serialized neural proposal-model parameters and compatibility metadata; distinct from a fitted tabular predictor
Context	Authenticated numerical dataset/target representation supplied to the proposal policy
Dataset encoder	Learned or specified mapping from permitted dataset structure/rows to context features
Target encoder	Representation of allowed target information; task-specific and restricted to permitted fitting rows
Vocabulary	Token identities, grammar and executable restrictions associated with a checkpoint
Generation zero	Initial candidate slots before evolutionary descendants
Exploration candidate	An admitted generic candidate in the neural arm; not falsely labelled neural-origin
Offspring	Mutation or crossover descendant with authenticated parent identities and actual fit evidence
OOF	Out-of-fold: predict rows using a fit that excluded those rows; adaptive recipe selection can still expose them at another stage
Cut identity	Frozen identity of the training/scoring row assignment and evaluator context
Development data	Data visible to optimization or recipe/model selection, regardless of a file's name
Sealed evaluation	Separately protected evaluation outside the learning and optimization closure
HWM	High-water mark: best retained result under a complete comparison identity

Term	Meaning in this repository
Teaching admission	Eligibility decision permitting a measured example into policy training
Motif / macro	Typed reusable program fragment with eligible provenance; not automatically a new primitive operator
Numerical policy	Declared arithmetic, probability, ordering, tolerance and runtime assumptions fixed before scoring
Alias	Alternate spelling of an existing recipe; not another independent method fit
Unsupported	A backend or task deliberately refuses a capability outside its admitted contract
Unavailable	A dependency or required runtime is missing; different from a model failure
Failed	An attempted operation ended unsuccessfully; retain its reason and measured/unknown costs
Skipped	A known test or operation did not execute; not passing coverage
Journal	Incremental durable started/finished/observation records for audit and original-window recovery
Receipt	Evidence binding command, source, inputs, runtime, resources, times, status and artifacts
Provenance	The chain of source, data, model, recipe and execution identities supporting an artifact
Elo	Relative Bradley–Terry rating on the arena's declared method universe and eligible pairwise scores
Warm demo	Demonstration after verified preparation; report separately from cold installation
Promotion	Explicit qualification of an exact artifact/model/backend for its intended serving use; a merge alone is insufficient

Questions that should remain answerable after a maintainer handoff

Which reviewed source and runtime produced this artifact? Which exact rows and original labels were available? Was this a recipe, a checkpoint or a saved fitted predictor? Which score selected it, under what comparison identity? Which failures, missing dependencies and cancelled jobs were retained? Did any nominal test labels influence selection? Did resume preserve the original deadline? Does the export predict from saved state, and which backend actually passed? Did training perform positive updates and change parameters? Can another maintainer locate and authenticate the receipt without reading credentials or sealed data?

The roadmap translates the unanswered parts into named acceptance items. Unknowns stay explicit until the corresponding evidence exists.

12 / Dataset statistics appendix

All statistics below use development rows only. Numeric values use finite observations; missing cells are counted separately. Standard deviation is the population statistic. Medians and the additional 1%, 5%, 25%, 75%, 95%, 99% quantiles in JSON linearly interpolate sorted values at `(n−1)p`. Category counts retain exact stored strings. Original class ordering and fold class counts are preserved. Upstream metadata, provenance URLs, source byte/hash pins, group-size histograms and all file identities remain in JSON.

197_cpu_act

Task `regression`; metric `rmse`; 6553 development rows; 1639 sealed rows counted from assignments; 21 numeric and 0 categorical features. Development groups: 6553; all groups: 8192; largest group: 1. Original fold sizes:[1311, 1311, 1311, 1310, 1310]; other-four-fold fit sizes:[5242, 5242, 5242, 5243, 5243]. Declared grouping columns: none.

Development file SHA-256: `c987b431681784b87f12f3c98e660fd0aa76f08d01fe005695f41470b3750019`. Assignment SHA-256: `624d44ee0a5e2214e8edf256f8d9c028ca3f3765237401e3c84cf9c232a60bc7`. Original source SHA-256: `d00fd6bac2eda0821a04ff10277663d211127745a19a474c0feee41a16914fdc`. Original license: unknown; repository license: MIT.

Development target: min 0, median 89, max 99, mean 84.01938, population SD 18.290468; missing 0.

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
lread	numeric	0	207	0	7	1845	19.349306	53.733421
lwrite	numeric	0	177	0	1	575	12.922173	29.496482
scall	numeric	0	3696	109	2053	12493	2307.3473	1631.4764
sread	numeric	0	752	6	166	5318	209.85686	199.84585
swrite	numeric	0	601	7	117	5456	149.81505	162.43571
fork	numeric	0	207	0	0.80000001	20.120001	1.8901007	2.4752517
exec	numeric	0	349	0	1.2	59.560001	2.7978773	5.2634111
rchar	numeric	0	6414	278	122879	2526649	195586.22	238487.78
wchar	numeric	0	6397	1498	46082	1801623	95334.943	140638.4
pgout	numeric	0	362	0	0	81.440002	2.2342881	5.2216254
ppgout	numeric	0	670	0	0	168.06	5.8141904	14.712169
pgfree	numeric	0	927	0	0	523	11.634935	31.594236
pgscan	numeric	0	1004	0	0	1237	21.142667	69.957532
atch	numeric	0	219	0	0	211.58	1.0821486	5.4985106
pgin	numeric	0	745	0	2.8	126.45	8.2493835	13.75562
ppgin	numeric	0	955	0	3.8099999	292.60999	12.334296	22.180491

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
pflt	numeric	0	2626	0	64.199997	899.79999	110.20706	114.35173
vflt	numeric	0	3310	0.2	121.56	1365	185.84375	190.4443
runqsz	numeric	0	253	1	2	2823	19.619182	128.21611
freemem	numeric	0	2757	55	579	12027	1768.3189	2497.6881
freeswap	numeric	0	6154	2	1137126	2042280	1328586	421591.38

225_puma8NH

Task `regression`; metric `rmse`; 6553 development rows; 1639 sealed rows counted from assignments; 8 numeric and 0 categorical features. Development groups: 6553; all groups: 8192; largest group: 1. Original fold sizes:[1311, 1311, 1311, 1310, 1310]; other-four-fold fit sizes:[5242, 5242, 5242, 5243, 5243]. Declared grouping columns: none.

Development file SHA-256: `31a4ce74a20613e192cab948518ab391feb0b70b0863fbe263d288d4a5df9c6e`. Assignment SHA-256: `f901eda7d22c08a8e7d6b8eb8455f21de4fc58eb4cb497657d2a56bd035cc8e1`. Original source SHA-256: `0be6203e167cc5e7b038368dbfe0a7790d5dc423c9d7e42887907c5c03f81c27`. Original license: unknown; repository license: MIT.

Development target: min-12.415322, median 1.054127, max 11.87619, mean 1.1382282, population SD 5.6448802; missing 0.

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
theta 1	numeric	0	6547	-1.884729	0.0060899998	1.884735	-0.0022002574	1.0823712
theta 2	numeric	0	6543	-1.8834831	0.016564	1.883306	0.010064784	1.0937572
theta 3	numeric	0	6548	-1.884449	0.023234	1.884046	0.0076206044	1.0832023
thetad 1	numeric	0	6548	-1.884632	-0.013532	1.884923	-0.007727921	1.0913692
thetad 2	numeric	0	6549	-1.8847851	-0.030689999	1.884077	-0.013800457	1.09539
thetad 3	numeric	0	6546	-1.884558	0.020455999	1.884746	0.007869032	1.0864734
tau 1	numeric	0	6534	-0.5999950 2	0.0047479998	0.599011	0.00079091185	0.34369033
tau 2	numeric	0	6533	-0.5998719 9	0.0048799999	0.59991199	0.0012599463	0.34712905

230_machine_cpu

Task `regression`; metric `rmse`; 169 development rows; 40 sealed rows counted from assignments; 6 numeric and 0 categorical features. Development groups: 152; all groups: 190; largest group: 5. Original fold sizes:[34, 32, 37, 33, 33]; other-four-fold fit sizes:[135,137,132,136,136]. Declared grouping columns: none.

Development file SHA-256: `07c9b448d18c0479cb817235efff5ec242fd3723329dcd69582d39043d150c87`. Assignment SHA-256: `68d86fbc518c72ee20dc5feefac8ad2e5a2f7c36ea9613b2755c7177f6f1bf82`. Original source SHA-256: `937b0a23076f948343b8b7ac005bc70ea7080c391218fb4896462699b8cb59c4`. Original license: unknown; repository license: MIT.

Development target: min 6, median 50, max 1150, mean 111.53254, population SD 169.51452; missing 0.

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
MYCT	numeric	0	56	17	110	1500	196.59172	248.93309
MMIN	numeric	0	20	64	2000	32000	2901.7751	3928.1762
MMAX	numeric	0	21	64	8000	64000	12481.811	12389.434
CACH	numeric	0	22	0	8	256	24.633136	38.093256
CHMIN	numeric	0	15	0	3	52	4.9940828	7.2483396
CHMAX	numeric	0	28	0	8	176	19.100592	27.884089

522_pm10

Task `regression`; metric `rmse`; 402 development rows; 98 sealed rows counted from assignments; 7 numeric and 0 categorical features. Development groups: 218; all groups: 273; largest group: 6. Original fold sizes:[81, 81, 91, 67, 82]; other-four-fold fit sizes:[321,321,311,335,320]. Declared grouping columns:['day'].

Development file SHA-256: `a33090c6e33323c48afd74338f72b1caa12376b6af4af00f7a48f6c03168df4c`. Assignment SHA-256: `13a00e1a058422110f6ff4caabdac7f648e5da976e10ed0a27be077534e469dd`. Original source SHA-256: `ecbabaa5a989a8c11f5a57e874c135f386da5379d6689ae6d5d4264a652fa5e0`. Original license: unknown; repository license: MIT.

Development target: min 0.69314998, median 3.3322001, max 5.3845, mean 3.3171902, population SD 0.85366052; missing 0.

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
cars_per_hour	numeric	0	374	3.8066599	7.4974751	8.3520803	7.0094372	1.0713765
temperature_at_2 m	numeric	0	197	-19	0.80000001	19.799999	0.51965173	6.3551238
wind_speed	numeric	0	71	0.40000001	2.8	9.3999996	3.1445274	1.7968553

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
temperature_diff_2m_25 m	numeric	0	56	-5	0	4	0.11666667	0.97697158
wind_direction	numeric	0	288	5	103.2	358	140.67065	84.378122
hour_of_day	numeric	0	24	1	12	24	12.266169	6.8120686
day	numeric	0	218	32	206.5	605	303.22139	197.99917

529_pollen

Task `regression`; metric `rmse`; 3078 development rows; 770 sealed rows counted from assignments; 4 numeric and 0 categorical features. Development groups: 3078; all groups: 3848; largest group: 1. Original fold sizes:[616,616,616,615,615]; other-four-fold fit sizes:[2462, 2462, 2462, 2463, 2463]. Declared grouping columns: none.

Development file SHA-256: `4afdc668b7c7bdde004b651e639ee62542518fa9629a4b9a9d18f42d2bfe5cf5`. Assignment SHA-256: `225ab47f4b84b8b29f808751a567dc06e49af57f77d918402dc98929494b9aad`. Original source SHA-256: `2e9414f0216e75296ba4211c567dd8a60d269273b38f8e00bd8fea3a37d2e654`. Original license: unknown; repository license: MIT.

Development target: min-12.0391, median-0.0282, max 10.4934, mean 0.013061306, population SD 3.1652511; missing 0.

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
RIDGE	numeric	0	3052	-23.283899	-0.1736	21.406601	-0.040160139	6.4118802
NUB	numeric	0	3047	-16.393499	-0.26875	17.258301	-0.017482649	5.1239751
CRACK	numeric	0	3057	-26.4652	-0.12075	25.6434	-0.07717969	7.9651552
WEIGHT	numeric	0	3067	-34.035198	-0.1534	35.802799	0.062779366	10.099992

579_fri_c0_250_5

Task `regression`; metric `rmse`; 200 development rows; 50 sealed rows counted from assignments; 5 numeric and 0 categorical features. Development groups: 200; all groups: 250; largest group: 1. Original fold sizes:[40, 40, 40, 40, 40]; other-four-fold fit sizes:[160,160,160,160,160]. Declared grouping columns: none.

Development file SHA-256: `9c2608cc2a14cb52e26f174518606fbc4bb7ed48b04aa4c62d840a7dc6cbd365`. Assignment SHA-256: `fa08e1237360b16d98c581d6c62695cb6c8631cca9940ba1671a6b8678e60790`. Original source SHA-256: `d32597bb1043df296ac3be1fff926646b26f268f89185310ea898e59b5fff896`. Original license: unknown; repository license: MIT.

Development target: min-2.2208376, median-0.050367098, max 2.1689749, mean-0.045378203, population SD 1.0368873; missing 0.

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
oz 1	numeric	0	200	-1.7117687	-0.040313881	1.7466414	-0.009772225	0.99221231
oz 2	numeric	0	200	-1.7675385	0.11001987	1.7166964	0.017360429	1.0225496
oz 3	numeric	0	200	-1.798439	-0.013159709	1.6682318	0.024478181	0.97661234
oz 4	numeric	0	200	-1.7556303	0.029955601	1.6519692	-0.027132471	0.97989126
oz 5	numeric	0	200	-1.8355519	0.022033866	1.7011706	-0.017172821	1.0096145

auto_insurance_price

Task `regression`; metric `rmse`; 162 development rows; 39 sealed rows counted from assignments; 13 numeric and 10 categorical features. Development groups: 154; all groups: 193; largest group: 2. Original fold sizes:[35, 32, 31, 31, 33]; other-four-fold fit sizes:[127,130,131,131,129]. Declared grouping columns: none.

Development file SHA-256: `6c0719a5e4433e653e8997fc1f19842fd3aae044531e7639484d24d879d7f971`. Assignment SHA-256: `9769b5f1510b1048afe81125f8f7dbb1272ab81825305af5414c8da6c52c7b1f`. Original source SHA-256: `244c05ca33384bcdd519ff88fad036ae7140a5438b76a8248401cdc065582ac7`. Original license: CC BY 4.0; repository license: MIT.

Development target: min 5118, median 10096.5, max 45400, mean 13127.222, population SD 8034.0417; missing 0.

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
make	categorical	0	21	n/a	n/a	n/a	n/a	n/a
fuel_type	categorical	0	2	n/a	n/a	n/a	n/a	n/a
aspiration	categorical	0	2	n/a	n/a	n/a	n/a	n/a
number_of_doors	categorical	0	2	n/a	n/a	n/a	n/a	n/a
body_style	categorical	0	5	n/a	n/a	n/a	n/a	n/a
drive_wheels	categorical	0	3	n/a	n/a	n/a	n/a	n/a
engine_location	categorical	0	2	n/a	n/a	n/a	n/a	n/a
wheel_base	numeric	0	49	86.6	96.9	120.9	98.664815	6.1873474
length	numeric	0	67	141.1	172.5	208.1	174.10123	12.321366
width	numeric	0	40	60.3	65.4	72	65.856173	2.1333027
height	numeric	0	46	47.8	54.1	59.8	53.730247	2.5110641
curb_weight	numeric	0	140	1488	2407.5	4066	2548.8457	518.36907

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
engine_type	categorical	0	6	n/a	n/a	n/a	n/a	n/a
number_of_cylinders	categorical	0	7	n/a	n/a	n/a	n/a	n/a
engine_size	numeric	0	40	61	110	326	126.39506	42.981816
fuel_system	categorical	0	7	n/a	n/a	n/a	n/a	n/a
bore	numeric	4	35	2.54	3.31	3.94	3.3218987	0.26890813
stroke	numeric	4	33	2.07	3.29	4.17	3.2515823	0.31914744
compression_ratio	numeric	0	28	7	9	23	10.003148	3.7645327
horsepower	numeric	1	52	48	95	262	103.68323	38.08872
peak_rpm	numeric	1	21	4150	5200	6600	5129.8137	480.66268
city_mpg	numeric	0	26	13	24.5	49	25.024691	6.2370365
highway_mpg	numeric	0	29	16	30	54	30.518519	6.6640942

Exact development category frequencies:

- `make`: `{"0": 3, "1": 6, "11": 11, "12": 16, "13": 9, "14": 5, "15": 4, "16": 1, "17": 3, "18": 12, "19": 25, "2": 5, "20": 11, "21": 8, "3": 2, "4": 6, "5": 11, "6": 2, "7": 3, "8": 13, "9": 6}`. Categories absent from the other four folds: `{"0": ["16"], "1": [], "2": [], "3": [], "4": []}`.
- `fuel_type`: `{"0": 14, "1": 148}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `aspiration`: `{"0": 134, "1": 28}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `number_of_doors`: `{"0": 93, "1": 69}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `body_style`: `{"0": 7, "1": 22, "2": 75, "3": 54, "4": 4}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `drive_wheels`: `{"0": 8, "1": 96, "2": 58}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `engine_location`: `{"0": 159, "1": 3}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `engine_type`: `{"0": 11, "1": 10, "2": 112, "3": 15, "4": 10, "5": 4}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `number_of_cylinders`: `{"0": 4, "1": 1, "2": 124, "3": 9, "4": 20, "5": 3, "6": 1}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": ["6"], "3": ["1"], "4": []}`.
- `fuel_system`: `{"0": 9, "1": 55, "2": 3, "3": 14, "5": 72, "6": 8, "7": 1}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": ["7"], "4": []}`.

690_visualizing_galaxy

Task `regression`; metric `rmse`; 258 development rows; 65 sealed rows counted from assignments; 4 numeric and 0 categorical features. Development groups: 258; all groups: 323; largest group: 1. Original fold sizes:[52, 52, 52, 51, 51]; other-four-fold fit sizes:[206,206,206,207,207]. Declared grouping columns: none.

Development file SHA-256: `5c060a8e2459df5e8a533003febab98755a531f3bce0426ac1b617edf2445659`. Assignment SHA-256: `d3f63cf22ce07441cdb3e01d77160d3c36a6630c99c5e7533f9b5563e06db892`. Original source SHA-256: `5fe909e08a622224135a2dcfd409bbe10abab4919f917a07b771c3fb00580852`. Original license: unknown; repository license: MIT.

Development target: min 1409, median 1581, max 1769, mean 1588.6085, population SD 91.295658; missing 0.

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
eastwest	numeric	0	258	-29.666929	-0.42766218	29.48414	-0.60256838	12.015002
northsouth	numeric	0	258	-49.107689	1.8613807	49.888725	2.4837183	22.042518
angle	numeric	0	7	12.5	102.5	133	81.486434	38.60098
radialposition	numeric	0	245	-51.099998	-1.8	55.700001	-1.5406977	25.187122

712_chscase_geyser1

Task `regression`; metric `rmse`; 172 development rows; 50 sealed rows counted from assignments; 2 numeric and 0 categorical features. Development groups: 135; all groups: 169; largest group: 4. Original fold sizes:[32, 34, 38, 35, 33]; other-four-fold fit sizes:[140,138,134,137,139]. Declared grouping columns: none.

Development file SHA-256: `a00344615df0ba8b22227753d6931e72c039bed451500c74c794a91038f63459`. Assignment SHA-256: `161ae799380a7919870d5ae493d2c026c6f36fda9fc2dae192ad5f9009fb205b`. Original source SHA-256: `9231e5c6bb08d6a7fab8509e4a8330063a5e3da77b75ff3b9dc412557ed3e334`. Original license: unknown; repository license: MIT.

Development target: min 43, median 75, max 95, mean 71.098837, population SD 12.668703; missing 0.

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
col_1	numeric	0	16	1	16	23	12.209302	7.6502261
col_2	numeric	0	33	1.7	4.05	5.1999998	3.5883721	1.0771861

503_wind

Task `regression`; metric `rmse`; 5238 development rows; 1336 sealed rows counted from assignments; 14 numeric and 0 categorical features. Development groups: 172; all groups: 216; largest group: 31. Original fold sizes:[1066, 1074, 1030, 1032, 1036]; other-four-fold fit sizes:[4172, 4164, 4208, 4206, 4202]. Declared grouping columns:['year', 'month'].

Development file SHA-256: `3bb9b2400ba47561c0df144b7a0d2b8583ad589425d3ada4d0c6313c331d4fd2`. Assignment SHA-256: `1cc0ecb4f63ebddab23f309899171f350164ff1b91b81c144e32ad57640cf281`. Original source SHA-256: `2cc1e040bdc20737674a00039cd80e16758c511e697e524ed17e4335ec93e9a6`. Original license: unknown; repository license: MIT.

Development target: min 1.75, median 14.88, max 42.540001, mean 15.517566, population SD 6.6603293; missing 0.

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
year	numeric	0	18	61	70	78	69.660939	5.171068
month	numeric	0	12	1	6	12	6.358152	3.3938973
day	numeric	0	31	1	16	31	15.737686	8.8047809
RPT	numeric	0	644	0.67000002	11.63	35.799999	12.284956	5.5789555
VAL	numeric	0	594	0.37	10.08	33.369999	10.579704	5.2260336
ROS	numeric	0	589	1.5	10.83	33.84	11.576588	4.9764516
KIL	numeric	0	439	0	5.71	28.459999	6.2592974	3.5713878
SHA	numeric	0	575	0.13	9.9200001	33.630001	10.369643	4.8910479
BIR	numeric	0	446	0	6.79	26.040001	7.0677301	3.9515225
DUB	numeric	0	565	0	9.0799999	30.370001	9.6849523	4.9379271
CLA	numeric	0	520	0	8.04	31.08	8.4148511	4.4694644
MUL	numeric	0	490	0	8.1199999	25.799999	8.4560596	4.1545162
CLO	numeric	0	520	0.039999999	8.21	28.209999	8.6498683	4.5013668
BEL	numeric	0	664	0.57999998	12.42	42.380001	13.044351	5.8033753

breast_cancer_wisconsin_diagnostic

Task `binary`; metric `auroc`; 455 development rows; 114 sealed rows counted from assignments; 30 numeric and 0 categorical features. Development groups: 455; all groups: 569; largest group: 1. Original fold sizes:[91, 91, 91, 91, 91]; other-four-fold fit sizes:[364,364,364,364,364]. Declared grouping columns: none.

Development file SHA-256: `cc8bdc81b49bef92ae2aee74a4e919ea5d2515465adef12b7ebae53733e33fc6`. Assignment SHA-256: `41c454f412740e2996d3b7894af2e6d7c75042bdae9db222d14427e1ca7fb64c`. Original source SHA-256: `b3845ddf3afff808cd6449daa546caf4d8b18f109b7ebb2344990e4c37fea291`. Original license: CC BY 4.0; repository license: MIT.

Frozen class order: `["0", "1"]`. Development counts: `{"0": 284, "1": 171}`. Counts by original fold: `{"0": {"0": 55, "1": 36}, "1": {"0": 58, "1": 33}, "2": {"0": 61, "1": 30}, "3": {"0": 57, "1": 34}, "4": {"0": 53, "1": 38}}`.

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
radius_mean	numeric	0	379	6.981	13.37	28.11	14.128822	3.5074831
texture_mean	numeric	0	395	9.71	18.83	39.28	19.337451	4.3409678
perimeter_mean	numeric	0	423	43.79	86.18	188.5	91.964747	24.188136
area_mean	numeric	0	437	143.5	551.1	2501	654.93604	347.93468
smoothness_mean	numeric	0	386	0.05263	0.09578	0.1634	0.096036352	0.014378444
compactness_mean	numeric	0	433	0.01938	0.09263	0.3454	0.10405275	0.053448375
concavity_mean	numeric	0	432	0	0.06154	0.4268	0.088762678	0.080580341
n_concave_mean	numeric	0	432	0	0.03326	0.2012	0.048554477	0.038664424
symmetry_mean	numeric	0	358	0.106	0.1784	0.304	0.18096022	0.028034982
fractal_dimension_mean	numeric	0	412	0.04996	0.06161	0.09744	0.062755077	0.00713061
radius_se	numeric	0	438	0.1144	0.3186	2.873	0.40416242	0.28494741
texture_se	numeric	0	421	0.3602	1.127	4.885	1.2224974	0.56657768
perimeter_se	numeric	0	432	0.757	2.244	21.98	2.8611211	2.0984043
area_se	numeric	0	428	6.802	24.25	542.2	40.441947	47.366591
smoothness_se	numeric	0	442	0.001713	0.006471	0.03113	0.007061817 6	0.003084035 3
compactness_se	numeric	0	438	0.002252	0.02048	0.1354	0.025522848	0.017892893
concavity_se	numeric	0	431	0	0.02586	0.396	0.03220903	0.031603666
n_concave_se	numeric	0	409	0	0.01083	0.05279	0.011810613	0.006361560 3
symmetry_se	numeric	0	407	0.007882	0.01868	0.07895	0.020614919	0.008722101 8
fractal_dimension_se	numeric	0	437	0.0008948	0.003224	0.02984	0.003788150 1	0.002449625 4
radius_worst	numeric	0	387	7.93	14.98	36.04	16.256743	4.7838454

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
texture_worst	numeric	0	420	12.02	25.4	49.54	25.688879	6.1068704
perimeter_worst	numeric	0	419	50.41	97.96	251.2	107.09422	33.240634
area_worst	numeric	0	440	185.2	688.6	4254	878.64	556.25614
smoothness_worst	numeric	0	352	0.07117	0.1312	0.2226	0.13195897	0.023257748
compactness_worst	numeric	0	429	0.02729	0.2113	1.058	0.25405004	0.15864571
concavity_worst	numeric	0	432	0	0.2267	1.252	0.27054573	0.20677617
n_concave_worst	numeric	0	396	0	0.09861	0.291	0.11400506	0.065796181
symmetry_worst	numeric	0	408	0.1565	0.281	0.6638	0.2903422	0.064265468
fractal_dimension_worst	numeric	0	430	0.05504	0.07999	0.2075	0.084133473	0.018414541

credit_approval_germany

Task `binary`; metric `auroc`; 800 development rows; 200 sealed rows counted from assignments; 3 numeric and 17 categorical features. Development groups: 800; all groups: 1000; largest group: 1. Original fold sizes:[160,160,160,160,160]; other-four-fold fit sizes:[640,640,640,640,640]. Declared grouping columns: none.

Development file SHA-256: `34ffaac6a51ea0ac2e153ba828ce6b2437f685f6d181bb175356a1285c36ef7d`. Assignment SHA-256: `42532d431f177698ca65270388a542d6a687c9634648d523d601ea7763e5a7c4`. Original source SHA-256: `93e819f4b2d26df1cd5eb9805bd2fe15767941c6c8d66c58eba86d9249cb8ce1`. Original license: CC BY 4.0; repository license: MIT.

Frozen class order: `["0", "1"]`. Development counts: `{"0": 552, "1": 248}`. Counts by original fold: `{"0": {"0": 109, "1": 51}, "1": {"0": 107, "1": 53}, "2": {"0": 118, "1": 42}, "3": {"0": 115, "1": 45}, "4": {"0": 103, "1": 57}}`.

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
credit_height	numeric	0	752	276	2330	18424	3317.4475	2867.9399
credit_duration	numeric	0	30	4	18	72	20.98125	12.204237
purpose	categorical	0	10	n/a	n/a	n/a	n/a	n/a
credit_guarantee	categorical	0	3	n/a	n/a	n/a	n/a	n/a
rate_fraction_income	categorical	0	4	n/a	n/a	n/a	n/a	n/a

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
credit_history	categorical	0	5	n/a	n/a	n/a	n/a	n/a
existing_credits_at_bank	categorical	0	4	n/a	n/a	n/a	n/a	n/a
other_credits	categorical	0	3	n/a	n/a	n/a	n/a	n/a
age	numeric	0	53	19	34	75	35.77375	11.346919
personal_status	categorical	0	4	n/a	n/a	n/a	n/a	n/a
dependents	categorical	0	2	n/a	n/a	n/a	n/a	n/a
employment_type	categorical	0	4	n/a	n/a	n/a	n/a	n/a
employment_length	categorical	0	5	n/a	n/a	n/a	n/a	n/a
foreign_worker	categorical	0	2	n/a	n/a	n/a	n/a	n/a
housing_status	categorical	0	3	n/a	n/a	n/a	n/a	n/a
residence_duration	categorical	0	4	n/a	n/a	n/a	n/a	n/a
checking_account_status	categorical	0	4	n/a	n/a	n/a	n/a	n/a
savings_account_status	categorical	0	5	n/a	n/a	n/a	n/a	n/a
assets	categorical	0	4	n/a	n/a	n/a	n/a	n/a
telephone	categorical	0	2	n/a	n/a	n/a	n/a	n/a

Exact development category frequencies:

- `purpose`: `{"0": 181, "1": 78, "2": 151, "3": 226, "4": 9, "5": 15, "6": 41, "7": 8, "8": 81, "9": 10}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `credit_guarantee`: `{"0": 726, "1": 37, "2": 37}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `rate_fraction_income`: `{"0": 108, "1": 177, "2": 131, "3": 384}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `credit_history`: `{"0": 31, "1": 38, "2": 424, "3": 72, "4": 235}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `existing_credits_at_bank`: `{"0": 499, "1": 274, "2": 23, "3": 4}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `other_credits`: `{"0": 112, "1": 38, "2": 650}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.

- `personal_status`: `{"0": 41, "1": 240, "2": 447, "3": 72}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `dependents`: `{"0": 671, "1": 129}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `employment_type`: `{"0": 16, "1": 166, "2": 492, "3": 126}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `employment_length`: `{"0": 51, "1": 131, "2": 261, "3": 145, "4": 212}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `foreign_worker`: `{"0": 768, "1": 32}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `housing_status`: `{"0": 144, "1": 571, "2": 85}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `residence_duration`: `{"0": 104, "1": 243, "2": 113, "3": 340}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `checking_account_status`: `{"0": 212, "1": 211, "2": 58, "3": 319}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `savings_account_status`: `{"0": 485, "1": 84, "2": 54, "3": 33, "4": 144}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `assets`: `{"0": 226, "1": 187, "2": 264, "3": 123}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `telephone`: `{"0": 477, "1": 323}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.

congressional_voting_records

Task `binary`; metric `auroc`; 347 development rows; 88 sealed rows counted from assignments; 0 numeric and 16 categorical features. Development groups: 274; all groups: 342; largest group: 8. Original fold sizes:[77, 57, 64, 84, 65]; other-four-fold fit sizes:[270,290,283,263,282]. Declared grouping columns: none.

Development file SHA-256: `77382af6acc208516bcf44d9ff2911848b90c9893eca5499353e79dd7aa98c0c`. Assignment SHA-256: `3a38db7f711af084a16a69625d5ac782b2591fd2aa0682f0074afaa39bdc45c1`. Original source SHA-256: `fbee2fea90f1e1ce3251869a84b87723e37ab6a512c515a4fb6b9f9d72e5ee0c`. Original license: CC BY 4.0; repository license: MIT.

Frozen class order: `["0", "1"]`. Development counts: `{"0": 209, "1": 138}`. Counts by original fold: `{"0": {"0": 58, "1": 19}, "1": {"0": 33, "1": 24}, "2": {"0": 36, "1": 28}, "3": {"0": 44, "1": 40}, "4": {"0": 38, "1": 27}}`.

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
handicapped_infants	categorical	0	3	n/a	n/a	n/a	n/a	n/a
water_project_cost_sharing	categorical	0	3	n/a	n/a	n/a	n/a	n/a
adoption_of_the_budget_resolution	categorical	0	3	n/a	n/a	n/a	n/a	n/a

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
physician_fee_freeze	categorical	0	3	n/a	n/a	n/a	n/a	n/a
el_salvador_aid	categorical	0	3	n/a	n/a	n/a	n/a	n/a
religious_groups_in_schools	categorical	0	3	n/a	n/a	n/a	n/a	n/a
anti_satellite_test_ban	categorical	0	3	n/a	n/a	n/a	n/a	n/a
aid_to_nicaraguan_contras	categorical	0	3	n/a	n/a	n/a	n/a	n/a
mx_missile	categorical	0	3	n/a	n/a	n/a	n/a	n/a
immigration	categorical	0	3	n/a	n/a	n/a	n/a	n/a
synfuels_corporation_cutback	categorical	0	3	n/a	n/a	n/a	n/a	n/a
education_spending	categorical	0	3	n/a	n/a	n/a	n/a	n/a
superfund_right_to_sue	categorical	0	3	n/a	n/a	n/a	n/a	n/a
crime	categorical	0	3	n/a	n/a	n/a	n/a	n/a
duty_free_exports	categorical	0	3	n/a	n/a	n/a	n/a	n/a
export_administation_act_south_africa	categorical	0	3	n/a	n/a	n/a	n/a	n/a

Exact development category frequencies:

- `handicapped_infants`: `{"0": 142, "1": 194, "2": 11}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `water_project_cost_sharing`: `{"0": 157, "1": 160, "2": 30}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `adoption_of_the_budget_resolution`: `{"0": 200, "1": 138, "2": 9}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `physician_fee_freeze`: `{"0": 147, "1": 191, "2": 9}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `el_salvador_aid`: `{"0": 177, "1": 157, "2": 13}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `religious_groups_in_schools`: `{"0": 224, "1": 113, "2": 10}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `anti_satellite_test_ban`: `{"0": 185, "1": 151, "2": 11}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.

- `aid_to_nicaraguan_contras`: `{"0": 192, "1": 144, "2": 11}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `mx_missile`: `{"0": 164, "1": 168, "2": 15}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `immigration`: `{"0": 176, "1": 166, "2": 5}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `synfuels_corporation_cutback`: `{"0": 123, "1": 211, "2": 13}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `education_spending`: `{"0": 142, "1": 182, "2": 23}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `superfund_right_to_sue`: `{"0": 169, "1": 159, "2": 19}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `crime`: `{"0": 207, "1": 127, "2": 13}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `duty_free_exports`: `{"0": 132, "1": 195, "2": 20}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `export_administation_act_south_africa`: `{"0": 219, "1": 53, "2": 75}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.

haberman

Task `binary`; metric `auroc`; 245 development rows; 61 sealed rows counted from assignments; 3 numeric and 0 categorical features. Development groups: 226; all groups: 283; largest group: 3. Original fold sizes:[49, 49, 46, 50, 51]; other-four-fold fit sizes:[196,196,199,195,194]. Declared grouping columns: none.

Development file SHA-256: `6261779f619ecbdf9184c49da9fae7bb4f8bbcf1ffadffd51637db2673407b3a`. Assignment SHA-256: `705d871416d7b284d96331ead05b1bf138ba0669923cbe7e28668cbc63c6d56e`. Original source SHA-256: `2f44ac12e09d2fd09d9ac70231cecbfbc333356339dec0509ad1ed96e36e94c0`. Original license: unknown; repository license: MIT.

Frozen class order: `["1", "2"]`. Development counts: `{"1": 181, "2": 64}`. Counts by original fold: `{"0": {"1": 36, "2": 13}, "1": {"1": 36, "2": 13}, "2": {"1": 33, "2": 13}, "3": {"1": 38, "2": 12}, "4": {"1": 38, "2": 13}}`.

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
Age_of_patient_at_time_of_operation	numeric	0	47	30	52	83	52.665306	10.685973
Patients_year_of_operation	numeric	0	12	58	63	69	62.857143	3.3036776
Number_of_positive_axillary_nodes_detected	numeric	0	28	0	1	46	3.9346939	6.6807537

ionosphere

Task `binary`; metric `auroc`; 281 development rows; 70 sealed rows counted from assignments; 32 numeric and 2 categorical features. Development groups: 280; all groups: 350; largest group: 2. Original fold sizes:[55, 57, 56, 57, 56]; other-four-fold fit sizes:[226,224,225,224,225]. Declared grouping columns: none.

Development file SHA-256: `9dffeb4dac21ce8104c8bb98f1f96968b1d6262c0c47bd2676293a615ef4bf16`. Assignment SHA-256: `73798005d06153b5109154cc728c53e95fe027f65fd548079e364750c38e7f50`. Original source SHA-256: `adb6fc877404f2b55f6d59fc7322265c733f5cf0ae6d7c2e52c3468733630023`. Original license: unknown; repository license: MIT.

Frozen class order: `["0", "1"]`. Development counts: `{"0": 102, "1": 179}`. Counts by original fold: `{"0": {"0": 21, "1": 34}, "1": {"0": 26, "1": 31}, "2": {"0": 18, "1": 38}, "3": {"0": 19, "1": 38}, "4": {"0": 18, "1": 38}}`.

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
0	categorical	0	2	n/a	n/a	n/a	n/a	n/a
1	categorical	0	1	n/a	n/a	n/a	n/a	n/a
2	numeric	0	174	-1	0.88103	1	0.66155744	0.47010906
3	numeric	0	215	-1	0.00892	1	0.028568683	0.44112987
4	numeric	0	160	-1	0.82683	1	0.60080947	0.53074655
5	numeric	0	206	-1	0.0127	1	0.089482954	0.46367215
6	numeric	0	180	-1	0.76001	1	0.55640886	0.49881365
7	numeric	0	204	-1	0.01128	1	0.099825658	0.52851044
8	numeric	0	194	-1	0.71875	1	0.54520381	0.48753108
9	numeric	0	211	-1	0.01681	1	0.17432249	0.47797457
10	numeric	0	197	-1	0.68627	1	0.49457996	0.55173223
11	numeric	0	212	-1	0.02825	1	0.16022107	0.49491547
12	numeric	0	193	-1	0.67028	1	0.4104532	0.61450019
13	numeric	0	215	-1	0.02105	1	0.080409858	0.50169559
14	numeric	0	188	-1	0.62936	1	0.35983819	0.64793776
15	numeric	0	214	-1	0	1	0.077333772	0.47171614
16	numeric	0	203	-1	0.62121	1	0.39399758	0.61580948
17	numeric	0	226	-1	0	1	0.0025851957	0.49900626
18	numeric	0	203	-1	0.61996	1	0.39353537	0.61746332
19	numeric	0	213	-1	0	1	-0.020650036	0.52258519

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
20	numeric	0	198	-1	0.52494	1	0.3334358	0.61650624
21	numeric	0	211	-1	0	1	0.014242954	0.51554817
22	numeric	0	200	-1	0.53246	1	0.35669872	0.61030843
23	numeric	0	209	-1	0	1	-0.036835658	0.53328823
24	numeric	0	206	-1	0.55389	1	0.39690103	0.57383195
25	numeric	0	218	-1	-0.01664	1	-0.065681388	0.50686043
26	numeric	0	206	-1	0.69811	1	0.53114655	0.51992115
27	numeric	0	227	-1	-0.02294	1	-0.051712669	0.5482146
28	numeric	0	196	-1	0.48348	1	0.36189975	0.58595804
29	numeric	0	212	-1	0	1	-0.03398274	0.51497855
30	numeric	0	199	-1	0.45114	1	0.34863491	0.56760128
31	numeric	0	209	-1	0	1	-0.014466975	0.51454688
32	numeric	0	198	-1	0.40445	1	0.33919872	0.52562623
33	numeric	0	210	-1	0	1	0.0074272242	0.4767589

Exact development category frequencies:

- `0`: `{"0": 30, "1": 251}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `1`: `{"0": 281}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.

sonar

Task `binary`; metric `auroc`; 166 development rows; 42 sealed rows counted from assignments; 60 numeric and 0 categorical features. Development groups: 166; all groups: 208; largest group: 1. Original fold sizes:[34, 33, 33, 33, 33]; other-four-fold fit sizes:[132,133,133,133,133]. Declared grouping columns: none.

Development file SHA-256: `413956c1d626d6671e4b2e99f06ef3c9dbeab52cb947aa2dda52c5eb4d590ec4`. Assignment SHA-256: `dce5e4eb3541c056cab75d72b0be5212d4788bfec249a54a6ffe6160e82853f9`. Original source SHA-256: `64378cabaecdb50e2e1e644c1d6bc359f0627ebe313870ad09e1e41ed972529b`. Original license: unknown; repository license: MIT.

Frozen class order: `["0", "1"]`. Development counts: `{"0": 88, "1": 78}`. Counts by original fold: `{"0": {"0": 19, "1": 15}, "1": {"0": 20, "1": 13}, "2": {"0": 16, "1": 17}, "3": {"0": 15, "1": 18}, "4": {"0": 18, "1": 15}}`.

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
A1	numeric	0	141	0.0015	0.0217	0.1371	0.028841566	0.023686115
A2	numeric	0	151	0.0006	0.0286	0.2339	0.037833735	0.034101163
A3	numeric	0	157	0.0015	0.0321	0.3059	0.042960843	0.037819594
A4	numeric	0	152	0.0058	0.04405	0.4264	0.053654217	0.046923156
A5	numeric	0	157	0.0067	0.0625	0.401	0.075328916	0.055578143
A6	numeric	0	159	0.0102	0.0898	0.3823	0.10617169	0.061505341
A7	numeric	0	159	0.0033	0.11245	0.3729	0.12627711	0.064775691
A8	numeric	0	162	0.0055	0.1117	0.459	0.13576145	0.088328103
A9	numeric	0	164	0.0075	0.15375	0.6828	0.18029096	0.12123601
A10	numeric	0	166	0.0113	0.1809	0.7106	0.2100741	0.138032
A11	numeric	0	162	0.0289	0.2273	0.7342	0.23626867	0.13285597
A12	numeric	0	165	0.0259	0.24905	0.706	0.24793976	0.13719085
A13	numeric	0	161	0.0184	0.25735	0.7131	0.27112349	0.13802798
A14	numeric	0	165	0.0273	0.2811	0.997	0.29665482	0.16533271
A15	numeric	0	163	0.0031	0.27805	0.9137	0.32257771	0.20514931
A16	numeric	0	162	0.0162	0.31505	0.9751	0.38494157	0.23299201
A17	numeric	0	161	0.0349	0.3242	1	0.42208735	0.26767643
A18	numeric	0	163	0.0375	0.3712	1	0.45387711	0.26423326
A19	numeric	0	164	0.0746	0.42215	1	0.49631566	0.25898032
A20	numeric	0	162	0.0656	0.4977	1	0.54843916	0.26497105
A21	numeric	0	162	0.0512	0.58605	1	0.59400783	0.25705453
A22	numeric	0	162	0.0219	0.65335	1	0.61085241	0.2582013
A23	numeric	0	157	0.0563	0.66805	1	0.63666627	0.24875836
A24	numeric	0	159	0.0239	0.68315	1	0.6634747	0.23756303
A25	numeric	0	159	0.024	0.7211	1	0.6719988	0.24365831
A26	numeric	0	155	0.0921	0.7545	1	0.69583916	0.23788158
A27	numeric	0	155	0.0481	0.7211	1	0.7011488	0.24248268
A28	numeric	0	155	0.0284	0.7337	1	0.69982289	0.2339089
A29	numeric	0	158	0.0144	0.6808	1	0.64515843	0.23642261
A30	numeric	0	162	0.1354	0.61945	1	0.58621928	0.21518403

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
A31	numeric	0	166	0.0482	0.5037	0.9657	0.51252169	0.21079284
A32	numeric	0	164	0.0404	0.44535	0.9306	0.44586867	0.20698438
A33	numeric	0	163	0.0477	0.39375	0.9708	0.42063614	0.19916545
A34	numeric	0	165	0.0212	0.3682	0.9647	0.40769639	0.23003905
A35	numeric	0	163	0.0223	0.3369	1	0.40306205	0.2611019
A36	numeric	0	164	0.0185	0.32115	1	0.3937	0.27059015
A37	numeric	0	166	0.0379	0.3103	0.9497	0.37263133	0.24345705
A38	numeric	0	164	0.0383	0.3153	1	0.34736988	0.21649059
A39	numeric	0	163	0.0407	0.28345	0.9857	0.33274458	0.20047974
A40	numeric	0	164	0.0202	0.28225	0.9297	0.32078976	0.17907208
A41	numeric	0	164	0.0438	0.26305	0.8995	0.29696867	0.17064681
A42	numeric	0	166	0.0056	0.2534	0.8246	0.27926265	0.17015944
A43	numeric	0	163	0.0159	0.2162	0.7733	0.24302651	0.14246534
A44	numeric	0	157	0.0255	0.1751	0.7762	0.21003795	0.13362077
A45	numeric	0	165	0.0046	0.148	0.7034	0.19582169	0.1514589
A46	numeric	0	161	0.008	0.1179	0.7292	0.16208976	0.13726397
A47	numeric	0	163	0.0073	0.10385	0.5522	0.12454398	0.089007856
A48	numeric	0	162	0.0052	0.0803	0.3339	0.09448494	0.063893903
A49	numeric	0	154	0.0073	0.0451	0.1981	0.053189157	0.035846172
A50	numeric	0	132	0.0006	0.0179	0.0825	0.020369277	0.013156943
A51	numeric	0	134	0.0009	0.014	0.1004	0.015896386	0.01129344
A52	numeric	0	127	0.0008	0.01125	0.0709	0.013522289	0.009936481 3
A53	numeric	0	117	0.0005	0.00955	0.039	0.010813855	0.007042289 7
A54	numeric	0	114	0.0011	0.0094	0.0352	0.010825904	0.007082550 5
A55	numeric	0	113	0.0006	0.0076	0.0447	0.009561445 8	0.007155254 7
A56	numeric	0	108	0.0004	0.00705	0.0394	0.008278915 7	0.005786719 4
A57	numeric	0	105	0.0003	0.00575	0.0355	0.007623494	0.005725802 1
A58	numeric	0	115	0.0003	0.0059	0.044	0.008021084 3	0.006577964 4

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
A59	numeric	0	103	0.0001	0.0061	0.0332	0.0079795181	0.0061732079
A60	numeric	0	98	0.0007	0.0053	0.0439	0.0065289157	0.0052317043

spambase

Task `binary`; metric `auroc`; 3723 development rows; 878 sealed rows counted from assignments; 57 numeric and 0 categorical features. Development groups: 3381; all groups: 4207; largest group: 69. Original fold sizes:[696,781,710,789,747]; other-four-fold fit sizes:[3027, 2942, 3013, 2934, 2976]. Declared grouping columns: none.

Development file SHA-256: `f02d0bf948106d71bdc4dd57a1acad05e36935396d3ce2450aa326e863d93748`. Assignment SHA-256: `d740e152f7c9942115e88fe36274779a4187bc3cdafa050d75d75ca207804991`. Original source SHA-256: `735e8d43f7ee78ac54ca17404510e95f4c8b364a721e4d9532a1b3221ed70961`. Original license: unknown; repository license: MIT.

Frozen class order: `["0", "1"]`. Development counts: `{"0": 2261, "1": 1462}`. Counts by original fold: `{"0": {"0": 413, "1": 283}, "1": {"0": 486, "1": 295}, "2": {"0": 422, "1": 288}, "3": {"0": 489, "1": 300}, "4": {"0": 451, "1": 296}}`.

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
0	numeric	0	136	0	0	4.34	0.10836691	0.31447626
1	numeric	0	158	0	0	14.28	0.23424926	1.4140581
2	numeric	0	208	0	0	5.1	0.28450175	0.50613967
3	numeric	0	39	0	0	42.81	0.076145582	1.53401
4	numeric	0	238	0	0	8.33	0.30181574	0.63247696
5	numeric	0	134	0	0	5.88	0.099076014	0.28457055
6	numeric	0	159	0	0	7.27	0.1084448	0.37299389
7	numeric	0	154	0	0	11.11	0.10242815	0.40766094
8	numeric	0	136	0	0	5.26	0.090405587	0.28227639
9	numeric	0	233	0	0	18.18	0.24231534	0.66893504
10	numeric	0	106	0	0	2.61	0.057700779	0.19908542
11	numeric	0	301	0	0.12	9.67	0.53767123	0.85542489
12	numeric	0	147	0	0	5.55	0.094139135	0.29447118
13	numeric	0	119	0	0	10	0.057160892	0.3424115

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
14	numeric	0	105	0	0	4.41	0.050021488	0.26409551
15	numeric	0	238	0	0	20	0.25256245	0.86568242
16	numeric	0	178	0	0	7.14	0.14358313	0.44016113
17	numeric	0	209	0	0	7.69	0.17846092	0.49965964
18	numeric	0	549	0	1.3	18.75	1.6304647	1.7444236
19	numeric	0	133	0	0	18.18	0.087045393	0.51437355
20	numeric	0	388	0	0.23	11.11	0.82221595	1.2353801
21	numeric	0	81	0	0	17.1	0.12898469	1.0758037
22	numeric	0	157	0	0	5.45	0.10662369	0.36628547
23	numeric	0	133	0	0	12.5	0.088568359	0.37568098
24	numeric	0	355	0	0	20.83	0.53969917	1.6402831
25	numeric	0	251	0	0	16.66	0.2710744	0.91077722
26	numeric	0	221	0	0	33.33	0.87795326	3.6989178
27	numeric	0	179	0	0	5.88	0.12094547	0.51186153
28	numeric	0	140	0	0	14.28	0.09994628	0.61141527
29	numeric	0	167	0	0	5.88	0.10789954	0.47740186
30	numeric	0	113	0	0	12.5	0.068775181	0.4277239
31	numeric	0	90	0	0	4.76	0.049648133	0.34482243
32	numeric	0	167	0	0	8.33	0.091326887	0.48361691
33	numeric	0	97	0	0	4.76	0.050577491	0.34585255
34	numeric	0	161	0	0	20	0.10762557	0.5614855
35	numeric	0	151	0	0	4.76	0.099151222	0.40425707
36	numeric	0	182	0	0	6.89	0.13856299	0.43320226
37	numeric	0	46	0	0	8.33	0.014488316	0.24219157
38	numeric	0	143	0	0	11.11	0.07558152	0.42405656
39	numeric	0	116	0	0	4.76	0.067894171	0.36984345
40	numeric	0	89	0	0	7.14	0.045412302	0.37629292
41	numeric	0	166	0	0	14.28	0.12858716	0.77234937

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
42	numeric	0	127	0	0	3.57	0.045417674	0.21994189
43	numeric	0	143	0	0	20	0.080515713	0.65927373
44	numeric	0	220	0	0	21.42	0.29374161	0.99304581
45	numeric	0	202	0	0	22.05	0.18154714	0.94847185
46	numeric	0	34	0	0	2.17	0.0059629331	0.082484296
47	numeric	0	93	0	0	5	0.025753425	0.19009714
48	numeric	0	280	0	0	4.385	0.03976014	0.24401967
49	numeric	0	594	0	0.065	9.752	0.14165941	0.28547001
50	numeric	0	203	0	0	4.081	0.017885039	0.11686678
51	numeric	0	867	0	0	32.478	0.26717996	0.80078459
52	numeric	0	456	0	0	6.003	0.07593285	0.24670478
53	numeric	0	279	0	0	19.829	0.046236637	0.46825088
54	numeric	0	1867	1	2.285	1102.5	5.2333	32.564591
55	numeric	0	244	1	15	9989	53.011281	208.61383
56	numeric	0	827	1	97	15841	282.47059	604.67923

phoneme

Task `binary`; metric `auroc`; 4330 development rows; 1074 sealed rows counted from assignments; 5 numeric and 0 categorical features. Development groups: 4279; all groups: 5349; largest group: 2. Original fold sizes:[866,867,866,870,861]; other-four-fold fit sizes:[3464, 3463, 3464, 3460, 3469]. Declared grouping columns: none.

Development file SHA-256: `bbb0adca013fc84e710901a2a7ad7523000c351e38c9e041cbd9bef125315c3e`. Assignment SHA-256: `0b377d73783b9e26995fe4a821dfefc745d84316f7f457bcecf4985bb832e0f2`. Original source SHA-256: `2e1c31a2c560f3539be5bda1084939765c80ddd814bd6919fd8a3465f9b8aee6`. Original license: unknown; repository license: MIT.

Frozen class order: `["0", "1"]`. Development counts: `{"0": 3048, "1": 1282}`. Counts by original fold: `{"0": {"0": 610, "1": 256}, "1": {"0": 623, "1": 244}, "2": {"0": 583, "1": 283}, "3": {"0": 623, "1": 247}, "4": {"0": 609, "1": 252}}`.

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
Aa	numeric	0	1851	-1.7	0.4865	4.107	0.82364804	0.86789971
Ao	numeric	0	2210	-1.327	1.068	4.378	1.2516467	0.85489827

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
Dcl	numeric	0	2277	-1.641	0.716	3.155	0.76292702	0.92614533
Iy	numeric	0	2152	-1.494	0.286	2.785	0.40085681	0.79622039
Sh	numeric	0	1607	-1.246	-0.044	2.719	0.079781524	0.57446239

tic_tac_toe

Task `binary`; metric `auroc`; 766 development rows; 192 sealed rows counted from assignments; 0 numeric and 9 categorical features. Development groups: 766; all groups: 958; largest group: 1. Original fold sizes:[154,153,153,153,153]; other-four-fold fit sizes:[612,613,613,613,613]. Declared grouping columns: none.

Development file SHA-256: `3d2766839ac876c2c3342d3340b2b10d78aae59d0663ec4ced605836f86fc643`. Assignment SHA-256: `5f25ffba77126e2c010512008c353731e2c65de4d33d7b47d2cad20a9366cf1b`. Original source SHA-256: `d00d1522a06b5b67a0aea355c268ffe575e4a28651fa4fbf0e204705be3ce47b`. Original license: unknown; repository license: MIT.

Frozen class order: `["0", "1"]`. Development counts: `{"0": 252, "1": 514}`. Counts by original fold: `{"0": {"0": 54, "1": 100}, "1": {"0": 53, "1": 100}, "2": {"0": 51, "1": 102}, "3": {"0": 47, "1": 106}, "4": {"0": 47, "1": 106}}`.

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
t1	categorical	0	3	n/a	n/a	n/a	n/a	n/a
t2	categorical	0	3	n/a	n/a	n/a	n/a	n/a
t3	categorical	0	3	n/a	n/a	n/a	n/a	n/a
t4	categorical	0	3	n/a	n/a	n/a	n/a	n/a
t5	categorical	0	3	n/a	n/a	n/a	n/a	n/a
t6	categorical	0	3	n/a	n/a	n/a	n/a	n/a
t7	categorical	0	3	n/a	n/a	n/a	n/a	n/a
t8	categorical	0	3	n/a	n/a	n/a	n/a	n/a
t9	categorical	0	3	n/a	n/a	n/a	n/a	n/a

Exact development category frequencies:

- `t1`: `{"0": 178, "1": 260, "2": 328}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `t2`: `{"0": 201, "1": 266, "2": 299}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `t3`: `{"0": 168, "1": 268, "2": 330}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `t4`: `{"0": 204, "1": 262, "2": 300}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.

- t5: {"0": 119, "1": 273, "2": 374}. Categories absent from the other four folds: {"0": [], "1": [], "2": [], "3": [], "4": []}.
- t6: {"0": 196, "1": 261, "2": 309}. Categories absent from the other four folds: {"0": [], "1": [], "2": [], "3": [], "4": []}.
- t7: {"0": 160, "1": 263, "2": 343}. Categories absent from the other four folds: {"0": [], "1": [], "2": [], "3": [], "4": []}.
- t8: {"0": 214, "1": 262, "2": 290}. Categories absent from the other four folds: {"0": [], "1": [], "2": [], "3": [], "4": []}.
- t9: {"0": 167, "1": 265, "2": 334}. Categories absent from the other four folds: {"0": [], "1": [], "2": [], "3": [], "4": []}.

horse_colic_surgery

Task binary; metric auroc; 292 development rows; 76 sealed rows counted from assignments; 7 numeric and 13 categorical features. Development groups: 283; all groups: 354; largest group: 4. Original fold sizes:[58, 61, 58, 58, 57]; other-four-fold fit sizes:[234,231,234,234,235]. Declared grouping columns: none.

Development file SHA-256: 4b92b18271d7a289565d1845df42b8d903b22c321954cc78dd9fba90800f92ff. Assignment SHA-256: 1cd242e37b4b64ebc39ba83e834f7822ebe1bc19746b8970c74574deb99f3539. Original source SHA-256: 5079ad94ed5d03057e754b97692f1675603d515ebd663fcf34b87c45103e449c. Original license: CC BY 4.0; repository license: MIT.

Frozen class order: ["0", "1"]. Development counts: {"0": 180, "1": 112}. Counts by original fold: {"0": {"0": 38, "1": 20}, "1": {"0": 35, "1": 26}, "2": {"0": 31, "1": 27}, "3": {"0": 38, "1": 20}, "4": {"0": 38, "1": 19}}.

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
age_category	categorical	25	1	n/a	n/a	n/a	n/a	n/a
rectal_temperature	numeric	59	38	35.4	38.1	40.8	38.139056	0.73254976
heart_rate	numeric	20	53	30	62	184	71.330882	29.13295
respiratory_rate	numeric	62	38	8	28	96	31.617391	18.790878
extremity_temperature	categorical	53	4	n/a	n/a	n/a	n/a	n/a
peripheral_pulse	categorical	67	4	n/a	n/a	n/a	n/a	n/a
mucous_membrane_appearance	categorical	39	6	n/a	n/a	n/a	n/a	n/a
capillary_refill_time	categorical	36	2	n/a	n/a	n/a	n/a	n/a
pain	categorical	47	5	n/a	n/a	n/a	n/a	n/a
peristalsis	categorical	41	4	n/a	n/a	n/a	n/a	n/a

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
abdominal_distension	categorical	53	4	n/a	n/a	n/a	n/a	n/a
nasogastric_tube_stomach_gas	categorical	115	3	n/a	n/a	n/a	n/a	n/a
nasogastric_tube_reflux_volume	categorical	112	3	n/a	n/a	n/a	n/a	n/a
nasogastric_tube_reflux_ph	numeric	237	22	1	5	8	4.6490909	1.9697598
rectal_feces_presence	categorical	100	4	n/a	n/a	n/a	n/a	n/a
rectal_examination	categorical	112	5	n/a	n/a	n/a	n/a	n/a
red_blood_cell_count	numeric	31	50	4	44	75	44.961303	10.838738
blood_protein	numeric	37	78	3.3	7.5	86	25.715294	27.945219
abdominocentesis_appearance	categorical	154	3	n/a	n/a	n/a	n/a	n/a
abdominocentesis_protein	numeric	183	38	0.1	2	10.1	2.8963303	1.841366

Exact development category frequencies:

- `age_category`: `{"0": 267}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `extremity_temperature`: `{"0": 74, "1": 33, "2": 107, "3": 25}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `peripheral_pulse`: `{"0": 5, "1": 118, "2": 91, "3": 11}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `mucous_membrane_appearance`: `{"0": 87, "1": 25, "2": 61, "3": 41, "4": 20, "5": 19}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `capillary_refill_time`: `{"0": 180, "1": 76}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `pain`: `{"0": 44, "1": 62, "2": 65, "3": 37, "4": 37}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `peristalsis`: `{"0": 41, "1": 16, "2": 123, "3": 71}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `abdominal_distension`: `{"0": 82, "1": 60, "2": 64, "3": 33}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `nasogastric_tube_stomach_gas`: `{"0": 61, "1": 92, "2": 24}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.

- `nasogastric_tube_reflux_volume`: `{"0": 109, "1": 37, "2": 34}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `rectal_feces_presence`: `{"0": 11, "1": 56, "2": 50, "3": 75}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `rectal_examination`: `{"0": 27, "1": 17, "2": 18, "3": 40, "4": 78}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `abdominocentesis_appearance`: `{"0": 43, "1": 43, "2": 52}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.

iris

Task `multiclass`; metric `log_loss`; 122 development rows; 28 sealed rows counted from assignments; 4 numeric and 0 categorical features. Development groups: 119; all groups: 147; largest group: 3. Original fold sizes:[25, 23, 25, 25, 24]; other-four-fold fit sizes:[97, 99, 97, 97, 98]. Declared grouping columns: none.

Development file SHA-256: `86b4b47bc14f4513838072b130aed1b9c4f5bd9056cc110215a8c66be3d9dcc3`. Assignment SHA-256: `4322393c97bb172340ad840055cbc0edb09b8321ece9d7ce5b0aed1934aadfad`. Original source SHA-256: `ce4982cba30bd5aaa352e51d55e1cf56c56ae6d573d644f36257de1177717228`. Original license: unknown; repository license: MIT.

Frozen class order: `["0", "1", "2"]`. Development counts: `{"0": 37, "1": 40, "2": 45}`. Counts by original fold: `{"0": {"0": 10, "1": 10, "2": 5}, "1": {"0": 5, "1": 11, "2": 7}, "2": {"0": 8, "1": 7, "2": 10}, "3": {"0": 7, "1": 8, "2": 10}, "4": {"0": 7, "1": 4, "2": 13}}`.

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
sepal-length	numeric	0	34	4.3	5.8	7.9	5.9008197	0.8357284
sepal-width	numeric	0	23	2	3	4.4	3.0491803	0.4435782
petal-length	numeric	0	40	1	4.5	6.9	3.8901639	1.7489767
petal-width	numeric	0	21	0.1	1.4	2.5	1.2540984	0.76170081

wine_recognition

Task `multiclass`; metric `log_loss`; 142 development rows; 36 sealed rows counted from assignments; 13 numeric and 0 categorical features. Development groups: 142; all groups: 178; largest group: 1. Original fold sizes:[29, 29, 28, 28, 28]; other-four-fold fit sizes:[113,113,114,114,114]. Declared grouping columns: none.

Development file SHA-256: `ff8de7ed12ef935c0d42be31c08bae6ca18ae909e60dee1d9c1887ce333e253f`. Assignment SHA-256: `0a5a84bbb63803ef0b31e7fde40c3fcd2d0e2246b62033cc820be3c907917614`. Original source SHA-256: `56f5ba3f3ba78f6e522a13e5a97e03ff28cf9fa4107021b22b932f6d4064c145`. Original license: unknown; repository license: MIT.

Frozen class order: `["1", "2", "3"]`. Development counts: `{"1": 47, "2": 57, "3": 38}`. Counts by original fold: `{"0": {"1": 10, "2": 11, "3": 8}, "1": {"1": 9, "2": 12, "3": 8}, "2": {"1": 10, "2": 11, "3": 7}, "3": {"1": 9, "2": 12, "3": 7}, "4": {"1": 9, "2": 11, "3": 8}}`.

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
1	numeric	0	106	11.45	13.05	14.83	13.017746	0.78176704
2	numeric	0	114	0.89	1.88	5.8	2.3453521	1.140337
3	numeric	0	74	1.36	2.36	3.23	2.3671831	0.27320313
4	numeric	0	55	10.6	19.25	30	19.509859	3.3591227
5	numeric	0	48	70	98	162	99.43662	14.362247
6	numeric	0	80	0.98	2.38	3.88	2.3109155	0.62909045
7	numeric	0	111	0.34	2.135	5.08	2.0360563	1.0064125
8	numeric	0	39	0.13	0.335	0.66	0.36577465	0.12517213
9	numeric	0	89	0.42	1.545	3.58	1.6009859	0.58075874
10	numeric	0	111	1.28	4.69	13	5.0982394	2.3589459
11	numeric	0	71	0.48	0.96	1.45	0.94574648	0.21709943
12	numeric	0	106	1.27	2.78	3.82	2.6256338	0.69527932
13	numeric	0	106	278	673.5	1680	751.73239	317.35709

car_evaluation

Task `multiclass`; metric `log_loss`; 1382 development rows; 346 sealed rows counted from assignments; 0 numeric and 6 categorical features. Development groups: 1382; all groups: 1728; largest group: 1. Original fold sizes:[277,277,276,276,276]; other-four-fold fit sizes:[1105, 1105, 1106, 1106, 1106]. Declared grouping columns: none.

Development file SHA-256: `014c5c81b356e617cac7daa8901914c12837f5458c221882466f2536f047fc46`. Assignment SHA-256: `c36e08e3b50f2c560f4a0d06cf8d719da1a1352870246ad04f12ec7b7a855fab`. Original source SHA-256: `9c18fe91ba5401cab754456a9bfbfe54a6c307b0d4b880259fb4218edef77998`. Original license: CC BY 4.0; repository license: MIT.

Frozen class order: `["0", "1", "2", "3"]`. Development counts: `{"0": 967, "1": 310, "2": 49, "3": 56}`. Counts by original fold: `{"0": {"0": 194, "1": 64, "2": 10, "3": 9}, "1": {"0": 195, "1": 59, "2": 12, "3": 11}, "2": {"0": 200, "1": 53, "2": 9, "3": 14}, "3": {"0": 186, "1": 69, "2": 10, "3": 11}, "4": {"0": 192, "1": 65, "2": 8, "3": 11}}`.

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
buying_price	categorical	0	4	n/a	n/a	n/a	n/a	n/a
maintenance_costs	categorical	0	4	n/a	n/a	n/a	n/a	n/a
doors	categorical	0	4	n/a	n/a	n/a	n/a	n/a
seats	categorical	0	3	n/a	n/a	n/a	n/a	n/a
luggage_boot	categorical	0	3	n/a	n/a	n/a	n/a	n/a

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
safety	categorical	0	3	n/a	n/a	n/a	n/a	n/a

Exact development category frequencies:

- `buying_price`: `{"0": 345, "1": 346, "2": 346, "3": 345}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `maintenance_costs`: `{"0": 345, "1": 346, "2": 346, "3": 345}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `doors`: `{"0": 341, "1": 351, "2": 346, "3": 344}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `seats`: `{"0": 454, "1": 466, "2": 462}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `luggage_boot`: `{"0": 448, "1": 468, "2": 466}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `safety`: `{"0": 467, "1": 468, "2": 447}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.

balance_scale

Task `multiclass`; metric `log_loss`; 500 development rows; 125 sealed rows counted from assignments; 0 numeric and 4 categorical features. Development groups: 500; all groups: 625; largest group: 1. Original fold sizes:[100,100,100,100,100]; other-four-fold fit sizes:[400,400,400,400,400]. Declared grouping columns: none.

Development file SHA-256: `9267117d816d008803477a791ca9cf346fefc17aba62afc9a9b5d444e9530329`. Assignment SHA-256: `40efdd8487cb7076e85a00d07e1d42d4902a84e1e9aef740d291e341b2509764`. Original source SHA-256: `8c87be0e86f362154b34bd608caf8c5804e357e437b7db91b14974b6afaffe32`. Original license: unknown; repository license: MIT.

Frozen class order: `["0", "1", "2"]`. Development counts: `{"0": 40, "1": 211, "2": 249}`. Counts by original fold: `{"0": {"0": 4, "1": 49, "2": 47}, "1": {"0": 11, "1": 42, "2": 47}, "2": {"0": 8, "1": 39, "2": 53}, "3": {"0": 11, "1": 43, "2": 46}, "4": {"0": 6, "1": 38, "2": 56}}`.

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
Left-Weight	categorical	0	5	n/a	n/a	n/a	n/a	n/a
Left-Distance	categorical	0	5	n/a	n/a	n/a	n/a	n/a
Right-Weight	categorical	0	5	n/a	n/a	n/a	n/a	n/a
Right-Distance	categorical	0	5	n/a	n/a	n/a	n/a	n/a

Exact development category frequencies:

- `Left-Weight`: `{"1": 100, "2": 100, "3": 100, "4": 100, "5": 100}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.

- `Left-Distance`: `{"1": 100, "2": 100, "3": 100, "4": 98, "5": 102}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `Right-Weight`: `{"1": 101, "2": 101, "3": 99, "4": 99, "5": 100}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `Right-Distance`: `{"1": 70, "2": 86, "3": 106, "4": 118, "5": 120}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.

dermatology

Task `multiclass`; metric `log_loss`; 292 development rows; 74 sealed rows counted from assignments; 1 numeric and 33 categorical features. Development groups: 292; all groups: 366; largest group: 1. Original fold sizes:[59, 59, 58, 58, 58]; other-four-fold fit sizes:[233,233,234,234,234]. Declared grouping columns: none.

Development file SHA-256: `a447d3a279d1f4c11f3b330935d362e3d8ca4c5e99afe5276f01e411a8e0795e`. Assignment SHA-256: `88ce0ea91c15e418c566d1a455a7b01b6cbc4335be5fbfca3efcf4549267a379`. Original source SHA-256: `c8852ba546e92249debecd9a897cda3f94fd10872fa13c62eb14ab40e279166e`. Original license: unknown; repository license: MIT.

Frozen class order: `["1", "2", "3", "4", "5", "6"]`. Development counts: `{"1": 89, "2": 48, "3": 61, "4": 39, "5": 40, "6": 15}`. Counts by original fold: `{"0": {"1": 16, "2": 9, "3": 11, "4": 10, "5": 8, "6": 5}, "1": {"1": 16, "2": 13, "3": 13, "4": 9, "5": 6, "6": 2}, "2": {"1": 15, "2": 9, "3": 16, "4": 8, "5": 7, "6": 3}, "3": {"1": 22, "2": 9, "3": 10, "4": 6, "5": 10, "6": 1}, "4": {"1": 20, "2": 8, "3": 11, "4": 6, "5": 9, "6": 4}}`.

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
erythema	categorical	0	4	n/a	n/a	n/a	n/a	n/a
scaling	categorical	0	4	n/a	n/a	n/a	n/a	n/a
definite_borders	categorical	0	4	n/a	n/a	n/a	n/a	n/a
itching	categorical	0	4	n/a	n/a	n/a	n/a	n/a
koebner_phenomenon	categorical	0	4	n/a	n/a	n/a	n/a	n/a
polygonal_papules	categorical	0	4	n/a	n/a	n/a	n/a	n/a
follicular_papules	categorical	0	4	n/a	n/a	n/a	n/a	n/a
oral_mucosal_involvement	categorical	0	4	n/a	n/a	n/a	n/a	n/a
knee_and_elbow_involvement	categorical	0	4	n/a	n/a	n/a	n/a	n/a
scalp_involvement	categorical	0	4	n/a	n/a	n/a	n/a	n/a
family_history	categorical	0	2	n/a	n/a	n/a	n/a	n/a

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
melanin_incontinence	categorical	0	4	n/a	n/a	n/a	n/a	n/a
eosinophils_in_the_infiltrate	categorical	0	3	n/a	n/a	n/a	n/a	n/a
PNL_infiltrate	categorical	0	4	n/a	n/a	n/a	n/a	n/a
fibrosis_of_the_papillary_dermis	categorical	0	4	n/a	n/a	n/a	n/a	n/a
exocytosis	categorical	0	4	n/a	n/a	n/a	n/a	n/a
acanthosis	categorical	0	4	n/a	n/a	n/a	n/a	n/a
hyperkeratosis	categorical	0	4	n/a	n/a	n/a	n/a	n/a
parakeratosis	categorical	0	4	n/a	n/a	n/a	n/a	n/a
clubbing_of_the_rete_ridges	categorical	0	4	n/a	n/a	n/a	n/a	n/a
elongation_of_the_rete_ridges	categorical	0	4	n/a	n/a	n/a	n/a	n/a
thinning_of_the_suprapapillary_epidermis	categorical	0	4	n/a	n/a	n/a	n/a	n/a
spongiform_pustule	categorical	0	4	n/a	n/a	n/a	n/a	n/a
munro_microabcess	categorical	0	4	n/a	n/a	n/a	n/a	n/a
focal_hypergranulosis	categorical	0	4	n/a	n/a	n/a	n/a	n/a
disappearance_of_the_granular_layer	categorical	0	4	n/a	n/a	n/a	n/a	n/a
vacuolisation_and_damage_of_basal_layer	categorical	0	4	n/a	n/a	n/a	n/a	n/a
spongiosis	categorical	0	4	n/a	n/a	n/a	n/a	n/a
saw-tooth_appearance_of_retes	categorical	0	4	n/a	n/a	n/a	n/a	n/a
follicular_horn_plug	categorical	0	4	n/a	n/a	n/a	n/a	n/a

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
perifollicular_parakeratosis	categorical	0	4	n/a	n/a	n/a	n/a	n/a
inflammatory_monoluclear_inflitrate	categorical	0	4	n/a	n/a	n/a	n/a	n/a
band-like_infiltrate	categorical	0	4	n/a	n/a	n/a	n/a	n/a
Age	numeric	0	58	0	26.5	60	27.921233	15.306591

Exact development category frequencies:

- `erythema`: `{"0": 3, "1": 45, "2": 175, "3": 69}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `scaling`: `{"0": 6, "1": 84, "2": 160, "3": 42}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `definite_borders`: `{"0": 45, "1": 72, "2": 133, "3": 42}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `itching`: `{"0": 90, "1": 62, "2": 80, "3": 60}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `koebner_phenomenon`: `{"0": 176, "1": 58, "2": 42, "3": 16}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `polygonal_papules`: `{"0": 234, "1": 1, "2": 32, "3": 25}`. Categories absent from the other four folds: `{"0": [], "1": ["1"], "2": [], "3": [], "4": []}`.
- `follicular_papules`: `{"0": 265, "1": 10, "2": 14, "3": 3}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `oral_mucosal_involvement`: `{"0": 236, "1": 9, "2": 36, "3": 11}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `knee_and_elbow_involvement`: `{"0": 201, "1": 21, "2": 50, "3": 20}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `scalp_involvement`: `{"0": 210, "1": 23, "2": 44, "3": 15}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `family_history`: `{"0": 257, "1": 35}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `melanin_incontinence`: `{"0": 233, "1": 8, "2": 40, "3": 11}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `eosinophils_in_the_infiltrate`: `{"0": 263, "1": 22, "2": 7}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `PNL_infiltrate`: `{"0": 190, "1": 57, "2": 39, "3": 6}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `fibrosis_of_the_papillary_dermis`: `{"0": 250, "1": 7, "2": 18, "3": 17}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.

- exocytosis: {"0": 92, "1": 47, "2": 109, "3": 44}. Categories absent from the other four folds: {"0": [], "1": [], "2": [], "3": [], "4": []}.
- acanthosis: {"0": 8, "1": 59, "2": 164, "3": 61}. Categories absent from the other four folds: {"0": [], "1": [], "2": [], "3": [], "4": []}.
- hyperkeratosis: {"0": 181, "1": 73, "2": 33, "3": 5}. Categories absent from the other four folds: {"0": [], "1": [], "2": [], "3": [], "4": []}.
- parakeratosis: {"0": 70, "1": 94, "2": 105, "3": 23}. Categories absent from the other four folds: {"0": [], "1": [], "2": [], "3": [], "4": []}.
- clubbing_of_the_rete_ridges: {"0": 202, "1": 17, "2": 44, "3": 29}. Categories absent from the other four folds: {"0": [], "1": [], "2": [], "3": [], "4": []}.
- elongation_of_the_rete_ridges: {"0": 161, "1": 20, "2": 74, "3": 37}. Categories absent from the other four folds: {"0": [], "1": [], "2": [], "3": [], "4": []}.
- thinning_of_the_suprapapillary_epidermis: {"0": 203, "1": 16, "2": 48, "3": 25}. Categories absent from the other four folds: {"0": [], "1": [], "2": [], "3": [], "4": []}.
- spongiform_pustule: {"0": 240, "1": 28, "2": 21, "3": 3}. Categories absent from the other four folds: {"0": [], "1": [], "2": [], "3": [], "4": []}.
- munro_microabcess: {"0": 231, "1": 26, "2": 26, "3": 9}. Categories absent from the other four folds: {"0": [], "1": [], "2": [], "3": [], "4": []}.
- focal_hypergranulosis: {"0": 232, "1": 12, "2": 36, "3": 12}. Categories absent from the other four folds: {"0": [], "1": [], "2": [], "3": [], "4": []}.
- disappearance_of_the_granular_layer: {"0": 216, "1": 23, "2": 40, "3": 13}. Categories absent from the other four folds: {"0": [], "1": [], "2": [], "3": [], "4": []}.
- vacuolisation_and_damage_of_basal_layer: {"0": 231, "1": 3, "2": 37, "3": 21}. Categories absent from the other four folds: {"0": [], "1": [], "2": [], "3": [], "4": []}.
- spongiosis: {"0": 155, "1": 23, "2": 81, "3": 33}. Categories absent from the other four folds: {"0": [], "1": [], "2": [], "3": [], "4": []}.
- saw-tooth_appearance_of_retes: {"0": 232, "1": 4, "2": 34, "3": 22}. Categories absent from the other four folds: {"0": [], "1": [], "2": [], "3": [], "4": []}.
- follicular_horn_plug: {"0": 276, "1": 8, "2": 5, "3": 3}. Categories absent from the other four folds: {"0": [], "1": [], "2": [], "3": [], "4": []}.
- perifollicular_parakeratosis: {"0": 276, "1": 4, "2": 9, "3": 3}. Categories absent from the other four folds: {"0": [], "1": [], "2": [], "3": [], "4": []}.
- inflammatory_monoluclear_inflitrate: {"0": 11, "1": 64, "2": 171, "3": 46}. Categories absent from the other four folds: {"0": [], "1": [], "2": [], "3": [], "4": []}.
- band-like_infiltrate: {"0": 226, "1": 3, "2": 19, "3": 44}. Categories absent from the other four folds: {"0": [], "1": [], "2": [], "3": [], "4": []}.

segmentation

Task multiclass; metric log_loss; 1848 development rows; 462 sealed rows counted from assignments; 18 numeric and 1 categorical features. Development groups: 1669; all groups: 2086; largest group: 3. Original fold sizes:[364,372,369,373,370]; other-four-fold fit sizes:[1484, 1476, 1479, 1475, 1478]. Declared grouping columns: none.

Development file SHA-256: `6691b654d75aa6ba4a66792fe7cbeef46738c9a02b5ca667cd41d9e141ecc67a`. Assignment SHA-256: `10bcfcc5634490f57e7b69f3b8071fd1e49d63e2a03a62e182772e6f3df321d6`. Original source SHA-256: `980ba4342d2fe489793cc3431678318f8d1467fe6bf6a687eb18bfc2267a049f`. Original license: unknown; repository license: MIT.

Frozen class order: `["0", "1", "2", "3", "4", "5", "6"]`. Development counts: `{"0": 261, "1": 262, "2": 265, "3": 262, "4": 269, "5": 260, "6": 269}`. Counts by original fold: `{"0": {"0": 47, "1": 49, "2": 51, "3": 54, "4": 51, "5": 59, "6": 53}, "1": {"0": 52, "1": 54, "2": 54, "3": 53, "4": 56, "5": 46, "6": 57}, "2": {"0": 51, "1": 52, "2": 51, "3": 53, "4": 56, "5": 53, "6": 53}, "3": {"0": 57, "1": 48, "2": 56, "3": 54, "4": 51, "5": 53, "6": 54}, "4": {"0": 54, "1": 59, "2": 53, "3": 48, "4": 55, "5": 49, "6": 52}}`.

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
region-centroid-col	numeric	0	253	1	121	254	124.82955	72.877771
region-centroid-row	numeric	0	237	11	122	251	123.68019	57.350115
region-pixel-count	categorical	0	1	n/a	n/a	n/a	n/a	n/a
short-line-density-5	numeric	0	4	0	0	0.33333334	0.014490139	0.040503723
short-line-density-2	numeric	0	3	0	0	0.22222222	0.0046897546	0.024343745
vedge-mean	numeric	0	582	0	1.2777773	27.944443	1.8955628	2.6719697
vedge-sd	numeric	0	1459	0	0.83148183	991.7184	5.9348586	45.97572
hedge-mean	numeric	0	637	0	1.4444447	44.722225	2.4776034	3.7718629
hedge-sd	numeric	0	1472	-1.5894573e-08	0.99814636	1386.3292	8.7672312	61.930518
intensity-mean	numeric	0	1087	0	21.611111	143.44444	36.80293	38.072465
rawred-mean	numeric	0	622	0	19.555555	137.11111	32.616883	34.956813
rawblue-mean	numeric	0	714	0	27.277778	150.88889	43.863998	43.415929
rawgreen-mean	numeric	0	637	0	20.333334	142.55556	33.92791	36.242479
exred-mean	numeric	0	414	-49.666668	-10.666667	9.888889	-12.558141	11.498634
exblue-mean	numeric	0	604	-12.444445	19.444445	82	21.183201	19.520045
exgreen-mean	numeric	0	361	-33.88889	-10.666667	24.666666	-8.6250601	11.569724
value-mean	numeric	0	715	0	28.555555	150.88889	44.813493	42.815875
saturatoin-mean	numeric	0	1519	0	0.37455574	1	0.42759796	0.23090285

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
hue-mean	numeric	0	1552	-3.0441751	-2.0481283	2.9124804	-1.3641642	1.5416936

Exact development category frequencies:

- `region-pixel-count`: `{"9.0": 1848}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.

vehicle

Task `multiclass`; metric `log_loss`; 677 development rows; 169 sealed rows counted from assignments; 18 numeric and 0 categorical features. Development groups: 677; all groups: 846; largest group: 1. Original fold sizes:[135,136,135,136,135]; other-four-fold fit sizes:[542,541,542,541,542]. Declared grouping columns: none.

Development file SHA-256: `117699e46fbd73fdfaa9996fc671a7a88463cc768f0f1e859b77ad7120a49b18`. Assignment SHA-256: `44645392494e0be6c4a4d3e708db4af46b0d4c7af52bc6cf3666bf407109030b`. Original source SHA-256: `dad03273414ae7127e30b6e4e66df90319a148b37ead5dd376dd140be5baedc4`. Original license: unknown; repository license: MIT.

Frozen class order: `["1", "2", "3", "4"]`. Development counts: `{"1": 169, "2": 175, "3": 174, "4": 159}`. Counts by original fold: `{"0": {"1": 34, "2": 35, "3": 36, "4": 30}, "1": {"1": 33, "2": 36, "3": 33, "4": 34}, "2": {"1": 37, "2": 34, "3": 34, "4": 30}, "3": {"1": 34, "2": 36, "3": 32, "4": 34}, "4": {"1": 31, "2": 34, "3": 39, "4": 31}}`.

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
COMPACTNESS	numeric	0	43	73	93	119	93.949778	8.1342405
CIRCULARITY	numeric	0	27	33	44	59	44.909897	6.1908953
DISTANCE CIRCULARITY	numeric	0	63	40	80	112	82.490399	15.889816
RADIUS RATIO	numeric	0	134	104	169	333	169.83161	33.806618
PR AXIS ASPECT RATIO	numeric	0	36	47	61	138	61.805022	8.2894697
MAX LENGTH ASPECT RATIO	numeric	0	19	2	8	55	8.6070901	4.8232398
SCATTER RATIO	numeric	0	126	112	157	265	169.65731	33.487679
ELONGATEDNESS	numeric	0	35	26	43	61	40.748892	7.8641824
PR AXISRECTANGULAR	numeric	0	13	17	20	29	20.645495	2.6103189

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
LENGTHRECTANGULAR	numeric	0	65	118	146	186	148.14032	14.560102
MAJORVARIANCE	numeric	0	125	130	179	320	189.4195	31.887562
MINORVARIANCE	numeric	0	384	184	366	1018	444.34269	178.10982
GYRATIONRADIUS	numeric	0	138	112	174	268	174.82422	32.80978
MAJORSKEWNESS	numeric	0	39	59	71	135	72.462334	7.7825212
MINORSKEWNESS	numeric	0	23	0	5	22	6.3353028	4.9982104
MINORKURTOSIS	numeric	0	41	0	11	41	12.82127	9.0213413
MAJORKURTOSIS	numeric	0	29	176	189	204	189.05761	6.1604295
HOLLOWSRATIO	numeric	0	31	181	197	211	195.79173	7.4621283

optdigits

Task `multiclass`; metric `log_loss`; 4496 development rows; 1124 sealed rows counted from assignments; 51 numeric and 13 categorical features. Development groups: 4496; all groups: 5620; largest group: 1. Original fold sizes:[900,899,899,899,899]; other-four-fold fit sizes:[3596, 3597, 3597, 3597, 3597]. Declared grouping columns: none.

Development file SHA-256: `afaa9819c5ac7059a64f7457e73444b3ba7c28ccb4cd3de27cdf5fe13f7751f6`. Assignment SHA-256: `0407e498a26a4cea16fe28c170e626d1338dc4bf8148d0e1be17ba3d87056018`. Original source SHA-256: `b70b7ac68a54d1ccd41bd24f11324c3af96b67541791fb9d8684cbdf151533c3`. Original license: unknown; repository license: MIT.

Frozen class order: `["0", "1", "2", "3", "4", "5", "6", "7", "8", "9"]`. Development counts: `{"0": 444, "1": 456, "2": 444, "3": 476, "4": 462, "5": 433, "6": 437, "7": 444, "8": 439, "9": 461}`. Counts by original fold: `{"0": {"0": 84, "1": 90, "2": 96, "3": 82, "4": 100, "5": 96, "6": 86, "7": 85, "8": 87, "9": 94}, "1": {"0": 88, "1": 90, "2": 74, "3": 109, "4": 95, "5": 83, "6": 96, "7": 93, "8": 93, "9": 78}, "2": {"0": 77, "1": 99, "2": 106, "3": 92, "4": 89, "5": 83, "6": 95, "7": 80, "8": 72, "9": 106}, "3": {"0": 93, "1": 88, "2": 92, "3": 89, "4": 90, "5": 79, "6": 72, "7": 99, "8": 94, "9": 103}, "4": {"0": 102, "1": 89, "2": 76, "3": 104, "4": 88, "5": 92, "6": 88, "7": 87, "8": 93, "9": 80}}`.

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
input 1	categorical	0	1	n/a	n/a	n/a	n/a	n/a
input 2	categorical	0	9	n/a	n/a	n/a	n/a	n/a
input 3	numeric	0	17	0	5	16	5.3854537	4.6694807

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
input 4	numeric	0	17	0	13	16	11.763345	4.2575085
input 5	numeric	0	17	0	13	16	11.573176	4.4513964
input 6	numeric	0	17	0	4	16	5.5584964	5.6073441
input 7	numeric	0	17	0	0	16	1.3727758	3.349232
input 8	numeric	0	16	0	0	15	0.13122776	0.99381714
input 9	categorical	0	4	n/a	n/a	n/a	n/a	n/a
input 10	numeric	0	16	0	0	15	1.9730872	3.0879654
input 11	numeric	0	17	0	12	16	10.52758	5.4372549
input 12	numeric	0	17	0	13	16	11.79782	4.01247
input 13	numeric	0	17	0	11	16	10.540036	4.7693414
input 14	numeric	0	17	0	9	16	8.261121	5.9790084
input 15	numeric	0	17	0	0	16	2.0991993	3.9437706
input 16	numeric	0	14	0	0	13	0.13411922	0.91377922
input 17	categorical	0	5	n/a	n/a	n/a	n/a	n/a
input 18	numeric	0	17	0	1	16	2.5849644	3.4827212
input 19	numeric	0	17	0	11	16	9.6209964	5.8251162
input 20	numeric	0	17	0	6	16	6.8256228	5.8753604
input 21	numeric	0	17	0	6	16	7.1846085	6.1510996
input 22	numeric	0	17	0	9	16	7.9857651	6.2479372
input 23	numeric	0	17	0	0	16	1.9759786	3.498392
input 24	categorical	0	8	n/a	n/a	n/a	n/a	n/a
input 25	categorical	0	2	n/a	n/a	n/a	n/a	n/a
input 26	numeric	0	17	0	1	16	2.363879	3.0806441
input 27	numeric	0	17	0	11	16	9.1581406	6.1416147
input 28	numeric	0	17	0	10	16	9.0745107	5.9029827
input 29	numeric	0	17	0	12	16	9.7602313	6.2595395
input 30	numeric	0	17	0	8	16	7.7955961	5.9558196
input 31	numeric	0	17	0	0	16	2.3709964	3.6629328
input 32	categorical	0	3	n/a	n/a	n/a	n/a	n/a
input 33	categorical	0	2	n/a	n/a	n/a	n/a	n/a
input 34	numeric	0	16	0	0	15	2.1419039	3.3179478
input 35	numeric	0	17	0	8	16	7.67371	6.2749206
input 36	numeric	0	17	0	11	16	9.155694	6.1866184
input 37	numeric	0	17	0	12	16	10.287811	5.9272272

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
input 38	numeric	0	17	0	10	16	9.0538256	5.8700142
input 39	numeric	0	15	0	1	14	2.9219306	3.5062095
input 40	categorical	0	1	n/a	n/a	n/a	n/a	n/a
input 41	categorical	0	8	n/a	n/a	n/a	n/a	n/a
input 42	numeric	0	17	0	0	16	1.4971085	3.0053319
input 43	numeric	0	17	0	5	16	6.6127669	6.5197965
input 44	numeric	0	17	0	7	16	7.1683719	6.4851243
input 45	numeric	0	17	0	8	16	7.8153915	6.2973257
input 46	numeric	0	17	0	10	16	8.5473754	5.7808416
input 47	numeric	0	17	0	1	16	3.518016	4.3473655
input 48	categorical	0	7	n/a	n/a	n/a	n/a	n/a
input 49	categorical	0	9	n/a	n/a	n/a	n/a	n/a
input 50	numeric	0	17	0	0	16	0.80649466	1.9683692
input 51	numeric	0	17	0	8	16	7.7595641	5.6567184
input 52	numeric	0	17	0	10	16	9.7504448	5.1729862
input 53	numeric	0	17	0	10	16	9.6554715	5.2848009
input 54	numeric	0	17	0	11	16	9.1548043	5.9832257
input 55	numeric	0	17	0	1	16	3.7446619	4.8885914
input 56	numeric	0	13	0	0	13	0.15814057	0.81911142
input 57	categorical	0	2	n/a	n/a	n/a	n/a	n/a
input 58	numeric	0	11	0	0	10	0.27935943	0.92382518
input 59	numeric	0	17	0	5	16	5.7382117	5.007576
input 60	numeric	0	17	0	13	16	11.981317	4.3434726
input 61	numeric	0	17	0	13	16	11.577625	4.9533614
input 62	numeric	0	17	0	6	16	6.7164146	5.7925554
input 63	numeric	0	17	0	0	16	2.0507117	3.990158
input 64	numeric	0	17	0	0	16	0.24154804	1.3632569

Exact development category frequencies:

- `input1`: `{"0.0": 4496}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `input2`: `{"0.0": 3810, "1.0": 343, "2.0": 175, "3.0": 89, "4.0": 41, "5.0": 19, "6.0": 10, "7.0": 6, "8.0": 3}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `input9`: `{"0.0": 4489, "1.0": 4, "2.0": 2, "5.0": 1}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": ["5.0"], "3": [], "4": []}`.

- input17: {"0.0": 4484, "1.0": 6, "2.0": 3, "3.0": 2, "5.0": 1}. Categories absent from the other four folds: {"0": [], "1": ["5.0"], "2": [], "3": [], "4": []}.
- input24: {"0.0": 4418, "1.0": 24, "2.0": 17, "3.0": 11, "4.0": 12, "5.0": 7, "6.0": 5, "8.0": 2}. Categories absent from the other four folds: {"0": [], "1": [], "2": ["8.0"], "3": [], "4": []}.
- input25: {"0.0": 4491, "1.0": 5}. Categories absent from the other four folds: {"0": [], "1": [], "2": [], "3": [], "4": []}.
- input32: {"0.0": 4484, "1.0": 11, "2.0": 1}. Categories absent from the other four folds: {"0": [], "1": [], "2": [], "3": ["2.0"], "4": []}.
- input33: {"0.0": 4492, "1.0": 4}. Categories absent from the other four folds: {"0": [], "1": [], "2": [], "3": [], "4": []}.
- input40: {"0.0": 4496}. Categories absent from the other four folds: {"0": [], "1": [], "2": [], "3": [], "4": []}.
- input41: {"0.0": 4455, "1.0": 14, "2.0": 11, "3.0": 6, "4.0": 5, "5.0": 3, "6.0": 1, "7.0": 1}. Categories absent from the other four folds: {"0": [], "1": ["7.0"], "2": [], "3": ["6.0"], "4": []}.
- input48: {"0.0": 4441, "1.0": 36, "2.0": 11, "3.0": 2, "4.0": 3, "5.0": 1, "6.0": 2}. Categories absent from the other four folds: {"0": [], "1": [], "2": [], "3": [], "4": ["5.0"]}.
- input49: {"0.0": 4465, "1.0": 18, "10.0": 1, "2.0": 4, "3.0": 4, "4.0": 1, "5.0": 1, "7.0": 1, "8.0": 1}. Categories absent from the other four folds: {"0": ["5.0"], "1": ["4.0"], "2": ["10.0", "7.0"], "3": [], "4": ["8.0"]}.
- input57: {"0.0": 4494, "1.0": 2}. Categories absent from the other four folds: {"0": [], "1": [], "2": [], "3": [], "4": []}.

penguins

Task multiclass; metric log_loss; 275 development rows; 69 sealed rows counted from assignments; 5 numeric and 2 categorical features. Development groups: 275; all groups: 344; largest group: 1. Original fold sizes:[55, 55, 55, 55, 55]; other-four-fold fit sizes:[220,220,220,220,220]. Declared grouping columns: none.

Development file SHA-256: cc9e242f3aa638cbaf2fc6951c37631b5928239aaa0c1dba03cc176629d0a4ed. Assignment SHA-256: 81a892fa47114e6462ad030e74c1dad1e3936fdd3c0aadfa8cacd354db82b332. Original source SHA-256: 416c13c9349eba384c5af4711f77a7761ffebcef23cb7e780f65172827b29087. Original license: unknown; repository license: MIT.

Frozen class order: ["0", "1", "2"]. Development counts: {"0": 121, "1": 55, "2": 99}. Counts by original fold: {"0": {"0": 25, "1": 11, "2": 19}, "1": {"0": 24, "1": 11, "2": 20}, "2": {"0": 24, "1": 11, "2": 20}, "3": {"0": 24, "1": 11, "2": 20}, "4": {"0": 24, "1": 11, "2": 20}}.

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
island	categorical	0	3	n/a	n/a	n/a	n/a	n/a
bill_length_mm	numeric	2	150	33.1	44.5	59.6	43.959341	5.5528954
bill_depth_mm	numeric	2	75	13.1	17.3	21.2	17.141026	1.9388646
flipper_length_mm	numeric	2	55	172	197	231	200.89744	14.092984

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
body_mass_g	numeric	2	89	2700	4050	6300	4188.4615	815.6716
sex	categorical	11	2	n/a	n/a	n/a	n/a	n/a
year	numeric	0	3	2007	2008	2009	2008.0218	0.81768874

Exact development category frequencies:

- `island`: `{"0": 135, "1": 100, "2": 40}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.
- `sex`: `{"0": 130, "1": 134}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.

tae

Task `multiclass`; metric `log_loss`; 124 development rows; 27 sealed rows counted from assignments; 3 numeric and 2 categorical features. Development groups: 86; all groups: 106; largest group: 4. Original fold sizes:[20, 25, 27, 30, 22]; other-four-fold fit sizes:[104, 99, 97, 94,102]. Declared grouping columns: none.

Development file SHA-256: `96b5a504b8ee490a66b5d4b01f8fed9f7753885bfc0b49e17c42099cc441517e`. Assignment SHA-256: `7e83b2b19fc5acb93d2afc04ce0a6430f5609b7380c806978d603562186e08ea`. Original source SHA-256: `5eb60cdd947ecb4ea476ba854299dbe18d522df3ed684fb8f60657fdd4132910`. Original license: unknown; repository license: MIT.

Frozen class order: `["1", "2", "3"]`. Development counts: `{"1": 41, "2": 40, "3": 43}`. Counts by original fold: `{"0": {"1": 6, "2": 10, "3": 4}, "1": {"1": 9, "2": 9, "3": 7}, "2": {"1": 10, "2": 5, "3": 12}, "3": {"1": 10, "2": 9, "3": 11}, "4": {"1": 6, "2": 7, "3": 9}}`.

Feature	Type	Missing	Distinct nonmissing	Minimum	Median	Maximum	Mean	Population SD
Whether_of_not_the_TA_is_a_native_English_speaker	categorical	0	2	n/a	n/a	n/a	n/a	n/a
Course_instructor	numeric	0	24	1	13	25	13.822581	6.9607753
Course	numeric	0	23	1	3.5	25	8.0806452	6.9683579
Summer_or_regular_semester	categorical	0	2	n/a	n/a	n/a	n/a	n/a
Class_size	numeric	0	42	5	27	66	28.193548	12.968123

Exact development category frequencies:

- `Whether_of_not_the_TA_is_a_native_English_speaker`: `{"1": 23, "2": 101}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.

- `Summer_or_regular_semester`: `{"1": 19, "2": 105}`. Categories absent from the other four folds: `{"0": [], "1": [], "2": [], "3": [], "4": []}`.

Explicitly omitted or unknown statistics

Sealed target values, target distributions, class balance, feature distributions, missingness and category cardinalities are deliberately unread. Their row counts and expected file identities come from frozen metadata only. Raw full-dataset distribution statistics are therefore unavailable; development summaries must not be labeled full-dataset summaries. Checkpoint dataset exposure, unrecorded subject/time identifiers and original licenses marked unknown remain unknown. The broad S3 catalog is a stored metadata inventory, not fresh payload validation, admissible-teacher inventory or a count of trained models. Exact current panel ranks, per-category wins and Elo await the unchanged panel. Final cold installation, final warm target performance, actual GPU training hours and trained-checkpoint consumption await their own receipts.

13 / Source map and companion files

Current implementation references

These links bind the reviewed implementation commit, not a moving branch. The merged main has the same source tree. Hashes below are SHA256 of file bytes; Git commit/tree identifiers use Git's separate object identity. A link needs access to the private repository.

Current source	SHA256
benchmarks/panel.json	ba7dace4cd535687002cc95538dbc7f7aaf77ec973177c7159b3c6d22471eee0
benchmarks/prepare.py	7600de95f0769309d814d80c6723d65f0588124b3e0d1a9875d8155fcc4a91c9
crates/feat-math/src/advanced_v3/fitted_cpp.rs	c20d7e9e7d530102435bdfd07a7c5275de041742a38d681842b87147e6c82fef
crates/feat-math/src/emit_cpp.rs	531e4d006caac9c41f3934df8f0f84d1f62749a9380ab4b67d1ec21180e65157
crates/feat-math/src/emit_js.rs	2e18c3a0fa85ff4fb0013d610be9404079956d9331c9988486b828bc7ac805cf
crates/feat-math/src/emit_py.rs	138a8c2a279da1b9df7b3c155606ac7e18c145982400ca7ace40f120cf386ca6
crates/feat-math/src/emit_rs.rs	92d1277a90886073ba5b3aa19aa9fb461f97ad3290c2c4a91a1ccaeec7c1d405
crates/feat-math/src/transpile.rs	374d7c1e54c9199b932963f564306deec8ffcc1f0ac7add5554fdcee04e0892a
crates/jope-bench/src/panel/campaign.rs	95e162538f7601f40bd2e7019639a6622259e516fa38cd1f92e83a0d4117b20c
crates/jope-bench/src/panel/campaign_cli.rs	419262eab7516ba0ede8432a5d1f162fd4d64047681764e55edab9e438bb2ad1
crates/jope-bench/src/panel/cli.rs	f1f9879d0b2de5473f8fcd9b3722e5c41147d5e33b26adac0948d0f4541a7b21
crates/jope-bench/src/panel/methods.rs	170b3fcd3fadc140c5d3006e34372a8042d6c8939395a3f633085b884e1cdaa6
crates/jope-bench/src/panel/optimize_cli.rs	17af561a039fd4832dd1bd23c226bbde5065b09b5273eb6e0b1353899042bd80
crates/jope-bench/src/panel/receipt_timing.rs	0a479d36f84c9c22381310e9dd83ca4cf4c48594a7541c666587c0226f4a7367
crates/jope-bench/src/panel/score.rs	10093ed25a81940c7ab985843e742efc49990f3d361c92764d6d6dcae07e81e9
crates/jope-bench/src/panel/selection_exposure.rs	a591787e6eaa95bb2e11d4c46776707076016b06785c4fad1dbc71d1a4d9fc8e
crates/jope-cli/src/kernel.rs	d8f26b2eb49652d61600c699155e7b69664bd0e868e1fc76229da012e6e547c8

Current source	SHA256
crates/jope-cli/src/main.rs	4b2990e5294d0668a8d5f7a6c1e809d91fbb93be1418fc3fb54f291c88251265
crates/jope-cli/src/schema.rs	307ce6a4fa3c22bef7c15e7272bd834100ba78715eba7b1c0547a2e1b6b04d1d
crates/jope-paths/src/lib.rs	abc95bfa0e258642e0101b2f82c07706492dfe1d6c8a6fd99043a475893aa120
crates/jope-pipeline/src/cli.rs	1bd10526c2d58c233f5bb1e9bd575bb5c7afa82691b2238ad2f6993f88bd6ae8
crates/jope-pipeline/src/cpu_context_bridge.rs	48dbe30216bf42520df668ca5b3d4a4138d21ae2731ba07c349fe6ec232729ff
crates/jope-pipeline/src/dataset.rs	3f34436e6d0bbb747c846d63dfe3bcbc4e1ce4e34863caa37d7dce0c7183e7ba
crates/jope-pipeline/src/emit.rs	bb351af0947653caf9bfb5d276efd8b2645599503a101f1100b9fc20bb07edb9
crates/jope-pipeline/src/execution.rs	e38f68bbb765918f421e69d82fa697d822d60a53eb10c51e30cc9f1ec8066a76
crates/jope-pipeline/src/experiment.rs	cec0f36073f73de8c09f508512c18e0098bb3f4028af08081d1b6a8639ab2e1e
crates/jope-pipeline/src/front_door.rs	71dc0402ff413f020a25983a5ba98386324c16af03363f5deae4fafceafd856a
crates/jope-pipeline/src/neural_search.rs	a8cd55c9c4f7b5a1060eab5979deb76db24484360389f2072f6b906c4bae0390
crates/jope-pipeline/src/search_context.rs	e423e9676a75426cdb6fdbbaf908e80c4bc1c89c5d93007aedcdfc0189cd382e
crates/jope-pipeline/src/supervisor.rs	a61265692d61580fe174c44a5f1cc2971e6512b803d91bc480d38c90a73a16b6
crates/jope-policy/src/config.rs	a740ca64edde353c9624f5fdc3b63878c1200ee95c159a71bca1556abc28ed37
crates/jope-policy/src/context_producer.rs	f218d0539e273df4f53263c8ca4f06639d68456ab2057a195bf3253e27aa32d2
crates/jope-policy/src/context_producer/admission.rs	2e98364dfe3994a21ccc2d6115adb7d7396c3de3a9babf15bcae473d4d59c441
crates/jope-policy/src/context_producer/input.rs	0fc3df5ed519439dd47fd27022e4be11594924a46fff4c926901cb514f5e4609
crates/jope-policy/src/lib.rs	2b0da78dcd569de31fdd0922c90e56e19e404ba9f1e184a1eb61cf634ac733a8
crates/jope-policy/src/model.rs	7d0f464fe4bd51ec4b2afa2257678565f959760c74bc347a7d1cebff3efe1e25
crates/jope-policy/src/ranking.rs	92c31cc55f3943aea2246bc0191325fbd625d2019fd49ce33c54f505b751843c
deploy/ci/e2e-smoke.sh	6becfc97830d87907bfb594aae247a117fb23162382d77bc75fbd770b4668354

Current source	SHA256
deploy/ci/emitters.sh	beb99faf01c60b4f39b0179eab46696bcd5cc10658b8cdd690962bbd35254e6b
deploy/ci/emitters/compare-bits.py	eaee5cc96d3d3161c85fefc02d636874adb4d399c340863534d83db5e8fe6249
deploy/ci/emitters/driver.cpp	899ad86994c93ac37312a66acde6900948948b22bfae1dd0be1516c7e99ca416
deploy/ci/emitters/driver.js	754b98bc8d694ec482e23826412e7dcbde76717e7d2f72d18a9adeb76ad12f75
deploy/prepare-panel-runtime.py	f9a0d1d9ca97dfd7664a39a8df6b1e364e738fdfaaa13a0e327402c073ee46f5
python/src/jope/policy/model_v3.py	11ab74d12bda8a2fb817b073a685d128eff06479f769e4301488339424314ffe

Pending source references

These are stopped custody patch views. They are not installed capabilities and are not linked as if they existed at the reviewed Git commit. The training/product chapters and detailed checklist name their patch identities and acceptance gates.

Pending path	SHA256 of inspected view
agent-delivery-refresh-68f56404/replay-001/source-text/crates/jope-bench/src/panel/demo_asm.rs	dd1a287c4ff3c1447a20e81d2c2990913923cd48c667f3be600f5d6b3c972f62
agent-delivery-refresh-68f56404/replay-001/source-text/crates/jope-bench/src/panel/fitted_export.rs	2f6e75d70e6d89bbfeac7db7ec67c585e6d7fc5c6287b134851db931ef5ffb30
agent-delivery-refresh-68f56404/replay-001/source-text/crates/jope-bench/src/panel/fitted_export_run.rs	3b007f44198aaa7b96fbca4936ed7ad551feea086f6018f1ff0458ddee6c7295
agent-delivery-refresh-68f56404/replay-001/source-text/docs/private-fitted-qualification.md	abf27b0dd7fabc0ee4a1f095d37d2ee4f77c5ffdfd2988699990521089f61ef8
agent-delivery-refresh-68f56404/replay-001/source-text/docs/private-warm-target.md	4234468c7f9b5348f7a264f815d10d6f62baf1f4b5397c6cf1fed644a12d97a8
agent-product-final-68f56404/replay-001/source-text/crates/feat-math/src/advanced_v3/fitted_portable.rs	f2f8998251f2e7e97900563a7032b7d8b8c80bf43e080b7b622b2e195f75737f
agent-product-final-68f56404/replay-001/source-text/crates/feat-math/src/advanced_v3/fitted_portable/scalar.rs	34823e72c9576c7025d6bbb31a44203f397e1d571642d516a63b5ef328413970
agent-product-final-68f56404/replay-001/source-text/crates/jope-bench/src/panel/demo.rs	00751a3094b42c6f742cffed5f7804803977f47d9f2837b161421e884f7efdfc
agent-product-final-68f56404/replay-001/source-text/crates/jope-bench/src/panel/demo_asm.rs	74c2c6164015bc65f53d021c77f78d9a9ae7238d6c68bb007da424328707064d
agent-product-final-68f56404/replay-001/source-text/crates/jope-bench/src/panel/inference.rs	c4929b553dd6893a48b836c2c3838b62b9d96fbd2c5f76634e235eaeed98e626
agent-product-final-68f56404/replay-001/source-text/deploy/install-private.py	3e4e16d5d7e224624c970aca7217952339c8e679b12752aa8317452c27120e9c

Pending path	SHA256 of inspected view
agent-product-final-68f56404/replay-001/source-text/docs/private-install-demo.md	05d728bba5fdc78c1568650ef8179d71ce19cf5c8860ad23317c8cea53dfebb6
crates/jope-bench/src/panel/experimental_proposal.rs	d03c765ac08d1ecc039a82f3dbea0c40bc5158baac042e8df348f0c91a90a018
crates/jope-bench/src/panel/optimize_cli.rs	487a299045d9a24604511cdecd8d4fa0b64ff055e4557339ce34120858549eca
crates/jope-bench/src/panel/origin_proof.rs	229d26685d60646887e5ec30ec4734aa78dfdb2f37788ffbf30f878ce593b675
crates/jope-cli/src/train.rs	486235575fc972306a6294233d94f8a37179945931e74bfb298e342752c634a2
crates/jope-cli/src/train/corpus.rs	a239fd8633848178214270339 63d8c0c5ebc54642130b087a9848aa51f2d93c5
crates/jope-learning/src/eligibility.rs	839e1de7b5f2c869817564ed39f3c30843208f7a00a1a277d27fe50857ce1e8d
crates/jope-learning/src/eligibility/inner.rs	b5fb468fcc554ec7da9227244d435d84593910c254678fb3f4174b5d7a4e9215
crates/jope-learning/src/record.rs	4360d96a4a6c0a27f5ba78fbcab4f5be10a3a800f56acee6379ef805b3815d61
crates/jope-pipeline/src/experimental_producer.rs	9a15e7077651ded2adf364a26a579720ae21aa92f0362c0122d132e34162411c
crates/jope-pipeline/src/search_plan.rs	fe8e402357b9b61359b03760401da9a568b8a1e7092b1222947fc30d2c8fb817
crates/jope-policy/src/context_producer/classification.rs	465af30abcdc7bcc02c046d92a71e0b7a22be910320e22158ef16b216adc4fa6
metadata-receipts/veox-jopedime-models-use2--bundles--hyperion-v7--manifest.json	fdaf938fc7f7957c06959f12c130b2bd81cdaef5e5bb2198ba75cf35da1c5233
metadata-receipts/veox-jopedime-models-use2--bundles--jope-prime-gen31-v2--manifest.json	c8d8d13a2f4cb2eeb21eea364c33e14429c84d344e6b50eb38a472173b97b6f2
metadata-receipts/veox-jopedime-models-use2--bundles--policy-step-60000--manifest.json	d8e259b26ef8d580ddb99ae5826c9501bcdc0c32045945eef9010926fd796e9e
python/src/jope/policy/experimental_proposal.py	f0338f0505a27c9f2f5eac7579e2c506da5c116b3a206f44c52a1cf87cf9c0aa
python/src/jope/policy/training_campaign.py	9d9cfb3257ecf2b6072cab8adc6d830f3df096605a06ff0677e6338c2aad5a2c

Machine-readable companions

The manual directory contains the following locally reproducible evidence. The original metadata/custody paths inside receipts describe where observations were obtained; use the copies under this directory when available. Historical preparation receipts intentionally retain their pre-execution wording. The separate acceptance receipt supersedes that status without rewriting history.

Companion	Purpose
`data/dataset-inventory.json`	Thirty complete development dataset records and 522 feature records
`data/catalog-summary.json`	Corpus task, row, feature, origin and license metadata totals
`data/metadata-receipts/`	Catalog JSONL and collection/model metadata; no payload bodies
`data/s3-estate.json`, `data/local-estate.json`	Dated current-object and local logical-byte inventories
`data/performance-evidence.json`	Current proof, historical score cells, reports, ladder identities and scope
`data/plot-*.csv`	Compact plot inputs with snapshot and source pins
`data/roadmap.json`	Detailed 38-item dependency/acceptance checklist
`docs/asm-maintainer-status.plist`	XML property-list representation of the same checklist
`data/example-acceptance/`	Actual fourteen-command result, audit and handoff
`examples/`	Private synthetic fixture, exact argument packet and complete-output validator
`figures/`	Reproducible charts and associated historical data inputs
`render-receipt.json`	PDF digest, pages, consumed input hashes and rendering environment

Rebuilding the book

From the repository root, use the documented pinned rendering environment and these local-only commands:

```
OPENBLAS_NUM_THREADS=1 OMP_NUM_THREADS=1 python3 docs/internal/asm-manual/tools/
make_inventory_figures.py
OPENBLAS_NUM_THREADS=1 OMP_NUM_THREADS=1 python3 docs/internal/asm-manual/tools/make_figures.py
python3 docs/internal/asm-manual/tools/build_manual.py
pdfinfo docs/asm-maintainer-manual.pdf
pdftotext -layout docs/asm-maintainer-manual.pdf /path/to/durable/review/manual.txt
```

Rendering uses Python Markdown, BeautifulSoup and WeasyPrint with system DejaVu fonts; their observed versions and font hashes are retained in the render receipt. Plotting uses Matplotlib and NumPy. The renderer rejects missing/nonlocal images and failed resource fetches, and checks that inputs did not change during rendering. It does not run models, fit predictors, fetch datasets or open sealed labels. Inspect the resulting pages and extracted text before treating a successful render as a complete document.